

Escaping Uncertainty Traps*

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Abstract

Which policies escape *uncertainty traps*, i.e., demand recessions with self-fulfilling uncertainty? We show that uncertainty traps are endemic to standard New Keynesian models, provided one preserves households' optimal response to risk. Moreover, no conventional Taylor rule can eliminate uncertainty traps. We identify three classes of policies which are effective. The first is an enriched monetary rule targeting risk premia in recessions. Second, a regime of fiscal dominance, which involves partially-unbacked spending alongside an accommodative monetary policy, prevents uncertainty traps. The third effective policy does unbacked spending only during non-fundamental recessions, which works without leading to the traditional fiscal equilibrium.

JEL Codes: E30, E40, G01.

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Policymakers worry that changes in expectations can drive economic fluctuations, even with no underlying shifts in fundamentals. Seeking to mitigate such expectation-driven dynamics, central bankers often turn to countercyclical stabilization policies. For instance, conventional wisdom says that the central bank should respond to demand recessions by aggressively lowering the nominal interest rate. Without this monetary response, demand recessions can become self-fulfilling: households' expectations of lower demand and lower inflation push up the real interest rate, which dampens households' actual demand and thereby confirms the original expectations. By lowering interest rates sufficiently, the central bank prevents the initial pessimism from becoming self-fulfilling.

This paper presents a novel analysis—centered on the role of aggregate risk—and shows that eliminating non-fundamental fluctuations requires a new policy paradigm. We begin by unveiling what we dub *uncertainty trap* equilibria in New Keynesian (NK) models. Given our focus on uncertainty, we deviate from the textbook analysis in one respect: we preserve agents' optimal response to risk, which precludes linearization, and are therefore concerned with global rather than local determinacy. In this setting, we provide a relatively general characterization of uncertainty traps, namely self-fulfilling equilibria featuring risk that rises in recessions. Such equilibria exist even when risk is mildly countercyclical, permitting equilibrium dynamics that conform with empirical evidence. Importantly, no matter how aggressively monetary policy responds to inflation and output, uncertainty traps can arise. In this sense, the Taylor principle fails.

We then provide three types of policies—one monetary and two fiscal—that succeed in eliminating uncertainty traps. The first is a monetary rule that explicitly targets risk. The second is a regime of fiscal dominance, namely unbacked government spending. The third also does unbacked spending but only in non-fundamental recessions. At a high level, all three policies are effectively ways to commit to sufficient stimulus in the event of an uncertainty trap. In that sense, they extend the conventional wisdom by dictating that the size of recessionary stimulus be greater when risk also spikes. Yet the policies are not equivalent; we discuss their pros and cons.

Uncertainty traps. To understand the key forces of our equilibria, suppose households suddenly expect the economy to enter a recession. Moreover, suppose aggregate demand volatility is countercyclical. In that case, the onset of the recession raises households' uncertainty about the future. Uncertainty being higher, households increase precautionary savings and reduce consumption, thereby justifying the initial expectation of a recession. In addition to inducing a recession today, precautionary savings shifts consumption into the future, setting the economy up for a later recovery.

Critical to the mechanism described above is that endogenous aggregate demand volatility is countercyclical. This can happen in NK models because demand volatility is not fully pinned down by equilibrium restrictions and can be shaped by coordination. But not any volatility profile is admissible. Instead, a volatility profile is consistent with equilibrium only if the implied output dynamics remain *globally stable*—that is, neither explosive nor implosive. If uncertainty rises sufficiently in our story above, then future demand recovers after the initial recession, which is precisely what ensures global stability. In fact, global stability, which is a statement about the tail behavior of the economy, really only requires that volatility rise sufficiently in *deep* recessions. Thus, one should think of sufficiently countercyclical volatility as a result, given the global stability requirement. Or put differently, households’ belief in countercyclical volatility is self-fulfilled by the recovery path it induces. Using stability as a guide, we characterize a broad class of volatility profiles that give rise to uncertainty traps.

So far, we have abstracted from monetary policy. A conventional Taylor rule tries to thwart non-fundamental recessions by lowering the interest rate. Without uncertainty, this policy works. But with uncertainty, the policy faces headwind: while lower interest rates provide stimulus that raise today’s consumption, higher uncertainty induces precautionary savings that reduce consumption. If uncertainty rises sufficiently, precautionary savings dominates in the sense that, on net, aggregate demand falls today but is expected to recover later. A demand recession emerges today, despite the Taylor rule. Consequently, *no conventional Taylor rule can eliminate all uncertainty traps*: for any such rule, sufficiently countercyclical uncertainty can make precautionary saving dominate monetary stimulus. Theorems 1-2 formalize this logic. The upshot is that uncertainty traps are endemic to NK models.

Are uncertainty traps plausible? We argue they are, for a few reasons. First, uncertainty traps display realistic dynamics. Not only do they feature countercyclical volatility, as in the real world, but our numerical examples show how such equilibria can be sustained by empirically-plausible levels of output-growth volatility and inflation. Second, uncertainty trap equilibria are non-explosive, meaning they do not require any hyperinflations or economic collapse, unlike the indeterminacies raised by [Cochrane \(2011\)](#). Non-explosiveness means that our equilibria are robust to popular “escape clauses” in the monetary economics literature, such as policies that eliminate explosive inflation paths ([Obstfeld and Rogoff, 1983, 1986](#); [Atkeson, Chari, and Kehoe, 2010](#); [Sims, 2013](#)).

Policy remedies. Given that conventional Taylor rules are insufficient, which policies can help an economy escape uncertainty traps? We identify three classes of policies.

First, we explore augmented monetary rules that directly target risk. This can work: strong *risk targeting* during downturns (i.e., an interest rate that responds more than one-for-one to the risk premium in recessions) eliminates the uncertainty trap (Theorem 3). Intuitively, the problem with the conventional Taylor rule is that it only provides sufficient stimulus in a conventional demand recession but not one that also features an uncertainty increase. Instead, risk targeting enforces sufficient stimulus in every demand recession, including uncertainty-driven ones. Suppose that in any uncertainty trap, monetary policy always reduces nominal rates enough to lower the risk-adjusted real rate—the one relevant for intertemporal consumption decisions. This creates demand stimulus today and is thus inconsistent with a recession. An obvious issue with all such monetary rules is the presence of an effective lower bound. Unless unconventional monetary policy (not modeled in this paper) can fill the void left by a constrained interest rate, an alternative policy is needed.

Second, uncertainty traps are eliminated by *fiscal dominance*—a policy regime characterized by “active fiscal” and “passive monetary” policies (Theorem 4). In this setup, the fiscal authority does not act to ensure debt sustainability under any and every possible shock—some shocks can be “unbacked.” Think of this as stimulus: unbacked government borrowing creates demand via wealth effects. To understand this, consider a non-fundamental recession that then reduces government surpluses (e.g., if tax revenue is increasing in output). The fiscal authority does not raise future taxes or reduce future spending to make up for the current deficits, so the market value of debt exceeds the present value of future surpluses. Government debt becomes net wealth for households, who spend the excess until this net wealth disappears. All uncertainty traps are eliminated by this policy stance. The key cost of fiscal dominance is that aggregate demand inherits fiscal shocks. Although the volatility from uncertainty traps vanishes, fiscal volatility emerges, reminiscent of a trade-off in [Bianchi and Melosi \(2017\)](#). In some sense, this policy is heavy-handed in that all spending is unbacked, and so fiscal policy steers aggregate demand at all times, not just in uncertainty traps.

Third, we analyze a more refined type of fiscal policy, which we call *fiscal volatility targeting*, that resolves some issues with the previous two policies. Fiscal volatility targeting does unbacked spending, hence a type of stimulus, but only in non-fundamental recessions. In other cases, when times are normal or a recession is for fundamental reasons, spending is completely backed and fiscal policy ensures debt sustainability. We prove that fiscal volatility targeting also eliminates uncertainty traps while keeping the economy insulated from fiscal shocks (Theorem 5). Meanwhile, monetary policy obeys the Taylor principle and steers the economy toward the “fundamental equilibrium.” In-

terestingly, and potentially powerfully, fiscal volatility targeting can be just as effective even if it is structured as an “escape clause,” namely an intervention that only occurs in deep recessions. Fiscal volatility targeting is thus a clean theoretical benchmark to illustrate how selective fiscal backstops can eliminate non-fundamental volatility without moving fully to fiscal dominance. That said, an implementation challenge is that policymakers must identify which recessions are non-fundamental. By misclassifying normal recessions as non-fundamental, the policy still selects an equilibrium, but its sub-optimal design introduces some excess volatility into the economy.

Overall, one can think of our policy menu in one of two ways. On the one hand, policymakers seeking to eliminate non-fundamental recessions should weigh the pros and cons of these policies. On the other hand, academic researchers often think of policies as determining the equilibrium in NK models. From this perspective, we are providing a recipe for equilibrium determinacy that differs from the literature.

Related literature. Beginning with [Sargent and Wallace \(1975\)](#), a literature explores indeterminacies in monetary models. Several papers have established indeterminacies in NK models related to the ZLB ([Benhabib, Schmitt-Grohé, and Uribe, 2001](#); [Benigno and Fornaro, 2018](#); [Heathcote and Perri, 2018](#)). By contrast, our uncertainty traps do not rely on the ZLB and hold under any conventional Taylor rule.¹ Related to our mechanism, heterogeneous-agent NK models can feature indeterminacy if idiosyncratic income risk is countercyclical ([Acharya and Dogra, 2020](#); [Ravn and Sterk, 2021](#); [Bilbiie, 2024](#); [Acharya and Benhabib, 2024](#)). Our uncertainty traps are novel in that they vitally depend on *aggregate* risk.

We go beyond most of the literature by studying the global, stochastic dynamics of NK models. Similar to our paper, [Caballero and Simsek \(2020\)](#) also analyze global dynamics to show how aggregate risk reduces aggregate demand, but restricting attention to the “fundamental equilibrium.”² Closely related to our study, contemporaneous work by [Lee and Dordal i Carreras \(2026\)](#) also studies a nonlinear IS curve with aggregate risk driving multiplicity. Our analysis departs from theirs in several important ways. First, we move beyond examples by characterizing a general class of self-fulfilling volatility profiles and identifying the key economic force behind them: countercyclical volatility

¹Thus, our paper adds an additional critique of conventional Taylor rules, beyond the fact that they may permit hyperinflations ([Cochrane, 2011](#)) or may be time-inconsistent ([Neumeyer and Nicolini, 2025](#)).

²To our knowledge, the nonlinear, stochastic NK literature focuses on the MSV solution and ignores the determinacy issues we raise. For example, [Rudebusch and Swanson \(2012\)](#), [Kung \(2015\)](#), and [Basu and Bundick \(2017\)](#) all study a (particular) third-order perturbation around steady state. [d’Avernas, Hubert de Fraisse, Ning, and Vandeweyer \(2024\)](#) go beyond perturbation methods and study the risk effects of quantitative easing in a *global solution* of an NK model.

in deep recessions. Second, we prove that self-fulfilling uncertainty can survive *any* Taylor rule, including nonlinear ones. Third, we demonstrate that such equilibria can deliver quantitatively-reasonable dynamics. These generalities help illustrate that uncertainty traps are not simply an unrealistic theoretical curiosity but are, in fact, endemic to NK models: a large class of empirically-realistic volatility profiles can emerge and survive the conventional monetary policy paradigm. Having mentioned those differences, our most important novel analysis is our exploration of policies that eliminate uncertainty traps. Whereas [Lee and Dordal i Carreras \(2026\)](#) identify a single knife-edge monetary rule that can eliminate self-fulfilling volatility, we demonstrate that a large class of policies, including various fiscal regimes, can be successful.

A literature on the fiscal theory of the price level (FTPL) also explores determinacy in NK models under various policy profiles.³ Unlike most of this literature, we analyze the nonlinear stochastic dynamics of the model. Some papers have extended FTPL to nonlinear stochastic environments, but almost exclusively with *flexible prices*. For example, [Bassetto and Cui \(2018\)](#) and [Brunnermeier, Merkel, and Sannikov \(2023, 2024\)](#) study stochastic and nonlinear flexible-price models, focusing on the determinacy of a bubble in government debt valuation.⁴ Two exceptions that do allow sticky prices, but sidestep our determinacy questions, are [Mehrotra and Sergeyev \(2021\)](#) and [Li and Merkel \(2025\)](#).⁵ Overall, we provide the first nonlinear, stochastic FTPL analysis in textbook NK models. Our fiscal volatility targeting policy is also new to the literature.

Finally, our paper connects to a broader set of models that predict self-fulfilling aggregate uncertainty. Some examples are [Bacchetta, Tille, and Van Wincoop \(2012\)](#), [Benhabib, Wang, and Wen \(2015\)](#), [Fajgelbaum, Schaal, and Taschereau-Dumouchel \(2017\)](#), and [Khorrami and Mendo \(2024, 2025a\)](#). Whereas these papers build on financial and information frictions, our setting only features nominal rigidities.

1 Baseline Model

We present a canonical New Keynesian (NK) economy with complete markets and nominal rigidities (e.g., [Galí, 2015](#)). We study the model in continuous time for tractability.

³Seminal contributions include [Leeper \(1991\)](#), [Sims \(1994\)](#), [Woodford \(1994\)](#), [Kocherlakota and Phelan \(1999\)](#), [Cochrane \(2001\)](#), and [Bassetto \(2002\)](#).

⁴Others studying the FTPL in nonlinear, but *deterministic*, environments with liquidity premia include [Berentsen and Waller \(2018\)](#), [Williamson \(2018\)](#), [Andolfatto and Martin \(2018\)](#), and [Miao and Su \(2024\)](#).

⁵[Mehrotra and Sergeyev \(2021\)](#) study fiscal sustainability with real debt, but with exogenous output. [Li and Merkel \(2025\)](#) study FTPL in a NK model with idiosyncratic risk, which can induce a government debt bubble; however, they avoid determinacy questions by assuming inflation and the output gap are Markovian in exogenous states and government bonds outstanding (i.e., they assume an MSV solution).

Appendix D provides analogous results in the discrete-time version.

Uncertainty. Our baseline model features no fundamental uncertainty in preferences or technologies. To transparently incorporate some uncertainty, we introduce a sunspot shock: a one-dimensional Brownian motion Z . All random processes will be adapted to Z . It is not critical that the shock be a pure sunspot. Appendix B shows that analogous multiplicity arises when Z is a fundamental monetary-policy shock. The deep point is that uncertainty is not pinned down—the identity of the shock is less important.

Preferences. The representative agent has rational expectations and time-separable utility with discount rate ρ , unitary EIS, and labor disutility parameter φ :

$$\mathbb{E} \left[\int_0^\infty e^{-\rho t} \left(\log(C_t) - \frac{L_t^{1+\varphi}}{1+\varphi} \right) dt \right]. \quad (1)$$

Consumption C_t has the nominal price P_t and labor L_t earns the nominal wage W_t .

Technology. The consumption good is produced by a linear technology $Y_t = L_t$. We abstract from fundamental uncertainty (e.g., productivity shocks) for maximal clarity.

Behind the aggregate production function is a structure common to most of the NK literature. In particular, a continuum of firms produce intermediate goods using labor in a linear technology. These intermediate goods are aggregated by a competitive final goods sector. The elasticity of substitution across intermediate goods is a constant ε . The intermediate-goods firms set prices strategically, described next.

Price setting. Intermediate-goods firms set prices strategically, taking into consideration the impact prices have on their demand. Price setting is not frictionless: firms changing their prices are subject to quadratic adjustment costs, a la Rotemberg (1982). (For simplicity, we assume these adjustment costs are non-pecuniary, so that resource constraints are not directly affected by price adjustments.) In the interest of exposition, we relegate the statement of and solution to this standard problem to Appendix G.

Definition: inflation and output gap. Let P_t denote the aggregate price level and $\pi_t := \dot{P}_t / P_t$ its inflation rate.⁶ Note also that the flexible-price level of output is given by $Y^* = (\frac{\varepsilon-1}{\varepsilon})^{\frac{1}{1+\varphi}}$. Following the literature, define the output gap $x_t := \log(Y_t/Y^*)$. Conjecture

⁶This notation reflects the standard sticky-price implication that the aggregate price level is locally deterministic in continuous time. Appendix G verifies this property in the Rotemberg environment.

that x_t and π_t have dynamics of the form

$$dx_t = \mu_{x,t}dt + \sigma_{x,t}dZ_t \quad (2)$$

$$d\pi_t = \mu_{\pi,t}dt + \sigma_{\pi,t}dZ_t \quad (3)$$

for some $\mu_x, \sigma_x, \mu_\pi, \sigma_\pi$ to be determined in equilibrium.

Monetary policy. Let ι_t denote the nominal short-term interest rate, which is controlled by the central bank. Monetary policy follows a Taylor rule that targets the output gap and inflation with

$$\iota_t = \bar{\iota} + \Phi(x_t, \pi_t), \quad (\text{MP})$$

for some long-run target rate $\bar{\iota}$ and some response function that satisfies $\Phi(0, 0) = 0$. In this model without fundamental shocks, the standard optimal target rate is $\bar{\iota} = \rho$. For the response function, a common example is the linear rule

$$\Phi(x, \pi) = \phi_x x + \phi_\pi \pi. \quad (\text{linear MP})$$

The Taylor principle, needed for local determinacy, says that Φ is increasing sufficiently fast in both x and π .⁷ We address the zero lower bound (ZLB) in an extension below. For now, think of negative interest rates as a proxy for unconventional monetary policy that can work even when the short rate is zero.

Financial markets. Financial markets are complete. Let M_t be the real stochastic discount factor induced by the real interest rate $r_t := \iota_t - \pi_t$ and the equilibrium price of risk h_t associated to the shock Z_t . The equity market is a claim on the profits of the intermediate-goods producers. Alternatively, we can think of these profits as being rebated to the consumers lump-sum. The risk-free bond market is in zero net supply—this will be generalized in Section 4 when we introduce “active” fiscal policies.⁸

Definition 1. An *equilibrium* is processes $(C_t, Y_t, L_t, W_t, P_t, M_t, B_t, \iota_t, r_t, \pi_t)_{t \geq 0}$, such that

- (i) Taking (M_t, W_t, P_t) as given, consumers choose $(C_t, L_t)_{t \geq 0}$ to maximize (1) subject

⁷With a linear rule, the Taylor principle requires $\phi_x > -\rho$ and $\phi_\pi > 1 - \rho\phi_x/\kappa$.

⁸For our baseline results, it is not necessary to completely shut down fiscal policy and assume the absence of government debt. Instead, the reader can think of fiscal policy levying lump-sum taxes and transfers but acting “passively” to always ensure its own solvency. As we show in Section 4, this “passive fiscal” interpretation generates an identical set of equilibria to those we uncover in Section 2.

to lifetime budget and No-Ponzi constraints. The lifetime budget constraint is

$$\frac{B_0}{P_0} + \Pi_0 + \mathbb{E} \left[\int_0^\infty M_t \frac{W_t L_t}{P_t} dt \right] \geq \mathbb{E} \left[\int_0^\infty M_t C_t dt \right] \quad (4)$$

where Π represents the real present-value of producer profits and B represents the bond-holdings of the consumer. The No-Ponzi constraint is $\lim_{T \rightarrow \infty} M_T \frac{B_T}{P_T} \geq 0$.⁹

- (ii) Firms set prices optimally, subject to their quadratic adjustment costs.
- (iii) Markets clear, namely $C_t = Y_t = L_t$ and $B_t = 0$.
- (iv) The central bank follows the interest rate rule (MP) for some given long-run target rate $\bar{r}$ and some given response function $\Phi(\cdot)$.

Equilibrium characterization. We first provide a summary characterization of all equilibria. Labor supply and consumption satisfy the following optimality conditions:

$$e^{-\rho t} L_t^\varphi = \lambda M_t \frac{W_t}{P_t} \quad (5)$$

$$e^{-\rho t} C_t^{-1} = \lambda M_t, \quad (6)$$

where λ is the Lagrange multiplier on the lifetime budget constraint (4).

On the firm side, Appendix G shows that optimal firm price setting gives rise to aggregate inflation dynamics that satisfy

$$\mu_{\pi,t} = \rho \pi_t - \eta \varepsilon \frac{W_t}{P_t} + \eta(\varepsilon - 1), \quad (7)$$

where η is each firm's degree of price flexibility. Notice that as $\eta \rightarrow 0$ (prices changes become infinitely costly), one possible equilibrium is to have $\pi_t \rightarrow 0$ for all times. We will assume this "rigid-price limit" is the equilibrium that obtains as $\eta \rightarrow 0$.

We use these conditions to obtain an "IS curve" and a "Phillips curve." Applying Itô's formula to (6), we obtain the consumption Euler equation, which may be rewritten in terms of the output gap as

$$\mu_{x,t} = l_t - \pi_t - \rho + \frac{1}{2} \sigma_{x,t}^2. \quad (\text{IS})$$

⁹In addition, to prevent arbitrages like "doubling strategies" that can arise in continuous time, we must impose a uniform lower bound on borrowing, e.g., $M_t B_t / P_t \geq -\underline{b}$, although $\underline{b}$ can be arbitrarily large. This standard technical restriction plays no role in our multiplicity results below.

Equation (IS) is the IS curve. Next, divide the FOCs (5)-(6), and use goods and labor market clearing $C_t = Y_t = L_t$ to get $Y_t^{1+\varphi} = \frac{W_t}{P_t}$. Substitute this into (7) to obtain

$$\mu_{\pi,t} = \rho\pi_t - \kappa \left(\frac{e^{(1+\varphi)x_t} - 1}{1 + \varphi} \right), \quad (\text{PC})$$

where $\kappa := \eta(\varepsilon - 1)(1 + \varphi)$. Equation (PC) is the Phillips curve.

The most important novelty in our paper is the presence of precautionary savings due to aggregate risk. This force is captured by the term $\frac{1}{2}\sigma_x^2$ in (IS). We will often refer to this term as a “risk premium” because σ_x^2 is exactly the equilibrium risk premium on the aggregate consumption claim.¹⁰ When writing the IS curve in terms of log consumption, as is typically done, the Jensen correction of $\frac{1}{2}$ also shows up.

Transversality and non-explosiveness. Together with the monetary policy rule (MP), equations (IS) and (PC) form the nonlinear “three equation model” in standard NK models. An equilibrium is completely characterized by these three equations, along with conditions that ensure that any output gap or inflation explosions are consistent with optimization behavior. More specifically, agents’ transversality conditions must hold, and they must not obtain minus infinite utility. These are usually regarded as technical conditions, but they are important to recognize in our setting.

For example, since $C_t = e^{x_t}Y^* = L_t$, the representative agent would obtain minus infinite utility if $x_t = \pm\infty$ in finite time, or even if $x_t \rightarrow \pm\infty$ too quickly. Clearly, this is not compatible with optimizing behavior if the agent has an alternative that delivers finite utility. When $x_t \rightarrow \infty$, real wages diverge, and agents may obtain finite utility by supplying a finite amount of labor forever. When $x_t \rightarrow -\infty$, the existence of a finite-utility deviation is less immediate; the non-explosiveness criterion below simply excludes such impoverishment paths from the equilibrium class we study. Similarly, we show in Appendix G that firms’ optimization rules out situations when $\pi_t \rightarrow \pm\infty$ too quickly, because that would induce an infinite amount of price adjustment costs.

In order to emphasize that the multiplicity unveiled later does not rely on explosive behavior (unlike the multiplicity exposited in Cochrane, 2011), we only consider equilibria satisfying a simple condition that conforms with most existing literature and rules out both finite-time and asymptotic explosions. As shown in Appendix A.1, these non-explosiveness criteria are also sufficient to ensure all agents’ transversality conditions

¹⁰The aggregate risk premium is σ_x^2 for the following reason. Aggregate consumption growth volatility is σ_x . Due to log utility, agents optimally set their individual consumption growth volatility equal to the market price of risk (i.e., the sensitivity of $d \log M$ to dZ). Hence, the quantity of risk and price of risk are both equal to σ_x , implying the risk premium (as the product of risk quantity and risk price) is σ_x^2 .

hold. Obviously, relaxing these restrictions permits more equilibria than we unveil.

Condition 1. *A non-explosive allocation has*

$$\sup_T \mathbb{E}|x_T| < \infty, \quad \sup_T \mathbb{E}[\pi_T^2] < \infty, \quad \text{and} \quad \sup_T \mathbb{E}[e^{(1+\varphi)x_T}] < \infty. \quad (8)$$

We summarize our characterization in the following lemma. The proof is standard except for a careful treatment of potentially explosive behavior (see Appendix A.1).

Lemma 1. *Suppose processes $(x_t, \pi_t, \iota_t)_{t \geq 0}$ satisfy (IS), (PC), (MP), and Condition 1. Then, $(x_t, \pi_t, \iota_t)_{t \geq 0}$ corresponds to an equilibrium of Definition 1.*

All our results restrict attention to equilibria satisfying Condition 1. For that reason, we include the following definition.

Definition 2. *A non-explosive equilibrium is an equilibrium in which Condition 1 holds.*

Linearized Phillips curve approximation. The nonlinearity central to our mechanism is the precautionary-savings term in the IS equation (IS). To isolate this force and connect directly to the standard determinacy literature, the sticky-price analysis below uses the standard linearized Phillips curve. A further benefit is that this first-order Phillips curve is identical under common pricing frictions such as Rotemberg and Calvo pricing.¹¹ Appendix F.2 presents complementary analysis of deterministic dynamics based on the fully nonlinear Phillips curve (PC).

By the first-order approximation $\frac{e^{(1+\varphi)x}-1}{1+\varphi} \approx x$, the linearized Phillips curve is

$$\mu_{\pi,t} = \rho\pi_t - \kappa x_t. \quad (\text{linear PC})$$

Within this approximation, a non-explosive equilibrium is an adapted process $(x_t, \pi_t, \iota_t)_{t \geq 0}$ satisfying (IS), (linear PC), (MP), and Condition 1.

2 Uncertainty Trap Equilibria

This section contains our main results on uncertainty traps. For theoretical clarity, we assume the Taylor principle holds. As is well known, the Taylor principle renders the

¹¹That being said, the nonlinear Phillips curves in the Rotemberg and Calvo models are different. As one example, Ascari and Rossi (2012) show that Rotemberg and Calvo models can be quite different in settings with trend inflation. The key difference: the nonlinear Calvo model features a state variable capturing the cross-sectional dispersion in prices, which drops out in the linearization. We cannot guarantee that all of our sunspot equilibria would survive the inclusion of such a state variable, although we see no obvious reason why this would matter besides complicating the analysis.

deterministic equilibrium unique among non-explosive equilibria—we verify and extend such results in Appendix F. Nevertheless, a large class of non-explosive stochastic equilibria exists for *any* degree of monetary responsiveness to inflation or output. We begin with some intuition and examples to build understanding, followed by more general theoretical propositions. We also illustrate numerically that uncertainty trap equilibria can feature quantitatively-plausible amounts of volatility and inflation.

2.1 Understanding the basics of uncertainty traps

Before turning to the formal arguments, we outline the core intuition. Why can a non-fundamental recession today, $x_0 < 0$, be an equilibrium outcome once we incorporate *precautionary savings* into the analysis? For simplicity, we illustrate the core intuition in the rigid-price limit, $\kappa \rightarrow 0$. This reduces the analysis to a one-dimensional stochastic system for the output gap.

Using $\bar{r} = \rho$ as the policy target and imposing $\pi_t = 0$ in the rigid-price limit, combining (IS) with (MP) collapses the standard NK system to

$$\mu_{x,t} = \Phi(x_t) + \frac{1}{2} \sigma_{x,t}^2. \quad (9)$$

We assume monetary policy satisfies the Taylor principle, i.e., $\Phi'(x) > 0$. An equilibrium is a stochastic process for x_t that satisfies (9) and the non-explosiveness Condition 1.

To visualize the critical role of risk, Figure 1 plots a hypothetical recession with three different levels of uncertainty. Imagine $x_0 < 0$ occurs for completely non-fundamental reasons. The left panel shows what happens if the recession features no uncertainty: output implodes, i.e., $x_t \rightarrow -\infty$ as $t \rightarrow \infty$, and therefore the initial hypothetical recession cannot be part of a non-explosive equilibrium. By contrast, the middle and right panels show examples with uncertainty. Output now becomes stabilized during the recession. If uncertainty is high enough (right panel), then output in fact recovers. Since these are non-explosive paths, they are equilibria. That is, the initial non-fundamental recession is justified—this is an uncertainty trap.

Let us unpack the economics behind these examples. After the decline in output, the rule $\Phi'(x) > 0$ implies the central bank lowers the interest rate to stimulate the economy, generating an intertemporal-substitution effect that front-loads consumption. What happens after that completely depends on the amount of uncertainty.

Without uncertainty (left panel), this intertemporal-substitution effect is the only force governing the path of aggregate demand. Thus, for the recession to be part of

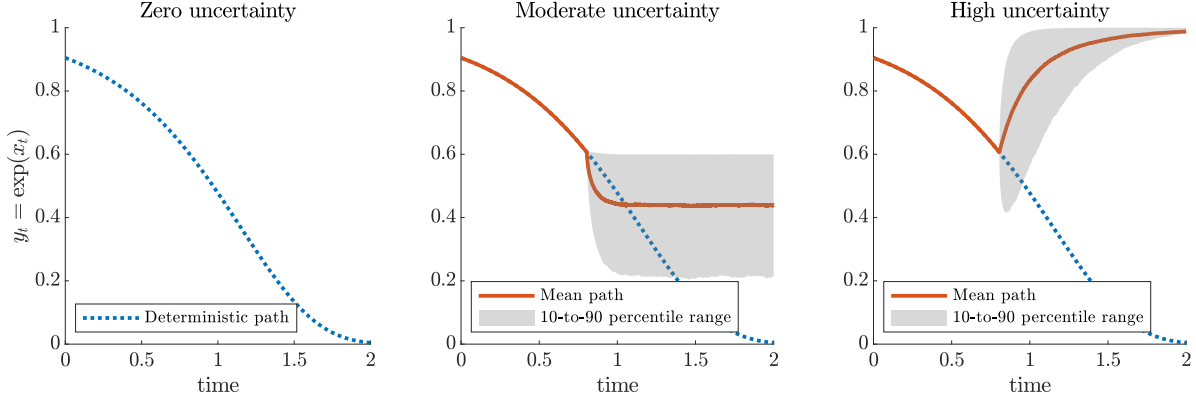


Figure 1: Output gap dynamics after a hypothetical non-fundamental recession ($x_0 = -0.1$) under an aggressive Taylor rule $\Phi(x) = \phi_x x$ with $\phi_x = 2$. The left panel has no uncertainty. The middle panel has just enough uncertainty to induce a permanent depression ($\frac{1}{2}\sigma_x^2 = \phi_x^2 x^2$ for $x < -0.5$). The right panel has higher uncertainty, which induces a speedy recovery ($\frac{1}{2}\sigma_{x,t}^2 = 2\phi_x^2 x_t^2$ after $\inf\{t : x_t < -0.5\}$).

an equilibrium, aggregate demand must keep declining over time, eventually leading the economy into a collapse. This implausible candidate path is ruled out by the non-explosiveness requirement. This is how the Taylor principle rules out non-fundamental recessions in conventional New Keynesian analysis.

If, instead, the recession is accompanied by higher uncertainty (middle and right panels), households engage in precautionary savings. Whereas the monetary stimulus is a force inducing front-loaded consumption, as discussed above for the deterministic path, precautionary savings is a force inducing back-loaded consumption. If the rise in uncertainty is large enough, then the net effect of these two competing forces—monetary stimulus versus precautionary savings—is a back-loaded consumption path. The initially conjectured recession thus becomes part of a non-explosive equilibrium.

The higher the uncertainty, the more precautionary savings, and the faster the recovery. In the middle panel of Figure 1, uncertainty rises just enough to keep the economy permanently trapped in a depression. In the right panel, the rise in uncertainty is greater, and the economy experiences a speedy recovery after the initial recession. Interestingly, in both cases, the uncertainty does not arise immediately as the economy enters recession, but only once the recession becomes deep.

2.2 General results with rigid prices

The intuition outlined in the previous section now needs to be formalized and extended. We continue to assume rigid prices ($\kappa \rightarrow 0$). There are three key questions that this section resolves. First, which volatility profiles are consistent with an uncertainty trap?

Second, can monetary policy kill these undesirable paths by following a more aggressive Taylor rule? Third, can uncertainty traps display empirically-reasonable properties and magnitudes, or should we discard them as unrealistic?

Which volatility profiles are consistent with an uncertainty trap? The examples in the previous section suggest that volatility must be sufficiently high and sufficiently counter-cyclical for our equilibria to take hold. The examples also suggest that volatility need not emerge immediately as the recession takes place, but only once the recession becomes deep. Indeed, there is some truth to these notions, but we can go further and provide a precise characterization. Recall that to confirm a non-explosive equilibrium, we need only to verify that x_t , whose dynamics are given by (9), remains within $(-\infty, +\infty)$ forever, and does not approach either boundary too rapidly in expectation. This disciplines which profiles σ_x are admissible.

On the one hand, the requirement that x_t not explode upwards implies that self-fulfilling uncertainty is inherently recessionary. Indeed, the risk-premium term σ_x^2 raises the drift $\mu_x = \Phi(x) + \frac{1}{2}\sigma_x^2$, thereby pushing output upward over time. When $x < 0$, this force moves the output gap back toward zero, preventing collapse and allowing the stochastic path to remain admissible as an equilibrium. When $x > 0$, however, the same force pushes the economy further away from the steady state, so that the candidate equilibrium eventually explodes. Consequently, volatility σ_x must vanish as $x \uparrow 0$ and be zero for all $x > 0$: *uncertainty traps are recessionary*.

On the other hand, the requirement that x_t not explode downwards implies that uncertainty must be sufficiently high in deep recessions. This critical point deserves careful development. We start by providing a general condition for non-explosiveness of stochastic processes. As shown in Lemma A.1 in the appendix, a sufficient condition for $\mathbb{E}[x_t]$ to never reach $-\infty$ is¹²

$$\lim_{x \rightarrow -\infty} \frac{\mu_x}{\sigma_x^2} > 0. \quad (10)$$

In other words, the expected growth rate μ_x must rise sufficiently as $x \rightarrow -\infty$ to offset the growth rate variance σ_x^2 . The idea is that the forces impacting dynamics in expectation must outweigh the shocks hitting the system.

To evaluate (10), we use the IS curve to replace μ_x , which is pinned down by monetary

¹²Lemma A.1 is actually stated in terms of positive processes like the level output gap $y_t = \exp(x_t)$. Writing the dynamics of y_t as $dy_t = \mu_y(y_t)dt + \sigma_y(y_t)dZ_t$, the relevant condition (H5) in the lemma requires $\lim_{y \rightarrow 0} 2y\mu_y(y)/\sigma_y^2(y) > 1$. Using the facts, from Itô's formula, that $\sigma_x = \sigma_y/y$ and $\mu_x = \mu_y/y - \frac{1}{2}\sigma_x^2$, we may substitute these results to rewrite condition (H5) as $\lim_{x \rightarrow -\infty} 2[\mu_x(x) + \frac{1}{2}\sigma_x^2(x)]/\sigma_x^2(x) > 1$, which can be simplified to obtain (10).

policy (i.e., $\Phi(x)$) and the risk premium (i.e., σ_x^2). After plugging in $\mu_x = \Phi(x) + \frac{1}{2}\sigma_x^2$, we have $\frac{\mu_x}{\sigma_x^2} = \frac{1}{2} + \frac{\Phi(x)}{\sigma_x^2}$. Therefore, condition (10) is equivalently

$$\lim_{x \rightarrow -\infty} \frac{\Phi(x)}{\sigma_x^2} > -\frac{1}{2}. \quad (11)$$

Under the Taylor principle, we have $\Phi(x) < 0$ for $x < 0$, so the left-hand-side is negative. Monetary policy lowers expected growth by doing stimulus in recessions. By contrast, risk σ_x^2 raises expected growth, because precautionary savings back-load aggregate demand. Hence, there is a race between monetary policy and risk, visible clearly in the ratio in (11). An uncertainty trap can arise only if the risk increases enough in recessions to dominate the efforts of monetary policy.

Importantly, condition (11) is only a relevant consideration in the tail, i.e., in deep recessions. Therefore, we have the key result that *the amount of volatility in recessions is completely unrestricted except in the extreme tail*. For one, this suggests that there is a large set of non-fundamental equilibria. Second, the amount of volatility in “normal” recessions can, in fact, be quite mild. As we will see below, this is a very useful feature for constructing example volatility profiles that are empirically-realistic.

Can monetary policy act even more aggressively to kill uncertainty traps? For a fixed volatility profile, a more aggressive Taylor rule can cause condition (11) to fail, thus potentially eliminating a candidate uncertainty trap. However, recall that equilibrium is derived instead by taking the policy rule as given and asking which volatility profiles are consistent with non-explosiveness. Given a monetary rule $\Phi(x)$, there is always a sufficiently countercyclical volatility profile σ_x such that (11) holds. For example, consider the volatility profile $\sigma_x^2(x) = -k\Phi(x)$ for $k > 2$, which effectively “undoes” the efforts of monetary policy; this volatility clearly satisfies $\Phi(x)/\sigma_x^2(x) = -1/k > -1/2$, and so (11) holds. This logic suggests that *no Taylor rule—no matter how aggressive—can eliminate all uncertainty traps*.

Theorem 1, proven in Appendix A.3, summarizes the theoretical results so far.

Theorem 1. *Suppose prices are rigid ($\kappa \rightarrow 0$). Consider any Taylor rule (MP) with $\bar{r} = \rho$ and $\Phi(x)$ increasing in x . Then,*

- (i) *There exist a continuum of non-explosive equilibria indexed by $x_0 < 0$ and the volatility function $\sigma_x(x)$. The volatility can be any mapping $\sigma_x : \mathbb{R} \mapsto \mathbb{R}$ that is finite for all $x \in (-\infty, 0)$ and satisfies suitable boundary conditions as $x \rightarrow -\infty$ and $x \rightarrow 0$. In particular, a sufficient set of requirements is that σ_x satisfies (11) as $x \rightarrow -\infty$, vanishes sufficiently quickly as $x \rightarrow 0$, and remains strictly positive on compact interior regions.*

- (ii) If $\inf_x \Phi'(x) > 0$, then all non-explosive equilibria have $x_t \leq 0$ forever, and hence any stochastic non-explosive equilibrium is recessionary.
- (iii) Tail excursions can be arbitrarily rare: for any $p > 0$, $\hat{x} < 0$, there exists a stationary non-explosive sunspot equilibrium with tail probability $\mathbb{P}\{x_t < \hat{x}\} < p$.

The final question tackled in this section is whether uncertainty traps can display empirically-realistic properties. We provide an example that leverages the crucial flexibility provided by Theorem 1, namely that $\sigma_x(x)$ can be essentially any function satisfying relevant boundary conditions. This is possible even if monetary-policy follows a super-aggressive rule attempting to eliminate non-fundamental recessions.

Example 1. Suppose monetary policy follows the aggressive rule

$$\Phi(x) = \phi_x(e^x - e^{-x}), \quad \phi_x > 0. \quad (12)$$

This rule satisfies the Taylor principle and, indeed, responds stronger than its linear approximation $2\phi_x x$. Consider the volatility function

$$\sigma_x^2 = \begin{cases} \nu^2 e^{-2x} - \Phi(x), & \text{if } x \leq \underline{x}; \\ \tilde{\sigma}_x^2(x), & \text{if } x \in (\underline{x}, \bar{x}) \\ 0, & \text{if } x \geq \bar{x}. \end{cases} \quad (13)$$

for any thresholds $\underline{x} < \bar{x} < 0$, any function $\tilde{\sigma}_x^2(x) \geq 0$, and any initial condition $x_0 < 0$. This volatility grows rapidly in a deep recession, is arbitrary when the economy is in an intermediate recession, and is zero in a small recession. We may directly check condition (11):

$$\lim_{x \rightarrow -\infty} \frac{\Phi(x)}{\sigma_x^2} = \lim_{x \rightarrow -\infty} \frac{-\phi_x e^{-x}}{\nu^2 e^{-2x}} = 0 > -\frac{1}{2}.$$

By Theorem 1, this volatility profile is an uncertainty trap. In fact, there exists a non-degenerate stationary distribution for x with full support in the region $x \leq \bar{x}$, and so the uncertainty trap features permanently depressed output and elevated volatility.

We use Example 1 in a numerical exercise to illustrate that uncertainty traps can have empirically-reasonable properties. Figure 2 displays the configuration and stationary distribution. We set $\bar{x} = \log(1 - 0.005)$ and $\underline{x} = \log(1 - 0.15)$, meaning the economy is always depressed by 0.5% below potential but only encounters extreme volatility when output is 15% below potential. The choice of $\tilde{\sigma}_x(x)$ is illustrated by the black solid line.

This function is specified to remain around 2% during ordinary recessions ($x_t \geq -0.03$) and to rise sharply only for deeper downturns. The discretized stationary distribution assigns approximately 20% probability to grid points below $\bar{x}$, whereas excursions into deep recessions are rare. Specifically, the probability of observing x below -0.05 is only 0.90%. These tail excursions are infrequent because even modest uncertainty is enough to create enough precautionary savings that induces a recovery on average. For example, the figure also shows that the drift is positive for all $x \leq -0.04$.

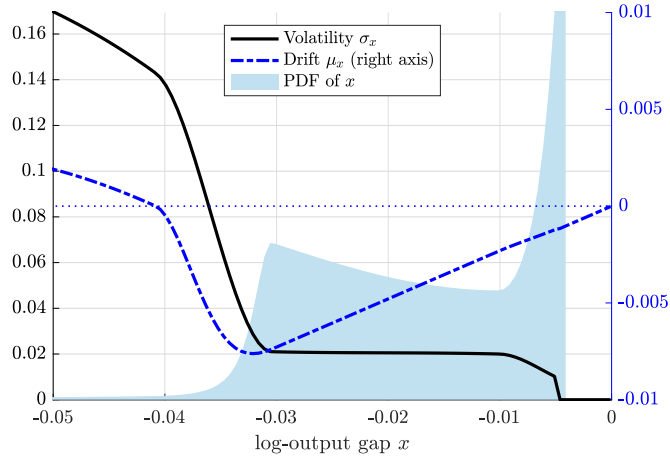


Figure 2: Uncertainty trap equilibrium from Example 1. The stationary distribution is obtained from a discretized Kolmogorov Forward Equation. With the 2,000-point state grid, the probability assigned to the upper-boundary grid cell is displayed as a boundary mass. Parameters: $\rho = 0.04$, $\nu = 0.50$, $\bar{x} = \log(0.995)$, $\underline{x} = \log(0.85)$, $\phi_x = 0.125$. The function $\tilde{\sigma}_x(x)$ is a piecewise cubic interpolation of the following points: $\tilde{\sigma}_x(\log(0.97)) = 0.021$, $\tilde{\sigma}_x(\log(0.99)) = 0.02$, $\tilde{\sigma}_x(\log(0.9949)) = 0.01$, and $\tilde{\sigma}_x(\log(0.96))$ chosen such that $\mu_x(\log(0.96)) = 0$. Continuity of $\sigma_x(x)$ is imposed at $x = \bar{x}$. Between grid points, the simulated paths interpolate volatility continuously between the final positive-volatility grid point and the zero-volatility region.

Because of infrequent tail excursions, this example features quantitatively reasonable output dynamics. The unconditional standard deviation of annual output growth is 2.56%, close to the data. This standard deviation increases to 5.87% conditional on output growth being at least half a standard deviation below its unconditional mean, so volatility is also countercyclical here.¹³

2.3 Inflation and uncertainty traps

The analysis so far has focused on the rigid-price limit. With partially-flexible prices, conventional wisdom suggests that a sufficiently strong reaction to inflation should re-

¹³To compute statistics at the annual frequency, we simulate the model at a daily frequency over a 5,000-year horizon and aggregate the data to the annual level.

store equilibrium uniqueness. It does not: a large class of uncertainty traps persist and retain the same qualitative structure as in the rigid-price case. Because the analysis is substantially more involved with partially-flexible prices, many results and proofs for this section are collected in Appendix C.

Our central result establishes the existence of these equilibria and characterizes their flexibility. We restrict attention to equilibria recursive in the output gap x , a choice made for analytical tractability that is not essential to the main points.¹⁴

Theorem 2 (Uncertainty traps with partially-flexible prices). *Consider the linearized Phillips curve (linear PC) with $\kappa > 0$ and the linear Taylor rule (linear MP) with $\phi_x > 0$, $\phi_\pi > 1$, $\bar{r} = \rho$.*

- (i) *Existence. There exist $\bar{x} < 0$ and functions $\pi(\cdot)$, $\sigma_x^2(\cdot)$ such that $\pi_t = \pi(x_t)$ and $\sigma_{x,t}^2 = \sigma_x^2(x_t)$ form a non-explosive sunspot equilibrium with x_t stationary and ergodic on $(-\infty, \bar{x})$. Volatility is countercyclical in the left tail ($\sigma_x^2(x)$ decreasing for $x \leq \underline{x}$, for some $\underline{x} < \bar{x}$) and vanishes at the upper boundary ($\sigma_x^2(x) \rightarrow 0$ as $x \uparrow \bar{x}$). Both deflationary ($\pi(x) < 0$) and inflationary ($\pi(x) > 0$) left tails are possible.*
- (ii) *Degrees of freedom. Fix any finite $\bar{x}$, and let $\pi : (-\infty, \bar{x}] \mapsto \mathbb{R}$ and $\sigma_x^2 : (-\infty, \bar{x}) \mapsto (0, \infty)$ be functions satisfying the regularity and boundary conditions of Theorem C.1. In particular, π is C^1 and piecewise C^2 , while σ_x^2 is piecewise continuous. As $x \rightarrow -\infty$, the boundary conditions rule out explosion and control the stationary left tail. As $x \uparrow \bar{x}$, they rule out escape by requiring volatility to vanish sufficiently quickly and the drift to point inward, $\phi_x \bar{x} + (\phi_\pi - 1)\pi(\bar{x}) < 0$. Together, these conditions ensure non-explosion and the required finite stationary moments.*

Suppose further that, wherever π is twice differentiable, the functions satisfy the ODE obtained by combining the NKPC with Itô's formula for $\pi_t = \pi(x_t)$:

$$\pi'(x) \left[\phi_x x + (\phi_\pi - 1)\pi(x) + \frac{1}{2}\sigma_x^2(x) \right] + \frac{1}{2}\sigma_x^2(x)\pi''(x) = \rho\pi(x) - \kappa x. \quad (14)$$

Then $\pi_t = \pi(x_t)$ and $\sigma_{x,t}^2 = \sigma_x^2(x_t)$ support a non-explosive sunspot equilibrium in which x_t is stationary and ergodic on $(-\infty, \bar{x})$.

The two parts serve distinct purposes. Part (i) is a clean existence result: within the recursive class, we show that uncertainty traps exist, with countercyclical volatility in

¹⁴We seek volatile equilibria distinct from the fundamental, or MSV, equilibrium $x_t = \pi_t = 0$, and we search for them only within equilibria recursive in x_t . This approach is conservative. Theorem 2 shows that uncertainty traps already exist within this class, but there could exist even more uncertainty trap constructions. We conjecture that such constructions are possible but do not pursue them here.

the left tail, vanishing volatility at the upper boundary, and either a deflationary or an inflationary left tail.

Part (ii) presents a characterization to illustrate the sense in which these equilibria are not isolated examples. Ultimately, a broad class of equilibria can exist for a simple mathematical reason: we are seeking to solve for two functions $\pi(\cdot)$ and $\sigma_x^2(\cdot)$ from the single ODE (14). Beyond that ODE, non-explosion and ergodicity require these functions to be suitably well behaved as $x \rightarrow -\infty$ and as $x \uparrow \bar{x}$. These requirements are encoded as boundary conditions.

Figure 3 displays two equilibria from Example C.1 in Appendix C.2, using the freedom outlined by Theorem 2. They are meant to illustrate the flexibility of the construction rather than to provide a quantitatively disciplined calibration; nevertheless, the implied stationary distributions place most of their mass on plausible ranges for the output gap and inflation. Both examples use the same model parameters.

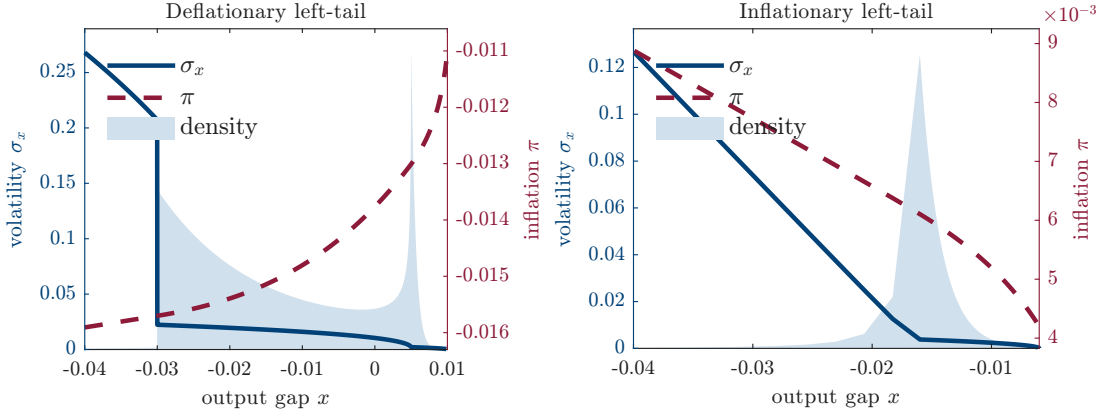


Figure 3: Two numerical examples following Theorem 2. In each panel, the solid line is $\sigma_x(x)$ on the left axis, the dashed line is $\pi(x)$ on the right axis, and the shaded area is the stationary density of x , rescaled to fit the left-axis range. Parameters are $\rho = 0.04$, $\kappa = 0.02$, $\phi_x = 0.5$, and $\phi_\pi = 1.5$. The functions used in constructing $\sigma_x^2(x)$ and $\pi(x)$ are detailed in Example C.1 of Appendix C.2.

With partially-flexible prices, uncertainty traps still look much as they did in the rigid-price case. Volatility $\sigma_x^2(x)$ rises in deep recessions and vanishes as x reaches the upper end of its accessible range; in between, the characterization in part (ii) leaves substantial freedom. These patterns are visible in Figure 3.

Two properties are new relative to the rigid-price case. First, although uncertainty traps remain recessionary phenomena—the output gap always stays below the threshold $\bar{x}$ —that threshold need not be negative. Part (ii) permits $\bar{x} > 0$ so long as $(\phi_\pi - 1)\pi(\bar{x}) + \phi_x \bar{x} < 0$, i.e., when there is sufficient deflation at the upper boundary. We verify numerically that such equilibria can be constructed (see the left panel of Figure 3).

Second, uncertainty traps can be either deflationary or inflationary. The left panel of Figure 3 features deflationary expansions alongside deflationary recessions; the right panel displays an inflationary recession, i.e., stagflation.

The stagflationary case is especially striking. A robust prediction of the linearized NK model is that demand recessions are deflationary and stagflation can arise only from supply shocks. To see why, suppose the economy were in an inflationary recession, with $x_0 < 0$ and $\pi_0 > 0$. The Phillips curve implies $\mu_\pi = \rho\pi - \kappa x > 0$, so inflation initially rises. Moreover, along any recovery path on which the output gap returns to its zero steady state, positive inflation remains unstable: as x_t approaches zero, the Phillips curve approaches $\mu_\pi = \rho\pi > 0$, so inflation cannot converge back to steady state and eventually violates non-explosiveness.

Our inflationary uncertainty traps can emerge precisely because they are stochastic and nonlinear. Suppose again that $x_0 < 0$ and $\pi_0 > 0$. The Phillips curve still requires expected inflation to rise, $\mu_\pi = \rho\pi - \kappa x > 0$, yet this positive drift can now coexist with inflation that falls along recovery paths. With $\pi_t = \pi(x_t)$ and $\pi'(x) < 0$, as in our construction, Itô's formula gives $\mu_\pi(x) = \pi'(x)\mu_x(x) + \frac{1}{2}\pi''(x)\sigma_x^2(x)$. When the output gap is expected to recover, $\mu_x(x) > 0$, the deterministic term $\pi'(x)\mu_x(x)$ pushes inflation down; but π is sufficiently convex and volatility sufficiently high that the Itô correction $\frac{1}{2}\pi''(x)\sigma_x^2(x)$ dominates and delivers $\mu_\pi(x) > 0$. Intuitively, declines in the output gap raise inflation disproportionately, so expected inflation can rise even though the output gap is expected to recover and inflation is a decreasing function of the output gap.

A final concern is that a more aggressive Taylor rule might restore uniqueness. Corollary C.2 in Appendix C shows that this is not the case: uncertainty traps persist even under nonlinear Taylor rules, including exponential and double-exponential rules.

2.4 Extensions and robustness

The baseline model features (1) only sunspot shocks; (2) a continuous-time setting; (3) log utility; and (4) a representative agent. We now examine how our multiplicity results extend beyond the baseline along each of these dimensions.

2.4.1 Fundamental shocks

For simplicity, the baseline model uses only sunspot shocks. This restriction is not central. The core issue is an indeterminacy in uncertainty within NK models: holding fixed the economy's fundamentals, the model may sustain different equilibrium responses to the same aggregate shock, despite the Taylor principle.

Appendix B illustrates this point in an NK economy in which uncertainty is driven by monetary-policy shocks. Theorem B.1 establishes the existence of a large class of equilibria that differ in the sensitivity of aggregate output to these shocks. We focus on monetary-policy shocks because the NK model was developed, in large part, to study their effects. The same logic suggests that analogous constructions should be possible with productivity, preference, or other aggregate shocks. This result qualifies the common practice of focusing on the MSV equilibrium under the Taylor principle.

2.4.2 Discrete-time model

Our continuous-time setting provides significant tractability to the analysis. However, the main results are not an artifact of the continuous-time formulation. As shown in Appendix D, uncertainty traps also arise in the standard discrete-time three-equation NK model once the linearized IS equation is replaced by its nonlinear counterpart. Indeed, under logarithmic preferences, the nonlinear IS equation can be written as

$$1 = \beta \mathbb{E}_t [e^{x_t - x_{t+1} + i_t - \pi_{t+1}}],$$

where β is the subjective discount factor, i_t is the log interest rate, x_t is the log output gap, and π_t is the log inflation rate. This nonlinear formulation preserves the Jensen, or precautionary-saving, effect that the usual linearization eliminates. The key challenge is providing a non-explosive stochastic process x_t that satisfies this nonlinear IS equation. Proposition D.1 in Appendix D provides a formal explicit construction.

2.4.3 Non-log preferences

Our baseline model adopts log utility for simplicity. We now show typical macro preferences permit uncertainty traps with even less extreme volatility. Start by letting the representative agent have a general time-additive utility function $u(c)$. The consumption FOC is then $e^{-\rho t} u'(C_t) \propto M_t$. Writing consumption dynamics as $dC_t = C_t[\mu_{C,t} dt + \sigma_{C,t} dZ_t]$, this then implies the Euler equation

$$\mu_{C,t} = \underbrace{\left(-C_t \frac{u''(C_t)}{u'(C_t)} \right)^{-1}}_{:=EIS_t} (\iota_t - \pi_t - \rho) + \frac{1}{2} \underbrace{\left(-C_t \frac{u'''(C_t)}{u''(C_t)} \right)}_{:=RP_t} \sigma_{C,t}^2.$$

Elasticity of intertemporal substitution EIS_t governs the consumption growth sensitivity to real rates $\iota_t - \pi_t$. Relative prudence RP_t governs the consumption growth sensitivity

to risk $\sigma_{C,t}^2$. The dynamics of output gap x_t , which coincide with those of $\log(C_t)$, are

$$\mu_{x,t} = EIS_t \times (\iota_t - \pi_t - \rho) + \frac{1}{2}(RP_t - 1)\sigma_{x,t}^2. \quad (15)$$

Preferences with higher prudence RP_t feature more precautionary savings, meaning that each unit of risk σ_x^2 creates more growth μ_x . Preferences with lower substitution elasticity EIS_t are less responsive to interest rates, meaning that monetary policy must act more aggressively to affect μ_x . Both mechanisms permit uncertainty traps with less risk.

The most popular preferences in macroeconomics work in both of these directions. For example, CRRA preferences $u(c) = \frac{c^{1-\gamma}-1}{1-\gamma}$ have $RP_t = 1 + \gamma$ and $EIS_t = \gamma^{-1}$, with standard macroeconomic calibrations assuming $\gamma > 1$. Higher γ increases RP_t , thus increasing precautionary savings, and reduces EIS_t , thus reducing the power of monetary policy. We obtain the following result, proved in Appendix A.3.

Proposition 1. *Assume the representative agent has CRRA utility $u(c) = \frac{c^{1-\gamma}-1}{1-\gamma}$ with $\gamma > 1$. Suppose prices are rigid ($\kappa \rightarrow 0$). Consider monetary policy rules with target rate $\bar{r} = \rho$. If the monetary rule $\Phi(x)$ and volatility function $\sigma_x^2(x)$ comprise a non-explosive equilibrium with log utility, then the more-aggressive monetary rule $\tilde{\Phi}(x) := \gamma\Phi(x)$ and smaller volatility $\tilde{\sigma}_x^2(x) := \gamma^{-1}\sigma_x^2(x)$ correspond to a non-explosive equilibrium with CRRA utility.*

2.4.4 Heterogeneous agents and idiosyncratic risk

Our model features a representative agent or, equivalently, perfect idiosyncratic risk sharing. Of course, this is unrealistic: idiosyncratic income risk is an order of magnitude larger than aggregate output volatility, so household heterogeneity likely matters. We now show that adding idiosyncratic risk to the model allows uncertainty traps to be sustained with even less aggregate volatility. Intuitively, idiosyncratic risk itself induces precautionary savings, meaning that, all else equal, less aggregate risk is needed to sustain an uncertainty trap.

Here, we pursue the more nuanced point that *countercyclical* of idiosyncratic risk—extensively documented in the literature (Storesletten et al., 2004; Guvenen et al., 2014)—reduces the burden of aggregate volatility for our equilibria. We demonstrate this point with a particularly simple and tractable modeling device that may serve as a useful technique for the literature.

The details are as follows. We introduce an uninsurable idiosyncratic Brownian motion $\tilde{Z}_t$ that redistributes consumption between two types of consumers—types A and B. Specifically, the flow $Y_t\sqrt{\omega_t(1-\omega_t)}v(x_t)d\tilde{Z}_t$ is transferred from type-B to type-A agents,

where ω_t is the equilibrium consumption share of the type-A agents. This exact functional form for redistribution is chosen in a way that permits a representative-agent characterization. The key exogenous function $\nu(x) > 0$, which depends on the output gap, governs the overall volatility of the uninsurable shocks. Note that agents cannot choose their consumption exposure to $d\tilde{Z}_t$ —rather, they are endowed with this exposure. Instead, agents optimally choose their savings as well as their exposure to the sunspot shock dZ_t . Finally, we assume the monetary target rate is set to $\bar{r} = \rho - \nu(0)^2$, which appropriately adjusts for the presence of idiosyncratic risk at steady state.

As shown in Appendix E, equilibrium output-gap dynamics satisfy

$$dx_t = \left(\Phi(x_t, \pi_t) - \pi_t + \nu(x_t)^2 - \nu(0)^2 + \frac{1}{2}\sigma_{x,t}^2 \right) dt + \sigma_{x,t} dZ_t. \quad (16)$$

Idiosyncratic shocks influence aggregate dynamics only through precautionary behavior, and countercyclical idiosyncratic volatility ($\nu'(x) < 0$) raises precautionary savings during recessions. Intuitively, equation (16) makes clear that $\nu(x)^2$ and $\frac{1}{2}\sigma_x^2$ are “substitutes,” in the sense that the total amount of precautionary savings is determined by $\nu(x)^2 + \frac{1}{2}\sigma_x^2$. If idiosyncratic risk is very countercyclical, then aggregate risk need not be as countercyclical to sustain an uncertainty trap. The proposition below, proved in Appendix A.3, makes this precise in the rigid-price limit.

Proposition 2. *Consider the rigid-price limit ($\kappa \rightarrow 0$). Suppose that the baseline model admits a non-explosive equilibrium on $\{x \leq 0\}$ with drift $\mu_x(x)$ and aggregate volatility $\sigma_x(x)$ satisfying $\lim_{x \uparrow 0} \sigma_x(x) = 0$ and the tail condition (10). Let $\nu(x) > 0$ with $\nu'(x) < 0$ for all $x \leq 0$, and suppose that $\frac{1}{2}\sigma_x^2(x) > \nu(x)^2 - \nu(0)^2$ for all $x < 0$ sufficiently negative. Then the model with idiosyncratic risk admits a non-explosive equilibrium with aggregate volatility $\tilde{\sigma}_x^2(x) := \max\{0, \sigma_x^2(x) - 2[\nu(x)^2 - \nu(0)^2]\}$. Moreover, for all $x < 0$ sufficiently negative, $\tilde{\sigma}_x^2(x) < \sigma_x^2(x)$, and the gap is larger in deeper recessions.*

3 A Monetary Rule with Risk Targeting

Uncertainty traps are robust to conventional Taylor rules based on inflation and the output gap. What about different types of monetary rules? We first demonstrate that a different type of policy rule, which targets risk, can eliminate uncertainty traps. Thus, monetary policy can indeed be successful at eliminating non-fundamental fluctuations. The main concern with this policy, as with many interest-rate policies, is that it cannot work if an effective lower bound exists for interest rates. The proofs for this section are collected in Appendix A.4.

3.1 Risk targeting

We now replace the plain-vanilla Taylor rule (MP) with a rule that explicitly responds to the risk premium. In particular, we use

$$\iota_t = \rho + \Phi(x_t, \pi_t) - (\alpha_- \mathbf{1}_{\{x_t < 0\}} + \alpha_+ \mathbf{1}_{\{x_t > 0\}}) \times \left(\frac{1}{2} \sigma_{x,t}^2 \right). \quad (\text{MP-vol})$$

Although conventional wisdom would suggest that targeting an asset price—which maps one-to-one into the output gap—suffices to target the risk premium, that is not true here. Intuitively, because uncertainty $\sigma_{x,t}$ is the mechanism that self-fulfils non-fundamental demand recessions, monetary policy must target this quantity, not only the level of demand x_t . Rule (MP-vol) directly targets this uncertainty.

Risk targeting can restore uniqueness among non-explosive equilibria. Substitute (MP-vol) into (IS) to get

$$dx_t = \left[\Phi(x_t, \pi_t) - \pi_t + \frac{1}{2}(1 - \alpha(x_t))\sigma_{x,t}^2 \right] dt + \sigma_{x,t} dZ_t, \quad (17)$$

where $\alpha(x) := \alpha_- \mathbf{1}_{\{x < 0\}} + \alpha_+ \mathbf{1}_{\{x > 0\}}$ is the state-dependent responsiveness to the risk premium. If $\alpha_- = \alpha_+ = 1$, then the risk premium vanishes from the drift, and we are back in the standard linearized situation where an aggressive reaction function Φ can restore determinacy. If $\alpha_+ < 1 < \alpha_-$, then the risk premium itself becomes destabilizing. During recessions, higher levels of $\sigma_{x,t}^2$ induce a strong monetary policy reaction that actually *reduces* savings on net. Thus, our core mechanism whereby volatility leads to precautionary savings is broken by this aggressive policy. For this reason, rule (MP-vol), with more aggressive risk targeting in bad times, can eliminate uncertainty traps.

The following theorem formalizes this intuition. It shows that a standard linear Taylor rule satisfying the Taylor principle restores uniqueness when augmented with a sufficiently strong response to risk in recessions. The result applies both to the standard partially flexible-price case and to the rigid-price limit.

Theorem 3 (Risk targeting). *Consider the linearized Phillips curve (linear PC) for any finite $\kappa \geq 0$ and the modified Taylor rule (MP-vol) with $\Phi(x, \pi) = \phi_x x + \phi_\pi \pi$. Suppose that $\phi_x > 0$, $\rho\phi_x + \kappa(\phi_\pi - 1) > 0$, $\alpha_+ \leq 1$, and that recessionary risk targeting is sufficiently strong: $\alpha_- \geq 1$. Then, the unique non-explosive equilibrium is $x_t = \pi_t = 0$ for all t .*

Lee and Dordal i Carreras (2026) present a related analysis in which $\alpha(x)$ is restricted to be constant. Under that restriction, they conclude that risk targeting is effective only in the knife-edge case $\alpha = 1$, and therefore that implementation is impractical—for

instance due to measurement challenges. Our results qualify this conclusion. Once the policy response to risk can vary with the state of the economy, a large set of policies can ensure determinacy. It is enough that monetary policy respond sufficiently strongly to risk in recessions, with $\alpha_- > 1$, whereas policy need not respond at all to risk in booms, e.g., $\alpha_+ = 0$. The policy asymmetry is linked to a robust and deep feature of uncertainty trap equilibria, namely that they are always recessionary. Policy must provide stimulus precisely in such recessionary times, when the economy is susceptible to non-fundamental deviations.

Similar to [Acharya and Benhabib \(2024\)](#), we can also rewrite the policy rule in terms of the “risk-adjusted rate of interest” $r_t^* := \rho - \frac{1}{2}\sigma_{x,t}^2$, which is the real interest rate consistent with $\mu_{x,t} = 0$. Written this way, our rule is

$$\iota_t = \rho + \Phi(x_t, \pi_t) + \alpha(x_t)(r_t^* - \rho), \quad (18)$$

and so one can interpret this policy as targeting the risk-adjusted rate r^* to be equal to its flexible-price level ρ . [Acharya and Benhabib \(2024\)](#) pursue this interpretation in an NK model with multiplicity due to countercyclical idiosyncratic risk. Using (18) with $\partial_\pi \Phi(x, \pi) > 1$ and $\alpha(x) \geq 1$, they show that this rule eliminates their multiplicity. We obtain a similar result, except the r^* targeting must necessarily be asymmetric: the rule must have $\alpha(x) \geq 1$ only when $x \leq 0$. Just importing their constant-coefficient rule directly would not yield determinacy except in the knife-edge case $\alpha = 1$.

3.2 Effective lower bounds

While risk targeting can restore uniqueness among non-explosive equilibria, it relies strongly on an unconstrained monetary policy. To understand this, recall that all the rules advocated above share the property that $\iota_t \rightarrow -\infty$ as $x_t \rightarrow -\infty$. If interest rates are lower bounded, this cannot work. In this section, we consider a situation where monetary policy is constrained: suppose ι_t must respect the lower bound $\iota_t \geq \underline{\iota}$. This lower bound does not need to be zero—it can be any level. With *any* effective lower bound, uncertainty traps remain. Therefore, unconventional monetary policy (not modeled in this paper) would need to step in at the lower bound.

To see how uncertainty traps can persist, consider the rigid-price equilibria and inspect output gap dynamics when ι_t is at its lower bound:

$$dx_t = \left[\underline{\iota} - \rho + \frac{1}{2}\sigma_{x,t}^2 \right] dt + \sigma_{x,t} dZ_t, \quad \text{when } x_t < 0 \quad \text{and} \quad \iota_t = \underline{\iota}. \quad (19)$$

A sufficiently high level of uncertainty can raise the drift and create stable dynamics by “pushing” x_t back toward steady state. Thus, high enough uncertainty once again permits non-fundamental demand recessions, since they will not lead to explosions.

Upon reflection, this logic is not about monetary policy being generically constrained; upper bounds on interest rates do not stymie monetary policy at all. An asymmetry arises because uncertainty traps are *recessionary and self-sustained by risk premia*. In a demand recession, the central bank would want to set low interest rates, but the lower bound $\iota_t \geq \underline{\iota}$ prevents such stimulus from becoming arbitrarily strong.

While the discussion above assumes the rigid-price limit, the points apply to the more general situation with partially-flexible prices. Suppose that, when not at the lower bound, interest rates follow a general policy rule $\Psi(x, \pi, \sigma_x^2)$ that can even target risk. An example is (MP-vol) with $\alpha_- > 1 > \alpha_+$, $\phi_x > 0$, and $\phi_\pi > 1$, which recall was able to eliminate uncertainty traps when not constrained by a lower bound.

Proposition 3 (Effective lower bounds). *Consider the IS equation (IS) and the linearized Phillips curve (linear PC) for any finite $\kappa \geq 0$. Let monetary policy satisfy*

$$\iota_t = \max\{\underline{\iota}, \Psi(x_t, \pi_t, \sigma_{x,t}^2)\}, \quad \underline{\iota} \leq \rho,$$

where $\Psi(0,0,0) = \rho$. Suppose that Ψ is strictly increasing in x and π , and that, on the recessionary region $\mathcal{R}_- := \{(x, \pi, s) : x < 0, \pi \leq 0, s \geq 0\}$, Ψ is continuous and the map $s \mapsto \Psi(x, \pi, s) + \frac{1}{2}s$ is weakly decreasing. Then, for every $x_0 < 0$, there exists a non-explosive equilibrium that features strictly positive output volatility while $x_t < 0$.

4 Fiscal Policies

We extend the model to include fiscal policy, formulated via lump-sum taxation and government transfers to the representative household. The real primary surplus of the government is the difference between these taxes and transfers, which we denote by S_t . Fiscal policy is characterized by a specification for S_t , to be described shortly.

Surpluses may be non-zero, so the government borrows by issuing short-term nominally riskless bonds B_t . The flow budget constraint of the government is

$$\dot{B}_t = \iota_t B_t - P_t S_t. \tag{20}$$

The nominal interest rate ι_t is controlled by monetary policy.

Equilibrium requires households' transversality condition for bond holdings,

$$0 = \lim_{T \rightarrow \infty} \mathbb{E}_0 \left[M_T \frac{B_T}{P_T} \right], \quad (\text{TVC})$$

where recall M denotes the real stochastic discount factor process. As is well known, the transversality condition implies the present-value formula for government debt:

$$\frac{B_t}{P_t} = \mathbb{E}_t \left[\int_t^\infty \frac{M_u}{M_t} S_u du \right]. \quad (\text{GD})$$

The real value of debt must equal the real present value of surpluses. In some arguments, it is easier to use (GD) than (TVC), although they are equivalent conditions.

For our main results in the text, we assume that surpluses follow a rule which features an exogenous component and a feedback from outstanding government debt back into surpluses. We specify the surplus-to-GDP ratio $s_t := S_t/Y_t$ as a function of an exogenous component $\hat{s}_t$ and a feedback from the real debt-to-GDP ratio $b_t := B_t/(P_t Y_t)$. The exogenous component $\hat{s}_t = \hat{s}(\Omega_t)$ is a bounded continuous function of Ω_t , which is an arbitrary vector of exogenous stationary state variables driven by a multivariate Brownian motion $\mathcal{Z}$ that is independent of the shock Z , i.e., $d\Omega_t = \mu_\Omega(\Omega_t)dt + \varsigma_\Omega(\Omega_t) \cdot d\mathcal{Z}_t$. Thus, all of our baseline fiscal rules take the form

$$s_t = \hat{s}(\Omega_t) + \alpha b_t, \quad (21)$$

where α is a policy parameter, to be specified in examples below.

The standard economic logic is that $\alpha > 0$ corresponds to spending that is completely “backed.” Imagine surpluses fall exogenously (i.e., $\hat{s}$ falls), say because spending increases while taxes remain temporarily fixed. To pay for the spending, the debt stock increases over time. But with $\alpha > 0$, the rise in debt induces future tax increases (or future spending cuts), so the surplus recovers over time. Thus, the initial spending increase is effectively financed by future taxes. By contrast, $\alpha \leq 0$ corresponds to “unbacked” spending: a spending increase that raises the debt stock is not accompanied by future tax increases (e.g., for $\alpha = 0$, the surplus-to-GDP ratio is exogenous).

In the language of [Leeper \(1991\)](#), fiscal policies with $\alpha > 0$ are regarded as “passive” and do not constrain equilibrium dynamics, because the government always ensures its own solvency. By contrast, policies with $\alpha \leq 0$ are considered “active” and can prune equilibria, at least in deterministic and linearized analyses. We also use this active/passive language and show that it still applies in the nonlinear, stochastic model.

Equilibrium definition. The dynamics of x and π now take the form

$$dx_t = \left[\iota_t - \pi_t - \rho + \frac{1}{2}\sigma_{x,t}^2 + \frac{1}{2}|\varsigma_{x,t}|^2 \right] dt + \sigma_{x,t}dZ_t + \varsigma_{x,t}d\mathcal{Z}_t \quad (22)$$

$$d\pi_t = \left[\rho\pi_t - \kappa x_t \right] dt + \sigma_{\pi,t}dZ_t + \varsigma_{\pi,t}d\mathcal{Z}_t. \quad (23)$$

Other than the generalization that these dynamics now include potential sensitivities to the surplus shocks $d\mathcal{Z}$, these dynamics are identical to the baseline IS curve (IS) and Phillips curve (PC). The reason is that lump-sum taxes and transfers do not alter the household first-order conditions. The representative household owns government bonds but also owes future taxes and receives future transfers. Fiscal policy therefore affects the analysis only through the additional equilibrium restrictions imposed by government solvency.¹⁵

Together with a nominal interest rate rule for ι_t and a surplus rule for s_t , equilibrium is fully characterized by the dynamical system (22)-(23), combined with government transversality (TVC), or equivalently the valuation equation (GD). We continue to require the non-explosion Condition 1. Proofs for this section are contained in Appendix A.5.

4.1 Baseline with passive fiscal policy

Sections 1-3 did not have government debt or taxes/transfers. However, everything we have said until now still holds with fiscal policies, so long as those policies are “passive.”

Indeed, consider a surplus rule with $\alpha = \bar{\alpha} > 0$. This policy pays back the debt by raising surpluses when debt rises. Such a fiscal policy is therefore sustainable in any equilibrium, in the sense that the transversality condition (TVC) automatically holds. Given the real SDF $M_t = e^{-\rho t}Y_t^{-1}$ and debt-to-GDP is $b_t := B_t/(P_tY_t)$, we are interested in the long-horizon properties of $\mathbb{E}_0[M_T \frac{B_T}{P_T}] = \mathbb{E}_0[e^{-\rho T}b_T]$. Under this fiscal policy,

$$\mathbb{E}_0[e^{-\rho T}b_T] = e^{-\bar{\alpha}T} \left(b_0 - \mathbb{E}_0 \left[\int_0^T e^{-(\rho-\bar{\alpha})t} \hat{s}_t dt \right] \right). \quad (24)$$

To derive (24), apply Itô's formula to $e^{(\bar{\alpha}-\rho)t}b_t$, using the flow government budget constraint (20), the price level dynamics $dP_t/P_t = \pi_t dt$, and the dynamics of x_t in (22). The

¹⁵The lifetime budget constraint of the representative household is

$$\mathbb{E}_t \left[\int_t^\infty \frac{M_u}{M_t} \frac{W_u L_u}{P_u} du \right] + \frac{B_t}{P_t} = \mathbb{E}_t \left[\int_t^\infty \frac{M_u}{M_t} S_u du \right] + \mathbb{E}_t \left[\int_t^\infty \frac{M_u}{M_t} C_u du \right].$$

By (GD), the lifetime budget constraint is equivalent to the budget constraint without any debt at all. And so the household optimality conditions are unchanged. This leads to (22)-(23).

terms on the right-hand-side of (24) vanish as $T \rightarrow \infty$,¹⁶ which implies that the transversality condition $\lim_{T \rightarrow \infty} \mathbb{E}_0[e^{-\rho T} b_T] = 0$ automatically holds, hence the government debt valuation equation automatically holds regardless of the equilibrium dynamics. All the uncertainty traps from Section 2 could emerge under such passive fiscal policies.

4.2 Active fiscal policy

The simplest example of active fiscal policy sets $\alpha = 0$ (exogenous surplus-to-GDP ratio). With this policy, and using (GD), we obtain

$$\frac{B_t}{P_t} = Y_t \times \mathbb{E}_t \left[\int_t^\infty e^{-\rho(u-t)} \hat{s}(\Omega_u) du \right]. \quad (25)$$

Equation (25) transparently encodes the equilibrium selection: whereas Y_t potentially exhibits sunspot shocks, B_t/P_t is pre-determined and $\hat{s}_t$ is exogenous. For this equation to hold, we must therefore have $\sigma_{x,t} = 0$.

In addition, a similar argument extends to all surplus rules with $\alpha < 0$ as well, because these are all surplus rules with unbacked spending and thus lead to a non-trivial debt valuation equation. We summarize the results in the following theorem.

Theorem 4 (Active fiscal). *The economy with $\alpha \leq 0$ necessarily has $\sigma_{x,t} = 0$ and pins down $\varsigma_{x,t}$ uniquely. Conversely, if $\sigma_{x,t} = 0$ and $\varsigma_{x,t}$ is the unique function implied by (GD), then the debt valuation equation (GD) automatically holds at every date, given it holds at $t = 0$.*

Remark 1 (Fiscal equilibrium). *Assuming monetary and fiscal policy are appropriately coordinated (i.e., passive money is paired with active fiscal), the entire equilibrium is unique even beyond $\sigma_{x,t}$ and $\varsigma_{x,t}$ —namely, π_t is pinned down (see Proposition H.1 in Appendix H).*

Thus, active fiscal policy eliminates self-fulfilling demand volatility, $\sigma_x = 0$, but introduces fiscal volatility $\varsigma_x \neq 0$. Bianchi and Melosi (2017) highlight a similar trade-off in which active fiscal policy can end a deflationary trap at the cost of introducing exposure to fiscal shocks. The key reason is that our baseline active fiscal policies are *always* doing unbacked spending and therefore always anchoring aggregate demand. Because of the omnipresent role of this fiscal policy in steering demand dynamics, some observers have referred to this class of policies as *fiscal dominance*.

¹⁶To show $e^{-\bar{\alpha}T} \mathbb{E}_0 \left[\int_0^T e^{-(\rho-\bar{\alpha})t} \hat{s}_t dt \right] \rightarrow 0$, use the boundedness of surpluses $|\hat{s}_t| \leq \hat{s}_{max}$ to get

$$\left| \mathbb{E}_0 \left[\int_0^T e^{\bar{\alpha}(t-T)} e^{-\rho t} \hat{s}_t dt \right] \right| \leq \hat{s}_{max} \int_0^T e^{\bar{\alpha}(t-T)} e^{-\rho t} dt = \frac{\hat{s}_{max}}{\bar{\alpha} - \rho} (e^{-\rho T} - e^{-\bar{\alpha}T}) \xrightarrow{T \rightarrow \infty} 0.$$

4.3 Discussion and intuition

How does fiscal policy select equilibria? And are these equilibrium selection results robust to some natural extensions of the model?

Consider what happens in a non-fundamental recession. If, all of a sudden, a sunspot demand shock arrives to the economy, the current surplus is necessarily impacted. To see this transparently, consider the simple case with $\hat{s}_t = \bar{s}$, so that the level of surpluses is

$$S_t = \bar{s}Y_t + \alpha \frac{B_t}{P_t}.$$

The real debt balance B_t/P_t is pre-determined, and so S_t falls when Y_t suddenly falls. Whether this is a problem depends on the stance of fiscal policy.

If $\alpha > 0$, the decline in S_t is no problem: the government finances the current deficit with a debt balance that grows over time, which eventually leads to higher surpluses. This type of surplus rule ensures that the debt valuation equation (GD) holds, despite the sunspot shock to demand. And so the initial non-fundamental recession is admissible.

But if $\alpha \leq 0$, the decline in S_t creates a tension: the government does not plan to raise future surpluses, despite a growing debt balance. There are two (equivalent) stories one can tell about how this policy stance eliminates the initial sunspot recession. The first one works through the debt valuation equation (GD). Given the decline in current and future surpluses, the present-value of surpluses declines and would, all else equal, fall short of the debt value B_t/P_t . If so, government debt acts as “net wealth” for households, who rush to sell their bonds and spend the excess—this demand surge is inconsistent with the conjectured demand recession. The second story leverages the transversality condition (TVC). If the government does not raise future surpluses, then its debt stock has to continually grow over time, not only to finance the initial decline in surpluses, but then later to finance the ever-growing stock of debt. But the explosive debt path is inconsistent with the economy’s appetite for debtholdings. Not surprisingly, since (TVC) and (GD) are equivalent conditions, these stories are two sides of the same coin.

A seemingly important point in the logic is that the debt stock B_t/P_t is pre-determined. Mathematically, this means that B_t/P_t evolves “locally deterministically,” i.e., it only has drift and no diffusion over small time intervals dt . Thus, the real debt value cannot inherit sunspot shocks. Two natural questions arise: why is B_t/P_t locally deterministic, and does this matter to the deeper equilibrium selection points?

Why are B_t and P_t locally deterministic? For B_t , this is due to our assumption of short-term debt: the price of debt is always 1, and so the value of debt “today” is always equal to the amount of debt issued “yesterday.” For P_t , this is a restatement of the fact

that prices are sticky. Nominal rigidities should mean that there is some high-enough frequency at which prices don't adjust; in continuous time, that high frequency is the Brownian one. For example, in our world with Rotemberg price stickiness, we can actually prove that the price *level* cannot have a diffusion component (see Appendix G). If firms had such fast-moving prices, they would incur too many price adjustment costs, and this is not optimal.¹⁷

At a deeper level, does it matter to our results that B_t and P_t are locally deterministic? This property is part of the baseline environment, but the economic logic is not merely a timing artifact. If debt values or the price level can move at high frequency, those movements must themselves satisfy additional equilibrium restrictions. For example, consider an extension with long-term debt, which introduces a per-unit debt price Q_t than can potentially move with shocks. Although the per-unit bond price Q_t can absorb demand shocks *in principle*, this is a forward-looking object. It must satisfy an asset-pricing equation and is not free to absorb any and all demand shocks.¹⁸ For the price level, one can imagine an extension where some firms' prices are flexible, which permits P_t to have high-frequency shocks. But, again, the price level is not some arbitrary equilibrium object that is free to absorb any and all demand shocks: prices still need to be consistent with firms' pricing strategies, which constrains how P_t moves.

Remark 2 (Discrete time). *One might suspect that the ability of active fiscal policy to prevent uncertainty traps is an artifact of the continuous-time formulation. Appendix D shows that this is not the case. It develops a discrete-time version of the baseline economy, and Proposition D.2 establishes that active fiscal policy rules out uncertainty traps there as well.*

4.4 Fiscal volatility targeting

Our benchmark fiscal policies, while transparent, may be unsatisfactory. The always-passive fiscal policy with $\alpha > 0$ has the issue that it permits uncertainty traps, unless monetary policy is able to do explicit risk targeting and is not constrained by an effective

¹⁷Similarly, in a world with Calvo price stickiness, where price-setting opportunities arrive idiosyncratically at some rate ω , a fraction ωdt of firms may change their price over a short time interval dt . This also implies that P_t is locally deterministic.

¹⁸For instance, if debt matures exponentially at rate β , then the bond price satisfies the pricing equation

$$Q_t = \tilde{\mathbb{E}}_t \left[\int_t^\infty \beta e^{-\int_t^s (\beta + \iota_u) du} ds \right],$$

where $\tilde{\mathbb{E}}$ denotes the risk-neutral expectation. This puts a strong constraint on how Q_t can move. For instance, if interest rates are pegged ($\iota_u = \bar{r}$), then this debt price is $Q_t = \beta / (\beta + \bar{r})$ which necessarily has zero volatility and cannot absorb any demand shocks. More generally, Khorrami and Mendo (2025b) show that the economy with long-term debt still has $\sigma_x = 0$ under active fiscal policy.

lower bound. The always-active fiscal policy with $\alpha \leq 0$ eliminates uncertainty traps but introduces fiscal volatility.

We now consider a class of fiscal policies, *fiscal volatility targeting*, that can eliminate uncertainty traps without going all the way to fiscal dominance. The policy specification builds on [Cochrane \(2023\)](#) in distinguishing between the debt-to-GDP path b_t that emerges under an arbitrary sequence of shocks versus a latent version of debt-to-GDP b_t^* that ignores contributions from certain subsets of shocks. In particular, we design b_t^* , which works like a debt-to-GDP target, by ignoring contributions from self-fulfilling demand and volatility shocks. Surplus policy depends on b_t^* rather than b_t , which intuitively disallows fiscal accommodation to arbitrary self-fulfilling shocks.

The details are as follows. Actual debt-to-GDP $b_t = \frac{B_t}{P_t Y_t}$ evolves as

$$db_t = (\rho b_t - s_t)dt - b_t(\sigma_{x,t}dZ_t + \zeta_{x,t}dZ_t) \quad (26)$$

where we have used the flow government budget constraint (20), the price level dynamics $dP_t/P_t = \pi_t dt$, and the dynamics of x_t in (22). By contrast, consider a latent variable that tracks debt-to-GDP without the non-fundamental demand shocks (i.e., with neither sunspot shocks nor fiscal shocks):

$$db_t^* = (\rho b_t^* - s_t)dt. \quad (27)$$

We assume that surpluses-to-GDP are given by the rule

$$s_t = \hat{s}_t + \bar{\alpha} b_t^*, \quad \bar{\alpha} > 0. \quad (28)$$

Thus, fiscal policy has passive traits, in the sense that it raises taxes or reduces spending in response to rising debt paths—but not in response to non-fundamental shocks.

For instance, if non-fundamental recession were to emerge, the decline in output Y_t would reduce current surpluses $S_t = \hat{s}_t Y_t$ and raise the debt stock B_t/P_t over time. Whereas a conventional passive fiscal policy would raise taxes in response, this fiscal volatility targeting does not, at least not for a while. In fact, fiscal volatility targeting does even more stimulus in the short run, only raising taxes later.¹⁹ In other words, fiscal volatility targeting effectively does unbacked spending during non-fundamental recessions, and it thus resembles active fiscal policy in such cases. An alternative way to

¹⁹To see this point, it is helpful to rewrite the surplus rule (28) in levels: $S_t = \hat{s}_t Y_t + \bar{\alpha} b_t^* Y_t$. Since both terms decline when Y_t declines, the surpluses decline more under this policy than if $b_t^* Y_t$ were replaced by the predetermined quantity $b_t Y_t = B_t/P_t$. Thus, one can think of fiscal volatility targeting as doing aggressive unbacked stimulus during non-fundamental recessions.

see this point is to write $s_t = \hat{s}_t + \bar{\alpha}b_t^* + 0 \times (b_t - b_t^*)$, which suggests that fiscal policy is passive with respect to b_t^* and active with respect to $b_t - b_t^*$. But it turns out that $b_t = b_t^*$ in equilibrium, so this policy does, in fact, look passive on the equilibrium path.

Theorem 5 (Fiscal volatility targeting). *Under fiscal policy (28), all equilibria have $\sigma_{x,t} = 0$, $\varsigma_{x,t} = 0$, and $b_t = b_t^*$. Conversely, if $b_t = b_t^*$, then (TVC) automatically holds.*

The new fiscal policy rules out self-fulfilling dynamics. This includes sunspot dynamics ($\sigma_x = 0$) as well as fiscal-related dynamics ($\varsigma_x = 0$). The latter are also, in some sense, a self-fulfilling fluctuation, because fiscal shocks are not part of the “minimum state variables” that matter for consumers. Why are all these shocks eliminated? The basic idea is that fiscal policy, by responding only to b_t^* and not to $b_t - b_t^*$, is essentially *targeting* zero self-fulfilling fluctuations.

The next question is what the resulting equilibrium looks like. The usual result when fiscal policy selects equilibria is that demand becomes anchored by fiscal considerations via the debt valuation equation. One symptom of this anchoring is that demand becomes susceptible to fiscal shocks. But here, the fiscal anchor is not present. Formally, Theorem 5 shows that $b = b^*$ implies the transversality condition automatically holds. Consequently, once fiscal policy establishes the equivalence $b = b^*$, the debt valuation equation becomes redundant and provides no anchor. Fiscal policy’s entire impact here is to eliminate non-fundamental fluctuations.

To fully determine the equilibrium, we can add an assumption on monetary policy. If monetary policy is active, following the Taylor principle, then the MSV solution is the unique equilibrium. Alternatively, if monetary policy is passive, all non-fundamental fluctuations are eliminated, but the initial condition remains indeterminate.

Proposition 4. *Suppose fiscal policy follows (28), while monetary policy follows (linear MP) with $\bar{\tau} = \rho$. With active monetary policy ($\rho + \phi_x > 0$, $\rho\phi_x + \kappa(\phi_\pi - 1) > 0$), the unique equilibrium is $x_t = \pi_t = 0$. With passive monetary ($\rho\phi_x + \kappa(\phi_\pi - 1) < 0$), the equilibrium is deterministic and has $\pi_t = \frac{1}{2(\phi_\pi - 1)} [\rho - \phi_x - \sqrt{(\rho - \phi_x)^2 - 4\kappa(\phi_\pi - 1)}] x_t$, with x_0 indeterminate.*

We provide one generalization. Whereas our baseline policy is always operational, similar results hold with interventions only in *deep recessions*. Specifically, assume

$$s_t = \begin{cases} \hat{s}_t + \bar{\alpha}b_t, & \text{if } x_t > \underline{x} \text{ and } b_t < \bar{b}, \\ \hat{s}_t + \bar{\alpha}b_t^*, & \text{otherwise,} \end{cases} \quad \text{with } \bar{\alpha} > 0, \quad (29)$$

for some output gap threshold $\underline{x}$ and some debt-to-GDP threshold $\bar{b}$. Deep recessions are possible but can be rare during an uncertainty trap (recall part (iii) of Theorem 1), and so

such a policy with low $\underline{x}$ only plans to intervene rarely. The inclusion of a debt-to-GDP threshold $\bar{b}$ is only needed for a technical argument in the result to follow, and $\bar{b}$ can be made arbitrarily large. Thus, the policy resembles “escape clauses” in the monetary economics literature (Obstfeld and Rogoff, 1983, 1986; Atkeson et al., 2010; Sims, 2013). The policy also resembles real-world policies: in normal times, governments responsibly pay back debts by raising taxes or reducing spending; but in emergencies, governments favor stimulus to lift the economy out of crisis. It turns out that such a policy also completely eliminates non-fundamental fluctuations, even if $\underline{x}$ is arbitrarily low, at least within a broad class of equilibria.

Definition 3. A *full-support equilibrium* is a non-explosive equilibrium with a recurrent tail, in the sense that $\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mathbf{1}_{\{x_t \leq x\}} dt > 0$ almost-surely for every x .

Full-support equilibria cover all the uncertainty traps studied in this paper. Indeed, Lemma A.3 shows that x_t is unbounded below in any non-explosive sunspot equilibrium that has $\sigma_{x,t} = \sigma_x(x_t)$ and $\pi_t = \pi(x_t)$, for some functions $\sigma_x(\cdot)$ and $\pi(\cdot)$. Therefore, any ergodic uncertainty trap in this class has $\mathbb{P}\{x_t \leq x\} > 0$ for every x under its stationary distribution.

Proposition 5 (Escape clause). *Assume fiscal policy (29). Then, there exist arbitrarily low thresholds $\underline{x}$ such that all full-support equilibria have $\sigma_{x,t} = 0$, $\varsigma_{x,t} = 0$, and $b_t = b_t^*$.*

We conclude by highlighting a policy challenge: fiscal volatility targeting is *informationally burdensome*. In particular, policymakers ideally should know which recessions are normal and which are non-fundamental, and they should only do unbacked spending during the non-fundamental recessions. Otherwise, the policy itself can create excess volatility. This is analogous to what happens if monetary policy designs a sub-optimal interest rate target $\bar{r}_t$ that fails to track fundamental shocks. The following remark explains this in more detail, using TFP shocks as an example.

Remark 3 (Sub-optimal policy design). *If the model also featured TFP shocks, “natural output” would be stochastic Y_t^* . Suppose its log dynamics featured diffusion $\sigma_{y^*,t} dZ_t$. Suppose actual log output $\log Y_t = \log Y_t^* + x_t$ featured the diffusion $(\sigma_{y^*,t} + \sigma_{x,t}) dZ_t$, where $\sigma_{x,t}$ is a non-fundamental exposure. Ideally, policy would implement $\sigma_{x,t} = 0$.*

Now, consider a debt target given by

$$db_t^* = (\rho b_t^* - s_t) dt - b_t^* \beta_t dZ_t,$$

for some β_t . By slightly modifying the proof of Theorem 5, we still find $b_t = b_t^*$. This implies that

$$\sigma_{x,t} = \beta_t - \sigma_{y^*,t}. \quad (30)$$

Consequently, to implement, policy should set $\beta_t = \sigma_{y^*,t}$. With a sub-optimal policy, $\beta_t \neq \sigma_{y^*,t}$, demand volatility is pinned down to a non-fundamental level.

In addition, monetary policy must be passive under this fiscal policy. Recall that, if monetary policy adheres to the Taylor principle, a volatility profile σ_x can only be consistent with a non-explosive equilibrium if it is sufficiently countercyclical in deep recessions. Given a policy choice for β_t , the resulting volatility in (30) generically fails to meet this bar.²⁰ Consequently, under imperfect fiscal volatility targeting and active monetary policy, equilibrium fails to exist. Monetary policy is forced to be passive, and excess demand volatility indeed coincides with (30).

5 Conclusion

Uncertainty traps are a novel type of global indeterminacy plaguing New Keynesian models, in which self-fulfilling recessions are sustained by endogenous countercyclical risk. Conventional interest-rate rules following the Taylor principle, no matter how aggressive, are no longer sufficient to eliminate self-fulfilling fluctuations. Instead, we identify three types of policies that can escape uncertainty traps, each with their own trade-offs. Our paper thus contributes to a deeper understanding of how policy regimes work in New Keynesian models and, more specifically, how their efficacy changes in the presence of endogenous risk.

The first effective policy is a monetary rule that, in addition to output and inflation, directly targets risk, lowering interest rates sufficiently when risk rises. A benefit of this approach is that it fits in naturally with the conventional Taylor-rule approach. An effective lower bound presents an obvious issue with such interest rate rules. Unless unconventional monetary policy (unmodeled in this paper) can fill the void left by a constrained interest rate, an alternative policy is needed.

The second effective policy is fiscal dominance, namely a regime in which government spending and transfers are only partially backed by future taxes. When unbacked spending occurs during an uncertainty trap recession, it devalues government debt, inducing agents to shift away from savings and towards consumption—this stimulative effect unravels the self-fulfilling aspect of the initial recession. This approach includes

²⁰For instance, imagine TFP volatility is constant $\sigma_{y^*,t} = \bar{\sigma}_{y^*}$. Suppose policy overestimates TFP volatility by 50% and sets $\beta_t = 1.5 \times \bar{\sigma}_{y^*}$, thereby inducing $\sigma_{x,t} = 0.5 \times \bar{\sigma}_{y^*}$. This constant output-gap volatility cannot be consistent with an uncertainty trap.

simple specifications such as exogenous proportional taxes and transfers, and it works at the effective lower bound or under an interest rate peg. A downside of fiscal dominance is that the economy inherits additional fluctuations due to fiscal shocks. The policy aims to eliminate one type of volatility—uncertainty traps—but introduces another.

A third effective policy—fiscal volatility targeting—combines virtues of the first two policies and mitigates their flaws. Under fiscal volatility targeting, tax backing depends on whether a recession is fundamental or non-fundamental. Government spending during normal recessions is completely backed. By contrast, spending during uncertainty traps is unbacked: the government does not raise taxes to pay back debt that was driven up by a previous non-fundamental recession. Such a policy eliminates non-fundamental volatility, even when monetary policy is constrained, without linking the economy to fiscal shocks. We view this result as providing a guide: a government seeking to eliminate uncertainty traps and related phenomena should, to the best of its ability, do unbacked spending only when those non-fundamental recessions emerge. Of course, such a policy is informationally-demanding, in the sense that policymakers must identify which are the non-fundamental recessions and which are normal downturns.

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Online Appendix:

Escaping Uncertainty Traps

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A Proofs of main results

A.1 Non-explosion and transversality conditions

For completeness, we briefly document the non-explosion requirements imposed by consumer and firm optimality. We then show that our non-explosion Condition 1 suffices to ensure these requirements hold. Thus, besides the standard derivations in the text, this will complete the proof of Lemma 1.

For the consumer side, using $C_t = L_t = Y^* e^{x_t}$, utility can be written as

$$U_0 = \frac{\log Y^*}{\rho} + \int_0^\infty e^{-\rho t} \mathbb{E} \left[x_t - \frac{(Y^*)^{1+\varphi}}{1+\varphi} e^{(1+\varphi)x_t} \right] dt.$$

Condition 1 implies

$$\int_0^\infty e^{-\rho t} \mathbb{E} \left[|x_t| + e^{(1+\varphi)x_t} \right] dt < \infty.$$

Indeed, (8) implies that the expectation is bounded uniformly for all sufficiently large t , so integrand vanishes exponentially, and hence the indefinite integral is finite. Therefore, the negative part of the consumer's payoff is integrable, and $U_0 > -\infty$.

The consumer's transversality condition holds if and only if the lifetime budget constraint (4) holds with equality. Now, note that since price adjustment costs are non-pecuniary, the real present value of aggregate profits are $\Pi_t = \mathbb{E}_t[\int_t^\infty \frac{M_s}{M_t} (Y_s - \frac{W_s L_s}{P_s}) ds]$. Using the resource constraint $C_t = Y_t$ and $B_0 = 0$, we therefore have that the consumer lifetime budget constraint (4) holds with equality, so long as all these integrals converge. Convergence of the integrals can be evaluated using the FOCs. The consumption FOC (6) implies $\mathbb{E}_0[\int_0^\infty M_t C_t dt] = (\rho\lambda)^{-1}$, so this integral converges. The labor FOC (5) and market clearing $C_t = L_t$ imply $\mathbb{E}_0[\int_0^\infty M_t \frac{W_t L_t}{P_t} dt] = \lambda^{-1} \mathbb{E}_0[\int_0^\infty e^{-\rho t} C_t^{1+\varphi} dt]$. Since $C_t^{1+\varphi} = (Y^*)^{1+\varphi} e^{(1+\varphi)x_t}$, this integral is finite by Condition 1.

For the firm side, note that Appendix G derives the optimality conditions from the firm's price setting problem. Condition 1 implies

$$\int_0^\infty e^{-\rho t} \mathbb{E} \left[e^{(1+\varphi)x_t} + \pi_t^2 \right] dt < \infty.$$

Thus the negative payoff terms associated with labor costs and price adjustment costs are integrable, so the firm's payoff is well-defined and strictly greater than $-\infty$. It also implies

$$\lim_{T \rightarrow \infty} e^{-\rho T} \mathbb{E}[e^{(1+\varphi)x_T}] = \lim_{T \rightarrow \infty} e^{-\rho T} \mathbb{E}[\pi_T^2] = 0,$$

because the expectations are bounded along the tail. Hence, the firm transversality conditions hold.

A.2 A tool to check for non-explosive equilibria

Here, we provide an abstract non-explosiveness lemma that is general enough to cover all the candidate stochastic equilibria in this paper, at least after an appropriate change-of-variables.

Let Z be a one-dimensional Brownian motion. Let $\mathcal{E}_t \in (e_L, e_H)$ be one-dimensional diffusion, following

$$d\mathcal{E}_t = \mu_e(\mathcal{E}_t)dt + \sigma_e(\mathcal{E}_t)dZ_t, \quad (\text{A.1})$$

where $-\infty < e_L < e_H < \infty$. Assume μ_e, σ_e are continuous on $[e_L, e_H]$. Let $V_t \geq 0$ satisfy

$$dV_t = \mu_v(V_t, \mathcal{E}_t)dt + \sigma_v(V_t, \mathcal{E}_t)dZ_t. \quad (\text{A.2})$$

Lemma A.1. *Consider the setting above. Assume that*

- (H1) *There is a strictly positive, bounded, and continuous function $\bar{v} : [e_L, e_H] \mapsto \mathbb{R}_+$ such that $V_t \leq \bar{v}(\mathcal{E}_t)$ at all times.*
- (H2) *$\mu_v(v, \mathcal{E})$ and $\sigma_v(v, \mathcal{E})$ are bounded on every set $\{(v, \mathcal{E}) : \varepsilon \leq v \leq \bar{v}(\mathcal{E}), \mathcal{E} \in (e_L, e_H)\}$, each $\varepsilon > 0$.*
- (H3) *$\sigma_v^2(v, \mathcal{E})$ is strictly positive for all $v > 0$ and $v < \bar{v}(\mathcal{E})$.*
- (H4) *For some $v_- > 0$, $\inf_{\mathcal{E} \in (e_L, e_H)} \inf_{v \leq v_-} (\frac{\sigma_v(v, \mathcal{E})}{v})^2 > 0$.*
- (H5) *$\theta > 1$, where*

$$\theta := \inf_{\mathcal{E} \in (e_L, e_H)} \lim_{v \rightarrow 0} \frac{2v\mu_v(v, \mathcal{E})}{\sigma_v^2(v, \mathcal{E})}. \quad (\text{A.3})$$

The limit exists for every $\mathcal{E} \in (e_L, e_H)$, and the convergence is uniform in $\mathcal{E}$.

Then, $\inf_{t \rightarrow \infty} \mathbb{E}[\log V_t] > -\infty$.

Proof of Lemma A.1. Let $\mathcal{D} := \{(v, \mathcal{E}) : 0 \leq v \leq \bar{v}(\mathcal{E}), e_L < \mathcal{E} < e_H\}$ denote the domain of the state dynamics. By assumption (H1), the process $(V_t, \mathcal{E}_t)$ remains in $\mathcal{D}$ with probability 1. The crux of the proof is to essentially show that V_t does not concentrate probability near the lower boundary $v = 0$.

Let θ be defined by (A.3). Choose $\alpha \in (0, \frac{\theta-1}{2})$. Define the Lyapunov function

$$f(v, \mathcal{E}) = v^{-\alpha}.$$

Letting $\mathcal{L}$ denote the infinitesimal generator of $(V, \mathcal{E})$, we have

$$\mathcal{L}f(v, \mathcal{E}) = -\alpha\mu_v(v, \mathcal{E})v^{-\alpha-1} + \frac{1}{2}\alpha(1+\alpha)\sigma_v^2(v, \mathcal{E})v^{-\alpha-2}. \quad (\text{A.4})$$

By assumption (H5), there exists $v_- < \inf_{\mathcal{E}} \bar{v}(\mathcal{E})$ such that for all $v \in (0, v_-]$,

$$\mu_v(v, \mathcal{E}) \geq \frac{1}{2}(1+2\alpha)\frac{\sigma_v^2(v, \mathcal{E})}{v}. \quad (\text{A.5})$$

Using (A.5) in (A.4), we have

$$\mathcal{L}f(v, \mathcal{E}) \leq -\frac{1}{2}\alpha^2 v^{-\alpha-2} \sigma_v^2(v, \mathcal{E}), \quad \text{for } v \in (0, v_-]. \quad (\text{A.6})$$

By assumptions (H3) and (H4), $c := \inf_{\mathcal{E}} \inf_{v \leq v_-} \frac{1}{2}\alpha^2 v^{-2} \sigma_v^2(v, \mathcal{E})$ is strictly positive, and so

$$\mathcal{L}f(v, \mathcal{E}) \leq -cf(v, \mathcal{E}), \quad \text{with } c > 0, \quad \text{for } v \leq v_-. \quad (\text{A.7})$$

On $\{v_- \leq v \leq v_+\}$, where $v_+ := \max_{\mathcal{E} \in [e_L, e_H]} \bar{v}(\mathcal{E})$, compactness and assumption (H2) implies that for some $b < \infty$, we have

$$\mathcal{L}f(v, \mathcal{E}) \leq b - cf(v, \mathcal{E}), \quad \text{for } v_- \leq v \leq v_+ \quad (\text{A.8})$$

Therefore, combining (A.7)-(A.8), and noting that $f(v, \mathcal{E}) < f(v_-, \mathcal{E})$ for all $v > v_-$, we obtain

$$\mathcal{L}f(v, \mathcal{E}) \leq b - cf(v, \mathcal{E}), \quad \text{on } \mathcal{D}. \quad (\text{A.9})$$

By Lemma A.2 below, (A.9) implies

$$\sup_{t \geq 0} \mathbb{E}[V_t^{-\alpha}] = \sup_{t \geq 0} \mathbb{E}[f(V_t, \mathcal{E}_t)] < \infty. \quad (\text{A.10})$$

Letting $x_t := \log(V_t)$ and applying Jensen's inequality on the convex map $u \mapsto e^{-\alpha u}$ gives

$$e^{-\alpha \mathbb{E}[x_t]} \leq \mathbb{E}[e^{-\alpha x_t}] = \mathbb{E}[V_t^{-\alpha}] < \infty,$$

so $\mathbb{E}[x_t] \geq -\frac{1}{\alpha} \log \mathbb{E}[V_t^{-\alpha}]$. Finally, take the infimum of both sides, using (A.10), to obtain $\inf_{t \rightarrow \infty} \mathbb{E}[x_t] > -\infty$. This completes the proof. $\square$

Lemma A.2. *Let $(X_t)_{t \geq 0}$ be a time-homogeneous diffusion process on a state space $\mathcal{X}$ with extended generator $\mathcal{L}$. Suppose a C^2 function $f : \mathcal{X} \mapsto \mathbb{R}_+$ satisfies*

$$\mathcal{L}f(x) \leq -cf(x) + b, \quad c > 0, \quad b < \infty.$$

Then, for all $t \geq 0$,

$$\mathbb{E}^x[f(X_t)] \leq (f(x) - b/c)e^{-ct} + \frac{b}{c}.$$

In particular, $\sup_{t \geq 0} \mathbb{E}^x[f(X_t)] < \infty$.

Proof of Lemma A.2. Using the generator bound, for any fixed $t > 0$, one gets

$$\mathbb{E}^x[f(X_t)] = f(x) + \int_0^t \mathbb{E}^x[\mathcal{L}f(X_s)] ds \leq f(x) - c \int_0^t \mathbb{E}^x[f(X_s)] ds + bt.$$

Set $F(t) = \mathbb{E}^x[f(X_t)]$. Then,

$$F(t) \leq f(x) - c \int_0^t F(s) ds + bt.$$

Note that F is time-differentiable, by Itô's formula, since f is a C^2 function. Differentiating yields

$$F'(t) \leq -cF(t) + b, \quad F(0) = f(x).$$

Solve this linear differential inequality by integrating against the factor e^{ct} :

$$\frac{d}{dt}[e^{ct}F(t)] \leq be^{ct} \implies e^{ct}F(t) - F(0) \leq \frac{b}{c}(e^{ct} - 1).$$

Rearranging gives the stated bound. $\square$

A.3 Proofs for Section 2

Proof of Theorem 1. Note that parts (i)-(ii) of this theorem are a special case of Theorem B.1, which also includes monetary policy uncertainty (see Appendix B.2).²¹ That said, we provide a complete proof of the present theorem for completeness.

Proof of statement (i). First, construct the volatility function $\sigma_x(x)$. Let $k > 2$ and $\bar{x} < 0$ be arbitrary. Let volatility be given by

$$\sigma_x^2(x) := \begin{cases} k[g(x) - \Phi(x)], & \text{if } x < \bar{x}, \\ 0, & \text{if } x \geq \bar{x}, \end{cases} \quad (\text{A.11})$$

where $g(\cdot)$ is any positive, finite-valued function satisfying $\lim_{x \rightarrow -\infty} g(x) = +\infty$. Since $\Phi(\cdot)$ is increasing and continuous, we have that $\sigma_x^2(x) \geq 0$, as required for there to be “excess volatility.” For the same reason, notice that $\sigma_x^2(x) > 0$ for all $x < \bar{x}$.

Next, consider the change of variables:

$$V_t := \exp[x_t].$$

Given $\bar{v} = \rho$, the dynamics of x_t are $dx_t = [\Phi(x_t) + \frac{1}{2}\sigma_x^2(x_t)]dt + \sigma_x(x_t)dZ_t$. Therefore, the dynamics of V_t are

$$\begin{aligned} dV_t &= \mu_v(V_t)dt + \sigma_v(V_t)dZ_t, \quad \text{where} \\ \mu_v(v) &:= v \left[\Phi(\log(v)) + \sigma_x^2(\log(v)) \right] \\ \sigma_v(v) &:= v\sigma_x(\log(v)). \end{aligned}$$

The plan is to show that V_t satisfies the hypotheses of Lemma A.1. Doing so will show that $\liminf_t \mathbb{E}[\log(V_t)] > -\infty$. Since $x^*(\mathcal{E}_t)$ is bounded and the construction keeps $x_t \leq x^*(\mathcal{E}_t)$, this implies $\limsup_{t \rightarrow \infty} \mathbb{E}|x_t| < \infty$. Moreover, the same upper bound on x_t implies $\limsup_{t \rightarrow \infty} \mathbb{E}[e^{(1+\varphi)x_t}] < \infty$. Hence, this will show that Condition 1 holds.

We obviously have that $V_t \geq 0$ forever. Defining $\bar{v} := e^{\bar{x}}$, let us show that $V_t \leq \bar{v}$ forever if V_0 starts below $\bar{v}$. For x_t such that $\bar{x} \leq x_t < 0$, or equivalently $\bar{v} \leq V_t < 1$, we have that $\sigma_v = 0$, and so $dV_t = \Phi(x_t)dt < 0$, by the fact that Φ is increasing and that $x < 0$. Therefore, V_t enters the region below $\bar{v}$ in finite time when starting from any

²¹Indeed, the MSV equilibrium with $\kappa \rightarrow 0$ and $\sigma \rightarrow 0$ is exactly $x^* = 0$ and $\sigma_{x^*} = 0$. Therefore, the “gap process” $x_t - x_t^*$ studied in that proof exactly coincides with x_t for the present situation.

point $V_0 < 1$. And if $V_t \leq \bar{v}$, it can never exit this region. So V_t satisfies hypothesis (H1) of Lemma A.1.

It is easy to see that μ_v and σ_v^2 are continuous on $\{v < \bar{v}\}$ and are only potentially infinite when $v = 0$. So V_t satisfies hypothesis (H2) of Lemma A.1.

Next, we showed earlier that $\sigma_x^2(x) > 0$ for all $x < \bar{x}$, and so $\sigma_v^2(v) > 0$ for all $v < \bar{v}$. This verifies that V_t satisfies hypothesis (H3) of Lemma A.1.

Next, we have that $\lim_{v \rightarrow 0} (\frac{\sigma_v(v)}{v})^2 = \lim_{x \rightarrow -\infty} \sigma_x^2(x) = +\infty > 0$. Hence, V_t satisfies hypothesis (H4) of Lemma A.1.

We next do a similar limiting analysis for the drift in order to check the ratio hypothesis (H5). Plugging (A.11) into the expression for μ_v , we have that

$$\begin{aligned} \theta &:= \lim_{v \rightarrow 0} \frac{2v\mu_v(v)}{\sigma_v(v)^2} = \lim_{x \rightarrow -\infty} \frac{2[\Phi(x) + \sigma_x^2(x)]}{\sigma_x^2(x)} = \lim_{x \rightarrow -\infty} \frac{2\Phi(x) + 2k[g(x) - \Phi(x)]}{k[g(x) - \Phi(x)]} \\ &= 2 + \frac{2}{k} \lim_{x \rightarrow -\infty} \frac{\Phi(x)}{g(x) - \Phi(x)} \geq 2 - \frac{2}{k} > 1, \end{aligned}$$

where we have used the fact that $\lim_{x \rightarrow -\infty} \frac{\Phi(x)}{g(x) - \Phi(x)} = -1 + \lim_{x \rightarrow -\infty} \frac{g(x)}{g(x) - \Phi(x)} \geq -1$, followed by the assumption $k > 2$. Consequently, hypothesis (H5) of Lemma A.1 holds.

This verifies all the hypotheses of Lemma A.1, proving that the construction is a valid non-explosive equilibrium.

Finally, we prove the claim that any volatility function is valid if it satisfies suitable boundary conditions. Instead of the σ_x^2 function in (A.11), consider any alternative function $\tilde{\sigma}_x^2$, which (a) coincides with σ_x^2 for $x \notin (-K, \bar{x} - K^{-1})$ for K arbitrarily large; and (b) is finite and strictly positive on $x \in (-K, \bar{x} - K^{-1})$. By inspection, the entire proof above remains valid. This proves statement (i) of the proposition.

Proof of statement (ii). We now prove that all stochastic equilibria are recessionary. Consider a general rule $\Phi(x)$ satisfying $\phi_x := \inf_x \Phi'(x) > 0$. Let $\sigma_{x,t}^2$ be any volatility process such that $\sigma_{x,t}^2 \geq 0$ and is not identically zero. Suppose, leading to contradiction that $x_0 > 0$ was part of a non-explosive equilibrium.

Recall the dynamics of Δ_t are

$$dx_t = \left[\Phi(x_t) + \frac{1}{2} \sigma_{x,t}^2 \right] dt + \sigma_{x,t} dZ_t, \quad x_0 > 0.$$

Consider the alternative process $\tilde{x}_t$ which drops the excess variance from the drift:

$$d\tilde{x}_t = \Phi(\tilde{x}_t) dt + \sigma_{x,t} dZ_t, \quad \tilde{x}_0 = x_0 > 0.$$

Define the stopping time

$$\tau_0 := \inf\{t > 0 : \tilde{x}_t = 0\},$$

and put $T_0 := T \wedge \tau_0$. Then,

$$\mathbb{E}_0[e^{-\phi_x T_0} \tilde{x}_{T_0}] = \tilde{x}_0 + \mathbb{E}_0 \left[\int_0^{T_0} e^{-\phi_x t} (\Phi(\tilde{x}_t) - \phi_x \tilde{x}_t) dt \right] \geq \tilde{x}_0 = x_0,$$

since $\Phi(\tilde{x}) \geq \phi_x \tilde{x}$ for all $\tilde{x} \geq 0$ (recall that $\phi_x > 0$ is the minimal slope of the general rule $\Phi(x)$). The left-hand-side can be written $e^{-\phi_x T} \mathbb{E}_0[\tilde{x}_T \mathbf{1}_{\{T < \tau_0\}}]$, since $\tilde{x}_{\tau_0} = 0$. Thus,

$$\mathbb{E}_0[\tilde{x}_T \mathbf{1}_{\{T < \tau_0\}}] \geq e^{\phi_x T} x_0,$$

which by taking $T \rightarrow \infty$ proves that $\limsup_{T \rightarrow \infty} \mathbb{E}_0[\tilde{x}_T] = +\infty$ with positive probability. Since standard diffusion comparison theorems imply that $x_T \geq \tilde{x}_T$ almost-surely, this implies that $\limsup_{T \rightarrow \infty} \mathbb{E}_0[x_T] = +\infty$, in violation of Condition 1.

Proof of statement (iii). Fix $p > 0$, $\hat{x} < 0$. Our task is to construct a volatility function so that the resulting dynamics are stationary, non-explosive, and satisfy $\mathbb{P}\{x_t < \hat{x}\} < p$.

We first set up the problem. For notational simplicity, denote $a(x) := \sigma_x^2(x)$. In the rigid-price limit, the candidate dynamics are

$$dx_t = \left(\Phi(x_t) + \frac{1}{2} a(x_t) \right) dt + \sqrt{a(x_t)} dZ_t.$$

In general, a one-dimensional diffusion is characterized by its scale density s and speed measure m (see, e.g., [Rogers and Williams \(2000, Ch. V, Secs. 46–53\)](#)). Without loss of generality, normalize the scale density at $\hat{x}$:

$$s(x) := \exp \left\{ - \int_{\hat{x}}^x \left(1 + \frac{2\Phi(z)}{a(z)} \right) dz \right\}, \quad m(x) := \frac{2}{a(x)s(x)}.$$

We show first that the integral of s diverges at both boundaries, which makes $-\infty$ and 0 unattainable. We then show that m has finite total mass, which implies that m (normalized) is an invariant probability density. Finally, we compute $\mathbb{P}\{x_t < \hat{x}\}$.

We construct the volatility function in two pieces:

$$a(x) = \begin{cases} h(x), & \text{if } x \in (\hat{x}, 0), \\ a_K(x), & \text{if } x \in (-\infty, \hat{x}], \end{cases}$$

where $h : (\hat{x}, 0) \mapsto (0, \infty)$ is any strictly positive function that vanishes sufficiently quickly as $x \uparrow 0$, and where

$$a_K(x) := K(1 - \Phi(x)), \quad K \geq 4.$$

Now, we prove the dynamics have the desired properties. First, the assumption that h vanishes sufficiently quickly as $x \uparrow 0$ implies

$$\int_{\hat{x}}^0 s(x) dx = +\infty, \quad 0 < M_+ := \int_{\hat{x}}^0 m(x) dx < \infty. \quad (\text{A.12})$$

The first of these conditions implies that the upper boundary is inaccessible. The second condition implies that mass is not concentrated near 0 in the long run. Note for later that M_+ is independent of K .

Next, we prove that $-\infty$ is inaccessible. Note that, for $x \leq \hat{x}$,

$$1 + \frac{2\Phi(x)}{a_K(x)} = 1 + \frac{2\Phi(x)}{K(1 - \Phi(x))} \geq 1 - \frac{2}{K} \geq \frac{1}{2},$$

where the first inequality uses $-\Phi/(1 - \Phi) < 1$ and the second uses $K \geq 4$. Therefore,

$$\frac{s(x)}{s(\hat{x})} = \exp \left\{ \int_x^{\hat{x}} \left(1 + \frac{2\Phi(z)}{a_K(z)} \right) dz \right\} \geq e^{(\hat{x}-x)/2}, \quad \text{for } x \leq \hat{x}. \quad (\text{A.13})$$

Thus, $s(x) \rightarrow +\infty$ exponentially as $x \rightarrow -\infty$, so that $\int_{-\infty}^{\hat{x}} s(x) dx = +\infty$, implying the lower boundary is inaccessible.

Next, we prove that the speed measure has finite mass in the lower tail. Using (A.13), as well as $a_K(x) = K(1 - \Phi(x)) \geq K$, we obtain

$$m(x) = \frac{2}{a_K(x)s(x)} \leq \frac{2}{Ks(\hat{x})} e^{(x-\hat{x})/2}, \quad \text{for } x \leq \hat{x}.$$

Integrating, we have

$$\int_{-\infty}^{\hat{x}} m(x) dx \leq \frac{4}{Ks(\hat{x})} = \frac{4}{K}, \quad (\text{A.14})$$

where we have used $s(\hat{x}) = 1$.

Finally, given the speed measure has finite total mass, it coincides with the scaled invariant density. Additionally, while m potentially has a discontinuity at $\hat{x}$, it remains

finite there, and so there is no point mass anywhere. Now, choose K large so that

$$M_- := \int_{-\infty}^{\hat{x}} m(x) dx < \frac{p}{1-p} M_+$$

and hence

$$\mathbb{P}\{x < \hat{x}\} = \frac{M_-}{M_- + M_+} < p. \quad (\text{A.15})$$

This concludes the proof. $\square$

Proof of Proposition 1. Into (15), we plug the volatility function $\tilde{\sigma}_x$, relative prudence $RP_t = 1 + \gamma$, elasticity of intertemporal substitution $EIS_t = \gamma^{-1}$, monetary policy rule $\iota = \rho + \tilde{\Phi}(x)$, and impose the rigid-price limit $\pi = 0$. The result is

$$\mu_x = \gamma^{-1} \tilde{\Phi}(x) + \frac{1}{2} \gamma \tilde{\sigma}_x^2(x).$$

This implies that the drift and diffusion of $y_t = e^{x_t}$ are given by

$$\begin{aligned} \mu_y &= y \left[\gamma^{-1} \tilde{\Phi}(x) + \frac{1}{2} (1 + \gamma) \tilde{\sigma}_x^2(x) \right] \\ \sigma_y &= y \tilde{\sigma}_x(x) \end{aligned}$$

For these dynamics, we compute the tail statistic in equation (A.3) of Lemma A.1 to be

$$\tilde{\theta} := \lim_{y \rightarrow 0} \frac{2y\mu_y}{\sigma_y^2} = 1 + \gamma + 2 \lim_{x \rightarrow -\infty} \frac{\gamma^{-1} \tilde{\Phi}(x)}{\tilde{\sigma}_x^2(x)}.$$

Repeat the same analysis for log utility version with reaction function $\Phi(x)$ and volatility function $\sigma_x(x)$ to obtain the tail statistic

$$\theta := \lim_{y \rightarrow 0} \frac{2y\mu_y}{\sigma_y^2} = 2 + 2 \lim_{x \rightarrow -\infty} \frac{\Phi(x)}{\sigma_x^2(x)}.$$

Assume $\Phi(x)$ and $\sigma_x^2(x)$ are an equilibrium with log utility, meaning $\theta > 1$. Using $\tilde{\Phi}(x) = \gamma \Phi(x)$ and $\tilde{\sigma}_x^2(x) = \gamma^{-1} \sigma_x^2(x)$, we then obtain that

$$\tilde{\theta} = 1 + \gamma + 2\gamma \lim_{x \rightarrow -\infty} \frac{\Phi(x)}{\sigma_x^2(x)} = \theta + (\gamma - 1)(\theta - 1)$$

Since $\theta > 1$ and $\gamma > 1$, we have $\tilde{\theta} > 1$, implying $\tilde{\Phi}(x)$ and $\tilde{\sigma}_x^2(x)$ constitute an equilibrium

with CRRA utility. $\square$

Proof of Proposition 2. Let $V(x) := \nu(x)^2 - \nu(0)^2$. In the baseline rigid-price model, $dx_t = \mu_x(x_t)dt + \sigma_x(x_t)dZ_t$, with $\mu_x(x) = \Phi(x) + \frac{1}{2}\sigma_x^2(x)$. In the model with idiosyncratic risk, define $\tilde{\sigma}_x^2(x) := \max\{0, \sigma_x^2(x) - 2V(x)\}$. The associated output-gap drift is

$$\tilde{\mu}_x(x) = \Phi(x) + V(x) + \frac{1}{2}\tilde{\sigma}_x^2(x) = \Phi(x) + \max\left\{V(x), \frac{1}{2}\sigma_x^2(x)\right\}.$$

We verify non-explosiveness by checking the two boundaries, as in the baseline proof. As $x \rightarrow -\infty$, the assumption $\frac{1}{2}\sigma_x^2(x) > V(x)$ implies that the max operator is inactive. Hence, for all $x < 0$ sufficiently negative, $\tilde{\sigma}_x^2(x) = \sigma_x^2(x) - 2V(x)$ and $\tilde{\mu}_x(x) = \Phi(x) + V(x) + \frac{1}{2}(\sigma_x^2(x) - 2V(x)) = \Phi(x) + \frac{1}{2}\sigma_x^2(x) = \mu_x(x)$. Thus, the transformed process has the same drift as the baseline process in the lower tail and a weakly smaller diffusion. Since (μ_x, σ_x) satisfies (10), so does $(\tilde{\mu}_x, \tilde{\sigma}_x)$. Therefore, the transformed process is non-explosive toward $-\infty$.

Next consider $x \uparrow 0$. Since $\nu(x) > 0$ and $\nu'(x) < 0$ for all $x \leq 0$, we have $V(x) > 0$ for $x < 0$ and $V(x) \rightarrow 0$ as $x \uparrow 0$. Therefore $0 \leq \tilde{\sigma}_x^2(x) \leq \sigma_x^2(x)$ for all $x < 0$, so $\lim_{x \uparrow 0} \tilde{\sigma}_x(x) = 0$. Moreover, since both $V(x)$ and $\sigma_x^2(x)$ vanish as $x \uparrow 0$, $\lim_{x \uparrow 0} \tilde{\mu}_x(x) = \lim_{x \uparrow 0} \mu_x(x)$. Hence the transformed dynamics inherit the upper-boundary behavior of the baseline equilibrium at 0. Since the baseline process is non-explosive on $\{x \leq 0\}$, the transformed process is also non-explosive on $\{x \leq 0\}$.

Finally, for all $x < 0$ sufficiently negative, the max operator is inactive, so $\tilde{\sigma}_x^2(x) = \sigma_x^2(x) - 2[\nu(x)^2 - \nu(0)^2] < \sigma_x^2(x)$. The gap is $\sigma_x^2(x) - \tilde{\sigma}_x^2(x) = 2[\nu(x)^2 - \nu(0)^2]$. Since $\nu(x) > 0$ and $\nu'(x) < 0$, this gap increases as x decreases, as claimed. $\square$

Proof of Theorem 2. The theorem leverages results in Appendix C. The hypotheses $\kappa > 0$, $\phi_x > 0$, $\phi_\pi > 1$, $\bar{l} = \rho$ are maintained throughout the results invoked below.

Proof of statement (i). For a deflationary left tail, choose $\pi_1 \in (0, \min\{\frac{\rho}{\phi_\pi - 1}, \frac{\kappa}{\rho}\})$ and choose $\pi_0 < 0$ satisfying (C.1). Such a π_0 always exists because the condition can be satisfied by taking π_0 sufficiently negative. Lemma C.4 then yields a non-explosive sunspot equilibrium, stationary and ergodic on $(-\infty, \bar{x})$, with $\bar{x} < 0$,

$$\pi(x) = \pi_0 + \pi_1 x, \quad \sigma_x^2(x) = 2D(\bar{x} - x)$$

for some $D > 0$. Hence σ_x^2 is strictly decreasing and converges to zero as $x \uparrow \bar{x}$. Moreover, $\pi(x) < 0$ for every $x \leq \bar{x}$, so this equilibrium has a deflationary left tail.

For an inflationary left tail, Lemma C.5 yields, for $\bar{x} < 0$ sufficiently negative, a non-explosive sunspot equilibrium that is stationary and ergodic on $(-\infty, \bar{x})$. By Lemma C.2, there exists $\underline{x} < \bar{x}$ such that $\pi(x) > 0$ and $\sigma_x^2(x)$ is decreasing for every $x \leq \underline{x}$. By Lemma C.3, $\sigma_x^2(x) \rightarrow 0$ as $x \uparrow \bar{x}$. This equilibrium therefore has an inflationary left tail and satisfies all the remaining claims in statement (i).

Proof of statement (ii). Set $v := \sigma_x^2$. The assumed regularity and boundary requirements verify conditions (i), (iii), and (iv) of Theorem C.1. Under the linear Taylor rule, equation (14) is precisely $\mathcal{T}[v, \pi](x) = 0$ wherever π is twice differentiable, and therefore verifies condition (ii) of that theorem. All the hypotheses of Theorem C.1 are thus satisfied. Its conclusion implies that $\pi_t = \pi(x_t)$ and $\sigma_{x,t}^2 = v(x_t)$ support a non-explosive sunspot equilibrium in which x_t is stationary and ergodic on $(-\infty, \bar{x})$. $\square$

A.4 Proofs for Section 3

Proof of Theorem 3. Substituting the policy rule into the IS equation gives $\mu_{x,t} = \phi_x x_t + (\phi_\pi - 1)\pi_t + \frac{1}{2}(1 - \alpha(x_t))\sigma_{x,t}^2$, where $\alpha(x) := \alpha_+ \mathbf{1}_{\{x > 0\}} + \alpha_- \mathbf{1}_{\{x < 0\}}$ is the state-dependent risk premium response, and with the convention that $\pi_t \equiv 0$ in the rigid-price case.

Suppose $\kappa \geq 0$, nesting both the rigid-price case and the partially-flexible price case. Let $a := \phi_\pi - 1$, so $a\kappa + \rho\phi_x > 0$. We claim that there exists $c > 0$ and $\lambda > 0$ such that

$$\phi_x x^2 + (a - c\kappa)x\pi + c\rho\pi^2 \geq \lambda(x^2 + c\pi^2) \quad (\text{A.16})$$

for all (x, π) . Equivalently, we want to show that $(x, \pi)A_{c,\lambda}(x, \pi)' \geq 0$ with matrix

$$A_{c,\lambda} = \begin{bmatrix} \phi_x - \lambda & \frac{1}{2}(a - c\kappa) \\ \frac{1}{2}(a - c\kappa) & c(\rho - \lambda) \end{bmatrix}$$

Matrix $A_{c,\lambda}$ is positive definite if and only if both eigenvalues are positive, or equivalently

$$\begin{aligned} \text{tr}(A_{c,\lambda}) &= \phi_x - \lambda + c(\rho - \lambda) > 0 \\ \det(A_{c,\lambda}) &= c(\phi_x - \lambda)(\rho - \lambda) - \frac{1}{4}(a - c\kappa)^2 > 0. \end{aligned}$$

It suffices to find c such that $\text{tr}(A_{c,0}) > 0$ and $\det(A_{c,0}) > 0$, i.e., when $\lambda = 0$. Indeed, under such a c , we may choose λ sufficiently small to ensure that $\text{tr}(A_{c,\lambda}) > 0$ and $\det(A_{c,\lambda}) > 0$ continue to hold.

If $\kappa = 0$, choose any $c > a^2/(4\rho\phi_x)$. Then, $\det(A_{c,0}) = c\rho\phi_x - a^2/4 > 0$ and $\text{tr}(A_{c,0}) =$

$\phi_x + c\rho > 0$. If $\kappa > 0$, choose $c := (2\rho\phi_x + \kappa a)/\kappa^2 > 0$, so that

$$\det(A_{c,0}) = c\rho\phi_x - \frac{1}{4}(a - c\kappa)^2 = \frac{\rho\phi_x(\rho\phi_x + \kappa a)}{\kappa^2} > 0.$$

Also, $\text{tr}(A_{c,0}) = \phi_x + c\rho > 0$. Thus, in either case, there exists $c > 0$ such that $A_{c,0}$ is positive definite. As explained above, this proves (A.16) holds for sufficiently small λ . We pick such a c and λ .

For $\varepsilon > 0$, define $f^\varepsilon(x, \pi) = (x^2 + c\pi^2 + \varepsilon)^{1/2}$, which is strictly convex as a function of (x, π) . Define $R_t^\varepsilon := f^\varepsilon(x_t, \pi_t)$. Applying Itô's formula to R_t^ε , we have

$$\begin{aligned} dR_t^\varepsilon &= \frac{1}{R_t^\varepsilon} \left(x_t dx_t + c\pi_t d\pi_t \right) \\ &\quad + \frac{1}{2} \left[f_{xx}^\varepsilon(x_t, \pi_t) \sigma_{x,t}^2 + f_{\pi\pi}^\varepsilon(x_t, \pi_t) \sigma_{\pi,t}^2 + 2f_{x\pi}^\varepsilon(x_t, \pi_t) \sigma_{x,t} \sigma_{\pi,t} \right] dt \end{aligned}$$

Because f^ε is strictly convex, the second line is non-negative for any vector of diffusions (σ_x, σ_π) , and so

$$dR_t^\varepsilon \geq \frac{1}{R_t^\varepsilon} \left(x_t dx_t + c\pi_t d\pi_t \right)$$

Using the IS equation and the Phillips curve to substitute dx_t and $d\pi_t$, we obtain

$$dR_t^\varepsilon \geq \frac{\phi_x x_t^2 + (a - c\kappa)x_t \pi_t + c\rho \pi_t^2 + \frac{1}{2}x_t(1 - \alpha(x_t))\sigma_{x,t}^2}{R_t^\varepsilon} dt + \frac{x_t \sigma_{x,t} + c\pi_t \sigma_{\pi,t}}{R_t^\varepsilon} dZ_t$$

The assumptions on $\alpha(x)$ imply the risk-premium term is nonnegative: $x(1 - \alpha(x)) \geq 0$. Using this inequality and (A.16),

$$dR_t^\varepsilon \geq \lambda \frac{x_t^2 + c\pi_t^2}{R_t^\varepsilon} dt + \frac{x_t \sigma_{x,t} + c\pi_t \sigma_{\pi,t}}{R_t^\varepsilon} dZ_t.$$

Integrating and taking expectations yields

$$\mathbb{E}[R_T^\varepsilon] \geq \mathbb{E}[R_0^\varepsilon] + \lambda \int_0^T \mathbb{E} \left[\frac{x_t^2 + c\pi_t^2}{(x_t^2 + c\pi_t^2 + \varepsilon)^{1/2}} \right] dt.$$

Taking $\varepsilon \rightarrow 0$, and denoting the limiting process by $R_t := \lim_{\varepsilon \rightarrow 0} R_t^\varepsilon = (x_t^2 + c\pi_t^2)^{1/2}$, yields

$$\mathbb{E}[R_T] \geq \mathbb{E}[R_0] + \lambda \int_0^T \mathbb{E}[R_t] dt.$$

Gronwall's inequality gives $\mathbb{E}[R_T] \geq e^{\lambda T} \mathbb{E}[R_0]$. This growth for $\mathbb{E}[R_T]$ implies that non-explosiveness conditions are violated unless $\mathbb{E}[R_t] = 0$ for every t . Thus $x_t = \pi_t = 0$ for all $t \geq 0$ a.s. $\square$

Proof of Proposition 3. Fix $x_0 < 0$. We construct a sunspot equilibrium in which $x_t \leq 0$ forever, and in which while $x_t < 0$, sunspot volatility is strictly positive, inflation is strictly negative, and the effective lower bound binds.

First, we define some constants. If $\kappa = 0$, set $\beta = 0$ and choose any $\lambda > 0$. If $\kappa > 0$, choose $\beta > 0$ such that $\beta^2 + \rho\beta < \kappa$, and set $\lambda := \kappa/\beta - \rho$. In both cases, define $\gamma := \lambda - \beta > 0$.

Next, define the inflation and volatility functions. Define

$$\pi(x) := \begin{cases} \beta x, & \text{if } x < 0; \\ 0, & \text{if } x \geq 0. \end{cases} \quad \text{and} \quad \sigma_x^2(x) := \begin{cases} 2(\rho - \underline{l} - \gamma x), & \text{if } x < 0; \\ 0, & \text{if } x \geq 0. \end{cases}$$

We set $\pi_t = \pi(x_t)$ and $\sigma_{x,t}^2 = \sigma_x^2(x_t)$. Notice that, $\pi(x_t) \leq 0$ and $\sigma_x^2(x_t) > 0$ when $x_t < 0$.

Additionally, note that the lower bound binds when $x_t < 0$. Since $s \mapsto \Psi(x, \pi, s) + s/2$ is weakly decreasing on $\mathcal{R}_-$ (the recessionary region), we have $\Psi(x, \beta x, \sigma_x^2(x)) + \frac{1}{2}\sigma_x^2(x) \leq \Psi(x, \beta x, 0)$. Moreover, monotonicity in x and π gives $\Psi(x, \beta x, 0) < \rho$ for $x < 0$. Hence,

$$\Psi(x, \beta x, \sigma_x^2(x)) < \rho - \frac{1}{2}\sigma_x^2(x) = \underline{l} + \gamma x < \underline{l}, \quad \text{if } x < 0.$$

Therefore, $\iota_t = \max\{\underline{l}, \Psi(x_t, \pi(x_t), \sigma_x^2(x_t))\} = \underline{l}$ when $x_t < 0$.

The task is to verify the dynamics in this construction lead to a non-explosive equilibrium. We first construct the dynamics and then verify the non-explosion condition.

Let $\tau_0 := \inf\{t \geq 0 : x_t = 0\}$. We first consider $t < \tau_0$, which implies $x_t < 0$ (since $x_0 < 0$). Using the construction above, the IS equation says

$$\begin{aligned} \mu_{x,t} &= \iota_t - \pi(x_t) - \rho + \frac{1}{2}\sigma_x^2(x_t) \\ &= \underline{l} - \beta x_t - \rho + (\rho - \underline{l} - \gamma x_t) \\ &= -\lambda x_t, \quad \text{for } t < \tau_0, \end{aligned}$$

where the second line substitutes the functions $\pi(\cdot)$ and $\sigma_x(\cdot)$ and uses the fact that the lower bound binds, while the third line uses $\lambda = \beta + \gamma$.

Next, guess-and-verify that $x_t = \pi_t = 0$ for all $t \geq \tau_0$ (i.e., the processes are absorbed at the origin). Indeed, this is consistent with the construction $\pi(0) = \sigma_x(0) = 0$. Furthermore, since $\Psi(0, 0, 0) = \rho$, we have that $\mu_{x,t} = \Psi(x_t, \pi_t, \sigma_{x,t}^2) - \pi_t - \rho + \frac{1}{2}\sigma_{x,t}^2 = 0$, so

the IS equation holds. Finally, we have $\mu_{\pi,t} = \rho\pi_t - \kappa x_t = 0$ and $\sigma_{\pi,t} = 0$, so the Phillips curve holds. This verifies the absorption at the origin.

Finally, we need to verify the non-explosiveness condition. Let $V_t := -x_t \geq 0$. Before absorption at zero, V_t satisfies

$$dV_t = -\lambda V_t dt + \sqrt{2(\rho - \underline{\iota} + \gamma V_t)} dZ_t.$$

Integrating and then taking expectations, we have $\mathbb{E}[V_t] \leq V_0$ for all $t \geq 0$. Applying Itô's formula to V_t^2 gives

$$dV_t^2 = \left[-2\lambda V_t^2 + 2\gamma V_t + 2(\rho - \underline{\iota}) \right] dt + 2V_t \sqrt{2(\rho - \underline{\iota} + \gamma V_t)} dZ_t.$$

Again integrating and taking expectations yields

$$\frac{d}{dt} \mathbb{E}[V_t^2] \leq -2\lambda \mathbb{E}[V_t^2] + 2\gamma \mathbb{E}[V_t] + 2(\rho - \underline{\iota}) \leq -2\lambda \mathbb{E}[V_t^2] + 2\gamma V_0 + 2(\rho - \underline{\iota})$$

where the second inequality follows from $\mathbb{E}[V_t] \leq V_0$. Thus, setting $y(t) := \mathbb{E}[V_t^2]$, we have $y'(t) \leq -2\lambda y(t) + B$, with $B := 2\gamma V_0 + 2(\rho - \underline{\iota}) < \infty$. Since $\lambda > 0$, Gronwall's inequality gives

$$y(t) \leq e^{-2\lambda t} y(0) + \frac{B}{2\lambda} (1 - e^{-2\lambda t}),$$

and therefore $\sup_{t \geq 0} \mathbb{E}[V_t^2] < \infty$. Since $V_t = -x_t$, this implies $\sup_{t \geq 0} \mathbb{E}[x_t^2] < \infty$, hence $\sup_{t \geq 0} \mathbb{E}|x_t| < \infty$. Moreover, because $\pi_t = \beta x_t$, we also have $\sup_{t \geq 0} \mathbb{E}[\pi_t^2] < \infty$. Finally, since $x_t \leq 0$, we have $\sup_{t \geq 0} \mathbb{E}[e^{(1+\varphi)x_t}] \leq 1$. Thus, Condition 1 holds. $\square$

A.5 Proofs for Section 4

Proof of Theorem 4. Consider fiscal rule (21) with $\alpha = \underline{\alpha} \leq 0$. First, we derive the dynamics of the real debt-to-GDP ratio $b_t := \frac{B_t}{P_t Y_t}$. By Itô's formula,

$$db_t = (\rho b_t - s_t)dt - b_t (\sigma_{x,t} dZ_t + \varsigma_{x,t} d\mathcal{Z}_t) \quad (\text{A.17})$$

where we have used the flow government budget constraint (20), the price level dynamics $dP_t/P_t = \pi_t dt$, and the dynamics of x_t in (22). Second, we purge surpluses of their endogenous component. So let us examine the dynamics of $e^{-(\rho-\alpha)t} b_t$, which are

$$d\left(e^{-(\rho-\alpha)t} b_t\right) = e^{-(\rho-\alpha)t} \left[-\hat{s}_t dt - b_t (\sigma_{x,t} dZ_t + \varsigma_{x,t} d\mathcal{Z}_t) \right]$$

Then,

$$\mathbb{E}_0[e^{-(\rho-\alpha)T}b_T] = b_0 - \mathbb{E}_0\left[\int_0^T e^{-(\rho-\alpha)t}\hat{s}_t dt\right] \quad (\text{A.18})$$

Recall that the transversality condition (TVC) can be equivalently stated in terms of debt-to-GDP as $\lim_{T \rightarrow \infty} \mathbb{E}_0[e^{-\rho T}b_T] = 0$. Using this and $\alpha \leq 0$, we then have that $\lim_{T \rightarrow \infty} \mathbb{E}_0[e^{-(\rho-\alpha)T}b_T] \rightarrow 0$. Thus, taking $T \rightarrow \infty$ in (A.18) yields $b_0 - \mathbb{E}_0[\int_0^\infty e^{-(\rho-\alpha)t}\hat{s}_t dt]$. Since an identical argument would hold for any future time after $t = 0$, we thus obtain the following valuation-like equation:

$$b_t = \mathbb{E}_t\left[\int_t^\infty e^{-(\rho-\alpha)(u-t)}\hat{s}(\Omega_u)du\right]$$

The right-hand-side is a present-value of the exogenous Markovian component of surpluses. This implies that b_t is purely determined by Ω_t , i.e., there exists a deterministic function $\hat{b}(\cdot)$ such that $b_t = \hat{b}(\Omega_t)$. In that case, b_t has no loading on the sunspot shock dZ_t , which implies by (26) that $b_t\sigma_{x,t} = 0$. Thus, $\sigma_{x,t} = 0$ for almost all t (except at the times when $b_t = 0$, which are zero Lebesgue measure almost-surely). For the statement about (GD) holding for every $t > 0$, given it holds at $t = 0$, simply note that the dynamics of db_t in (A.17) are derived without reference to the valuation equation (GD). $\square$

Remark A.1 (Non-Markovian surpluses). *From the proof of Theorem 4, it is clear that the same arguments hold even in the more general non-Markovian case where $(\hat{s}_t)_{t \geq 0}$ is independent of $(Z_t)_{t \geq 0}$, because in that case db_t still must not have any loading on dZ_t .*

Proof of Theorem 5. Integrating the dynamics of b^* from (27) and taking expectations, we obtain

$$e^{-\rho T}\mathbb{E}_0[b_T^*] = e^{-\bar{\alpha}T}\left(b_0^* - \mathbb{E}_0\left[\int_0^T e^{-(\rho-\bar{\alpha})t}\hat{s}_t dt\right]\right)$$

Given that $\hat{s}_t$ is exogenous and bounded, we have that the right-hand-side converges to zero as $T \rightarrow \infty$, provided $\bar{\alpha} > 0$. Therefore, $\lim_{T \rightarrow \infty} e^{-\rho T}\mathbb{E}_0[b_T^*] = 0$. At this point, we can already prove the final statement of the theorem: if $b = b^*$, then we have just proved that the transversality condition automatically holds.

Next, we may apply Itô's formula to $e^{-\rho t}(b_t - b_t^*)$ to obtain

$$b_0 - b_0^* = e^{-\rho T}\mathbb{E}_0[b_T] - e^{-\rho T}\mathbb{E}_0[b_T^*] \xrightarrow{T \rightarrow \infty} 0,$$

where the limit uses $\lim_{T \rightarrow \infty} e^{-\rho T}\mathbb{E}_0[b_T^*] = 0$, proved above, along with the transversality

condition (TVC), which says $\lim_{T \rightarrow \infty} e^{-\rho T} \mathbb{E}_0[b_T] = 0$. This proves that $b_0 = b_0^*$. The same holds for any arbitrary initial date, and so $b_t = b_t^*$.

Using this result and comparing the dynamics db_t with db_t^* in (26)-(27), we have that $\sigma_{x,t} = 0$ and $\varsigma_{x,t} = 0$. $\square$

Proof of Proposition 4. From Theorem 5, we have that $\sigma_{x,t} = 0$ and $\varsigma_{x,t} = 0$. Furthermore, we know that the debt valuation equation is redundant after $b_t = b_t^*$ is chosen. Thus, under fiscal policy (28), the necessary and sufficient conditions for equilibrium are

$$\begin{aligned} dx_t &= [\phi_x x_t + (\phi_\pi - 1)\pi_t] dt \\ d\pi_t &= [\rho\pi_t - \kappa x_t] dt + \sigma_{\pi,t} dZ_t + \varsigma_{\pi,t} d\mathcal{Z}_t \end{aligned}$$

plus the non-explosion condition. Clearly, one solution is $(x_t, \pi_t) = (0, 0)$ forever. To diagnose the solution set, notice that the drifts are linear in (x_t, π_t) . Thus, we may follow the standard practice by computing the eigenvalue decomposition of the transition matrix (i.e., the infinitesimal generator) to determine whether the dynamics are stable or unstable under the assumed monetary policy.

This is done in the proof of Proposition H.1 (set $\sigma = 0$ to eliminate the monetary policy shocks that are also considered in that proof). Case 1 of Proposition H.1 covers active money, while case 2 covers passive money. Under active money, the result is that if $(x_t, \pi_t) \neq (0, 0)$, then $\mathbb{E}_0[x_T]$ and $\mathbb{E}_0[\pi_T]$ explode exponentially as $T \rightarrow \infty$, in violation of Condition 1. Under passive money, the result is that $\pi_t \propto x_t$. Since x_t evolves deterministically, this implies that π_t also does. Hence, the only remaining indeterminacy is x_0 . $\square$

Proof of Proposition 5. Assume, leading to contradiction, a non-fundamental equilibrium with $\sigma_{x,t} \neq 0$ and/or $\varsigma_{x,t} \neq 0$. As stated, assume a full-support equilibrium, so that $\mathbb{P}\{\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mathbf{1}_{\{x_t \leq x\}} dt > 0\} = 1$ for any x .²²

Now, let $\underline{x} < 0$ and $\bar{b} > 0$ be any fixed constants. Write the surplus rule in an equivalent way to (29): $s_t = \hat{s}_t + \alpha_t b_t + \alpha_t^* b_t^*$, where

$$\alpha_t := \begin{cases} \bar{\alpha}, & \text{if } x_t > \underline{x} \text{ and } b_t < \bar{b}, \\ 0, & \text{otherwise,} \end{cases} \quad \text{and} \quad \alpha_t^* := \begin{cases} 0, & \text{if } x_t > \underline{x} \text{ and } b_t < \bar{b}, \\ \bar{\alpha}, & \text{otherwise.} \end{cases}$$

²²By showing that x_t is unbounded below, Lemma A.3 shows that this assumption necessarily holds for any equilibrium in a large class that also possesses a non-degenerate stationary distribution for $(x_t)_{t \geq 0}$.

By the fact that $\{x_t \leq \underline{x}\} \subset \{\alpha_t^* = \bar{\alpha}\}$, and the full-support assumption, we have

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mathbf{1}_{\{\alpha_t^* = \bar{\alpha}\}} dt \geq \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mathbf{1}_{\{x_t \leq \underline{x}\}} dt > 0 \quad \text{a.s.} \quad (\text{A.19})$$

We thus have

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \alpha_t^* dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \bar{\alpha} \mathbf{1}_{\{\alpha_t^* = \bar{\alpha}\}} dt > 0 \quad \text{a.s.,}$$

where we used $\bar{\alpha} > 0$ and (A.19) to obtain the inequality. This implies that

$$\int_0^\infty \alpha_t^* dt = +\infty \quad \text{a.s.} \quad (\text{A.20})$$

Apply Itô's formula to $e^{-\int_0^t (\rho - \alpha_u^*) du} b_t^*$, using the surplus rule, to get

$$d\left(e^{-\int_0^t (\rho - \alpha_u^*) du} b_t^*\right) = -e^{-\int_0^t (\rho - \alpha_u^*) du} \left(\hat{s}_t + \alpha_t b_t\right) dt$$

Integrating, rearranging, and taking expectations,

$$\mathbb{E}_0[e^{-\rho T} b_T^*] = \mathbb{E}_0\left[e^{-\int_0^T \alpha_t^* dt} \left(b_0^* - \int_0^T e^{-\rho t} e^{\int_0^t \alpha_u^* du} (\hat{s}_t + \alpha_t b_t) dt\right)\right] \quad (\text{A.21})$$

By $\alpha_t^* \geq 0$, we have $e^{-\int_0^T \alpha_t^* dt} \leq 1$. Also, recall that $\hat{s}_t$ is bounded. Finally, notice that the surplus rule implies $0 \leq \alpha_t b_t \leq \bar{\alpha} \bar{b}$ whenever $\alpha_t = \bar{\alpha}$ and $\alpha_t b_t = 0$ otherwise.

All these terms being bounded allows us to use the dominated convergence theorem to take $T \rightarrow \infty$ inside the expectation on the right-hand-side of (A.21). Doing so, and then using (A.20) to take the point-wise limit, we get

$$\begin{aligned} \lim_{T \rightarrow \infty} \mathbb{E}_0\left[e^{-\int_0^T \alpha_t^* dt} \left(b_0^* - \int_0^T e^{-\rho t} e^{\int_0^t \alpha_u^* du} (\hat{s}_t + \alpha_t b_t) dt\right)\right] \\ = \mathbb{E}_0\left[e^{-\int_0^\infty \alpha_t^* dt} \left(b_0^* - \int_0^\infty e^{-\rho t} e^{\int_0^t \alpha_u^* du} (\hat{s}_t + \alpha_t b_t) dt\right)\right] = 0. \end{aligned}$$

Therefore, the left-hand-side of (A.21) must also vanish as $T \rightarrow \infty$.

We use these facts to conclude. As in the proof of Theorem 5, we may apply Itô's formula to $e^{-\rho t} (b_t - b_t^*)$ in order to obtain

$$b_0 - b_0^* = e^{-\rho T} \mathbb{E}_0[b_T] - e^{-\rho T} \mathbb{E}_0[b_T^*] \xrightarrow{T \rightarrow \infty} 0,$$

where the limit uses $\lim_{T \rightarrow \infty} e^{-\rho T} \mathbb{E}_0[b_T^*] = 0$, proved above, along with the transversality

condition (TVC), which says $\lim_{T \rightarrow \infty} e^{-\rho T} \mathbb{E}_0[b_T] = 0$. This proves that $b_0 = b_0^*$. The same holds for any arbitrary initial date, and so $b_t = b_t^*$ for all t .

Using this result and comparing the dynamics db_t with db_t^* in (26)-(27), we have that $\sigma_{x,t} = 0$ and $\varsigma_{x,t} = 0$. $\square$

Lemma A.3. *In the rigid-price limit ($\kappa \rightarrow 0$), assume that $\Phi'(x) > 0$ for all x . Then, any sunspot equilibrium has x_t unbounded below. With partially-flexible prices ($\kappa > 0$), assume that $\partial_x \Phi > 0$ and $\partial_\pi \Phi > 1$ for all (x, π) . Then, any sunspot equilibrium in which $\pi_t = \pi(x_t)$ and $\sigma_{x,t} = \sigma_x(x_t)$ has x_t unbounded below.*

Proof of Lemma A.3. We prove separately the claims for rigid and flexible prices.

Proof for rigid-price limit ($\kappa \rightarrow 0$). The paper already proves that $x_t \leq 0$ is required, given $\Phi' > 0$, so we restrict attention to that case. Suppose that $x_t > \underline{x}$ with probability 1, where $0 > \underline{x} > -\infty$. We split the analysis into two possibilities: $\lim_{x \rightarrow \underline{x}} \sigma_x^2$ is finite or $\lim_{x \rightarrow \underline{x}} \sigma_x^2 = +\infty$.

If $\lim_{x \rightarrow \underline{x}} \sigma_x^2 < \infty$, analyze $y_t := e^{x_t}$ whose dynamics are

$$dy_t = \underbrace{y_t [\Phi(x_t) + \sigma_{x,t}^2]}_{:= \mu_{y,t}} dt + \underbrace{y_t \sigma_{x,t}}_{:= \sigma_{y,t}} dZ_t$$

The diffusion of y_t near its lower bound $\underline{y} := e^{\underline{x}}$ is $\sigma_{y,t}^2 \rightarrow \underline{y}^2 \sigma_{x,t}^2$, which is finite, while the drift is

$$\mu_{y,t} \rightarrow \underbrace{\underline{y} \Phi(\underline{x})}_{< 0} + \underbrace{\underline{y} \sigma_{x,t}^2}_{\text{finite}}$$

If the diffusion is zero at $\underline{y}$ then the drift will be negative, implying y_t will cross $\underline{y}$. If the diffusion is positive at $\underline{y}$, then because the drift is finite, once again y_t will cross $\underline{y}$. This contradicts $x_t > \underline{x}$.

If $\lim_{x \rightarrow \underline{x}} \sigma_x^2 = +\infty$, we analyze instead $u_t := \log(e^{x_t} - e^{\underline{x}})$, whose dynamics satisfy

$$du_t = \frac{e^{x_t}}{e^{x_t} - e^{\underline{x}}} \left[\Phi(x_t) + \left(1 - \frac{e^{x_t}}{e^{x_t} - e^{\underline{x}}} \right) \sigma_{x,t}^2 \right] dt + \frac{e^{x_t}}{e^{x_t} - e^{\underline{x}}} \sigma_{x,t} dZ_t$$

Notice that $\frac{e^x}{e^x - e^{\underline{x}}} \rightarrow \infty$ as $x \rightarrow \underline{x}$. Thus, as $x_t \rightarrow \underline{x}$ and hence $u_t \rightarrow -\infty$, the dynamics of u_t feature an infinite volatility and negative infinite drift, implying u_t will reach $-\infty$ and thus x_t will hit $\underline{x}$. This also contradicts $x_t > \underline{x}$.

Proof for partially-flexible prices ($\kappa > 0$). Consider partially-flexible prices ($\kappa > 0$) and a Markovian equilibrium: $\pi_t = \pi(x_t)$ and $\sigma_{x,t}^2 = \sigma_x^2(x_t)$. Applying Itô's formula to π and using the Phillips curve (linear PC) results in

$$\rho\pi - \kappa x = [\Phi(x, \pi) - \pi + \frac{1}{2}\sigma_x^2]\pi' + \frac{1}{2}\sigma_x^2\pi''. \quad (\text{A.22})$$

Let us try to construct an equilibrium for $x \in (\underline{x}, \bar{x})$, assuming only that $\underline{x} > -\infty$. We will prove that no function $\pi(\cdot)$ can satisfy (A.22) and coincide with dynamics for x_t that remain stable in this bounded region.

First, the dynamics need to be stable, meaning we need x_t to remain inside $(\underline{x}, \bar{x})$ forever. For this to hold, the diffusion $\sigma_x^2(x)$ must be some positive function that vanishes sufficiently rapidly as $x \rightarrow \underline{x}$ and $x \rightarrow \bar{x}$. Furthermore, the drift $\mu_x(x)$ must be pointing inward at these boundaries, i.e.,

$$\lim_{x \rightarrow \underline{x}} \Phi(x, \pi(x)) - \pi(x) \geq 0 \quad (\text{A.23})$$

$$\lim_{x \rightarrow \bar{x}} \Phi(x, \pi(x)) - \pi(x) \leq 0 \quad (\text{A.24})$$

Given assumptions that $\partial_x \Phi > 0$ and $\partial_\pi \Phi > 1$, this requires that

$$\pi(\underline{x}) > \pi(\bar{x}). \quad (\text{A.25})$$

From these requirements, we can deduce the slope of π at the boundaries. Taking the limit $x \rightarrow \underline{x}$ or $x \rightarrow \bar{x}$ in (A.22), we have

$$\pi'(\underline{x}) = \lim_{x \rightarrow \underline{x}} \frac{\rho\pi(x) - \kappa x}{\Phi(x, \pi(x)) - \pi(x)} \quad (\text{A.26})$$

$$\pi'(\bar{x}) = \lim_{x \rightarrow \bar{x}} \frac{\rho\pi(x) - \kappa x}{\Phi(x, \pi(x)) - \pi(x)} \quad (\text{A.27})$$

We can prove that π is non-monotonic. Indeed, if not, then π must be monotonically decreasing by (A.25). But using (A.25) and (A.23)-(A.24) in (A.26)-(A.27), we have that if $\pi'(\underline{x}) < 0$, then $\pi'(\bar{x}) > 0$, a contradiction.

Given the non-monotonicity, there exists some intermediate point $x^* \in (\underline{x}, \bar{x})$ such that $\pi'(x^*) = 0$. At any such x^* , we have

$$\rho\pi(x^*) - \kappa x^* = \frac{1}{2}\sigma_x^2(x^*)\pi''(x^*) \quad (\text{A.28})$$

There are three cases to consider: (i) $\pi'(\underline{x}) < 0 < \pi'(\bar{x})$; (ii) $\pi'(\underline{x}) > 0 > \pi'(\bar{x})$; (iii)

$\pi'(\underline{x}), \pi'(\bar{x}) > 0$. (Recall that $\pi'(\underline{x}), \pi'(\bar{x}) < 0$ has just been ruled out above.) The cases where any of the boundary slopes equal zero are handled easily by extension.

- Case (i): $\pi'(\underline{x}) < 0 < \pi'(\bar{x})$. In this case, the function π is decreasing initially before increasing, so the first such point x^* must satisfy $\pi''(x^*) > 0$, and so $\rho\pi(x^*) - \kappa x^* > 0$ by (A.28). At the same time, $\pi'(\underline{x}) < 0$ implies $\rho\pi(\underline{x}) - \kappa \underline{x} < 0$ by (A.23) and (A.26). But this is a contradiction, since π decreasing between $\underline{x}$ and x^* implies

$$\rho\pi(\underline{x}) - \kappa \underline{x} > \rho\pi(x^*) - \kappa x^*.$$

- Case (ii) $\pi'(\underline{x}) > 0 > \pi'(\bar{x})$. This case is symmetric to case (i), but the argument starts from the upper boundary $\bar{x}$ instead. Indeed, the last such point x^* must satisfy $\pi''(x^*) < 0$, and so $\rho\pi(x^*) - \kappa x^* < 0$ by (A.28). At the same time, $\pi'(\bar{x}) < 0$ implies $\rho\pi(\bar{x}) - \kappa \bar{x} > 0$ by (A.24) and (A.27). But this is a contradiction, since π decreasing between x^* and $\bar{x}$ implies

$$\rho\pi(x^*) - \kappa x^* > \rho\pi(\bar{x}) - \kappa \bar{x}$$

- Case (iii): $\pi'(\underline{x}), \pi'(\bar{x}) > 0$. The fact that π is non-monotonic and has positive slopes at the boundaries implies that there exist two global extrema points $x^* < x^{**}$ such that $\pi'(x^*) = \pi'(x^{**}) = 0$ and $\pi''(x^*) < 0 < \pi''(x^{**})$ —i.e., x^* is the global maximum, while x^{**} is the global minimum. By (A.28), $\pi''(x^*) < 0 < \pi''(x^{**})$ implies $\rho\pi(x^*) - \kappa x^* < 0 < \rho\pi(x^{**}) - \kappa x^{**}$. However, the fact that $\pi(x^*) > \pi(x^{**})$ implies the contradiction

$$\rho\pi(x^*) - \kappa x^* > \rho\pi(x^{**}) - \kappa x^{**}$$

Having ruled out all cases, and since $\underline{x}$ and $\bar{x}$ were arbitrary, there cannot exist any $\pi(\cdot)$ corresponding to a bounded equilibrium. Thus, $\underline{x} = -\infty$ must hold. $\square$

B Monetary policy shocks (rather than sunspots)

Our baseline model features only sunspot uncertainty, for theoretical clarity. However, this is not necessary. In environments in which sunspot equilibria arise, there is often also indeterminacy in the response of the economy to fundamental shocks, since agents can use the fundamental shock as a source of coordination. Here, we demonstrate this by adding fundamental uncertainty to the monetary policy rule.

The model is the same, except now the interest rate rule is given by

$$\iota_t = \bar{\iota} + \Phi(x_t, \pi_t) + \sigma \mathcal{E}_t, \quad (\text{B.1})$$

where $\mathcal{E}_t$ is an exogenous “monetary shock” process that is driven by Z_t and is mean-reverting to zero. The baseline model can be thought of as the $\sigma \rightarrow 0$ limit of this rule.

B.1 Benchmark: MSV Equilibrium

We start by expositing the benchmark “minimum state variable” (MSV) solution to the model. The sole state variable is $\mathcal{E}_t$, the monetary policy shock. An MSV solution assumes all other endogenous variables are purely a function of $\mathcal{E}_t$. Let us write the dynamics of $\mathcal{E}_t$ in the general form

$$d\mathcal{E}_t = \mu_e(\mathcal{E}_t)dt + \sigma_e(\mathcal{E}_t)dZ_t \quad (\text{B.2})$$

for some drift and diffusion coefficients $\mu_e(\mathcal{E})$ and $\sigma_e(\mathcal{E})$.

Assume $x_t = x(\mathcal{E}_t)$ and $\pi_t = \pi(\mathcal{E}_t)$ for some functions $x(\cdot)$ and $\pi(\cdot)$. By Itô’s formula, we may calculate $\mu_{x,t} = \mu_e(\mathcal{E}_t)x'(\mathcal{E}_t) + \frac{1}{2}\sigma_e(\mathcal{E}_t)^2x''(\mathcal{E}_t)$ and $\sigma_{x,t} = \sigma_e(\mathcal{E}_t)x'(\mathcal{E}_t)$. Plugging these into (IS), along with the monetary policy rule (MP), and dropping time subscripts, we have

$$\mu_e(\mathcal{E})x'(\mathcal{E}) + \frac{\sigma_e(\mathcal{E})^2x''(\mathcal{E})}{2} = \bar{\iota} + \Phi(x(\mathcal{E}), \pi(\mathcal{E})) + \sigma\mathcal{E} - \pi(\mathcal{E}) - \rho + \frac{(\sigma_e(\mathcal{E})x'(\mathcal{E}))^2}{2} \quad (\text{B.3})$$

Similarly, substituting $\mu_{\pi,t} = \mu_e(\mathcal{E}_t)\pi'(\mathcal{E}_t) + \frac{1}{2}\sigma_e(\mathcal{E}_t)^2\pi''(\mathcal{E}_t)$ into (PC), we have

$$\mu_e(\mathcal{E})\pi'(\mathcal{E}) + \frac{\sigma_e(\mathcal{E})^2\pi''(\mathcal{E})}{2} = \rho\pi(\mathcal{E}) - \kappa\left(\frac{e^{(1+\varphi)x(\mathcal{E})} - 1}{1 + \varphi}\right) \quad (\text{B.4})$$

Equations (B.3)-(B.4) is a system of second-order ODEs that characterizes the MSV solution. We assume an MSV solution exists. As will become clear, all the equilibria unveiled in this paper constitute deviations from any MSV solution, since x and π will not be solely functions of $\mathcal{E}$.

Example B.1 (Linear monetary policy). *To illustrate an example analytically, we adopt the following specification. The monetary policy shock is an Ornstein-Uhlenbeck process, i.e., $\mu_e(\mathcal{E}) = -\zeta\mathcal{E}$ and $\sigma_e(\mathcal{E}) = 1$. This is the continuous-time equivalent of an AR(1) process. In addition,*

the economy features a linearized Phillips curve (*linear PC*) and a linear monetary policy rule (*linear MP*). Substituting these linear functional forms into (B.3)-(B.4), we obtain

$$-\zeta \mathcal{E} x'(\mathcal{E}) + \frac{1}{2} x''(\mathcal{E}) = \bar{\iota} + \phi_x x(\mathcal{E}) + (\phi_\pi - 1) \pi(\mathcal{E}) + \sigma \mathcal{E} - \rho + \frac{1}{2} x'(\mathcal{E})^2 \quad (\text{B.5})$$

$$-\zeta \mathcal{E} \pi'(\mathcal{E}) + \frac{1}{2} \pi''(\mathcal{E}) = \rho \pi(\mathcal{E}) - \kappa x(\mathcal{E}) \quad (\text{B.6})$$

Let us guess a linear equilibrium $x(\mathcal{E}) = a_x + b_x \mathcal{E}$ and $\pi(\mathcal{E}) = a_\pi + b_\pi \mathcal{E}$. Plugging these into (B.5)-(B.6), we verify the form of the solution as long as

$$b_x = -\frac{(\zeta + \rho)\sigma}{(\zeta + \rho)(\zeta + \phi_x) + \kappa(\phi_\pi - 1)} \quad \text{and} \quad b_\pi = -\frac{\kappa\sigma}{(\zeta + \rho)(\zeta + \phi_x) + \kappa(\phi_\pi - 1)}$$

and

$$a_x = -\rho \frac{\bar{\iota} - \rho + \frac{1}{2} b_x^2}{\phi_x \rho + (\phi_\pi - 1) \kappa} \quad \text{and} \quad a_\pi = -\kappa \frac{\bar{\iota} - \rho + \frac{1}{2} b_x^2}{\phi_x \rho + (\phi_\pi - 1) \kappa}$$

So long as $(\zeta + \rho)(\zeta + \phi_x) + \kappa(\phi_\pi - 1) > 0$, the MSV solution has a particularly simple and intuitive form where x and π respond negatively to monetary tightening. If, furthermore, monetary policy picks the “natural” target rate of $\bar{\iota} = \rho - \frac{1}{2} b_x^2$, then $a_x = a_\pi = 0$, so that the time-series average values of x_t and π_t will be zero.

Example B.2 (No monetary policy uncertainty). Next, consider the $\sigma \rightarrow 0$ limiting case where monetary shocks vanish, so that $\mathcal{E}_t$ is an extrinsic process. Consequently, the unique MSV solution has $x_t = x^*$ and $\pi_t = \pi^*$ both equal to some constants, independent of $\mathcal{E}$. Plugging this into (B.3)-(B.4), we obtain the system of equations

$$\begin{aligned} 0 &= \bar{\iota} - \rho + \Phi(x^*, \pi^*) - \pi^* \\ 0 &= \rho \pi^* - \kappa \left(\frac{e^{(1+\varphi)x^*} - 1}{1 + \varphi} \right) \end{aligned}$$

If the target rate is set appropriately at $\bar{\iota} = \rho$, then the MSV solution becomes $x^* = \pi^* = 0$. Thus, this $\sigma \rightarrow 0$ limit indeed corresponds to our benchmark model.

B.2 Indeterminacy in Uncertainty

In the context of this model, we prove a general result about indeterminacy in the economy’s sensitivity to monetary shocks. This generalization of Theorem 1 shows that our results are, at their core, about an indeterminacy in uncertainty within NK models, not

about sunspot volatility *per se*. It could, instead, simply be the case that agents coordinate on a responsiveness to fundamental shocks in a manner which differs from the MSV solution. This may be important if one wonders about the practical relevance of our indeterminacy, because it means that no extrinsic sources of uncertainty are needed to induce coordination in NK models.

Theorem B.1. *Suppose prices are rigid ($\kappa \rightarrow 0$) and monetary shocks are present ($\sigma > 0$). Consider any Taylor rule (MP) with $\bar{\tau} = \rho$ and $\Phi(x)$ increasing in x . Let $x^*(\mathcal{E})$ denote the MSV solution for this economy, and let $\sigma_{x^*}(\mathcal{E})$ denote its volatility. Then,*

- (i) *If $\mathcal{E}$ is bounded, there exist a continuum of non-explosive equilibria indexed by $x_0 < x^*(\mathcal{E}_0)$ with a volatility function $\sigma_x(\mathcal{E}, x)$ possessing “excess volatility” in the sense that $\sigma_x^2(\mathcal{E}, x) \geq \sigma_{x^*}^2(\mathcal{E})$. The volatility can be any mapping $\sigma_x : \mathbb{R}^2 \mapsto \mathbb{R}$ that is finite on $\{(\mathcal{E}, x) : -\infty < x \leq x^*(\mathcal{E})\}$ and satisfies suitable boundary conditions as $x \rightarrow -\infty$ and $x \rightarrow x^*(\mathcal{E})$.*
- (ii) *If $\inf_x \Phi'(x) > 0$, then all non-explosive equilibria possessing excess volatility have $x_t \leq x^*(\mathcal{E}_t)$ forever, and hence excess volatility is recessionary.*

Proof of Theorem B.1. Let $x_t^* = x^*(\mathcal{E}_t)$ denote the MSV solution. Denote demand volatility in the MSV solution by $\sigma_{x^*}(\mathcal{E}_t) = \frac{d}{d\mathcal{E}} x^*(\mathcal{E}_t)$. Note that both x^* and σ_{x^*} must be bounded, since the MSV solution requires the second derivative of x^* to exist, and since the monetary state $\mathcal{E}$ is bounded.

Proof of statement (i). First, construct the volatility function $\sigma_x(\mathcal{E}, x)$. Let $k > 2$ and $\delta > 0$ be arbitrary. Define $\bar{x}(\mathcal{E}) := \min[0, x^*(\mathcal{E}) - \delta]$. Let volatility be given by

$$\sigma_x^2(\mathcal{E}, x) := \sigma_{x^*}^2(\mathcal{E}) + \begin{cases} k[g(x - x^*(\mathcal{E})) - \Phi(x) + \Phi(x^*(\mathcal{E}))], & \text{if } x < \bar{x}(\mathcal{E}), \\ 0, & \text{if } x \geq \bar{x}(\mathcal{E}), \end{cases} \quad (\text{B.7})$$

where $g(\cdot)$ is any positive, finite-valued function satisfying $\lim_{x \rightarrow -\infty} g(x) = +\infty$. Since we have only defined the squared volatility, we also pick the construction such that σ_x and σ_{x^*} always have the same sign. Since $\Phi(\cdot)$ is increasing and continuous, we have that $\sigma_x^2(\mathcal{E}, x) \geq \sigma_{x^*}^2(\mathcal{E})$, as required for there to be “excess volatility.” For the same reason, notice that $\sigma_x^2(\mathcal{E}, x) > \sigma_{x^*}^2(\mathcal{E})$ for all $x < \bar{x}(\mathcal{E})$.

Next, consider the change of variables:

$$V_t = \exp[x_t - x_t^*].$$

Given $\bar{t} = \rho$, the dynamics of x_t and $x_t^* = x^*(\mathcal{E}_t)$ are

$$\begin{aligned} dx_t &= \left[\Phi(x_t) + \sigma \mathcal{E}_t + \frac{1}{2} \sigma_x^2(\mathcal{E}_t, x_t) \right] dt + \sigma_x(\mathcal{E}_t, x_t) dZ_t \\ dx_t^* &= \left[\Phi(x_t^*) + \sigma \mathcal{E}_t + \frac{1}{2} \sigma_{x^*}^2(\mathcal{E}_t) \right] dt + \sigma_{x^*}(\mathcal{E}_t) dZ_t \end{aligned}$$

Therefore, the dynamics of V_t are

$$\begin{aligned} dV_t &= \mu_v(\mathcal{E}_t, V_t) dt + \sigma_v(\mathcal{E}_t, V_t) dZ_t, \quad \text{where} \\ \mu_v(\mathcal{E}, v) &:= v \left[\Phi(\chi(\mathcal{E}, v)) - \Phi(x^*(\mathcal{E})) + \frac{\sigma_x^2(\mathcal{E}, \chi(\mathcal{E}, v)) - \sigma_{x^*}^2(\mathcal{E})}{2} + \frac{(\sigma_x(\mathcal{E}, \chi(\mathcal{E}, v)) - \sigma_{x^*}(\mathcal{E}))^2}{2} \right] \\ \sigma_v(\mathcal{E}, v) &:= v \left(\sigma_x(\mathcal{E}, \chi(\mathcal{E}, v)) - \sigma_{x^*}(\mathcal{E}) \right) \end{aligned}$$

where $\chi(\mathcal{E}, v) := x^*(\mathcal{E}) + \log(v)$ inverts the change-of-variables and gives x in terms of v . The plan is to show that V_t satisfies the hypotheses of Lemma A.1. Doing so will show that $\liminf_t \mathbb{E}[\log(V_t)] > -\infty$. Since $x^*(\mathcal{E}_t)$ is bounded and the construction keeps $x_t \leq x^*(\mathcal{E}_t)$, this implies $\limsup_{t \rightarrow \infty} \mathbb{E}|x_t| < \infty$. Moreover, the same upper bound on x_t implies $\limsup_{t \rightarrow \infty} \mathbb{E}[e^{(1+\varphi)x_t}] < \infty$. Hence, this will show that Condition 1 holds.

We obviously have that $V_t \geq 0$ forever. Defining $\bar{v}(\mathcal{E}) := e^{-\max[x^*(\mathcal{E}), \delta]}$, let us show that $V_t \leq \bar{v}(\mathcal{E}_t)$ forever if V_0 starts below $\bar{v}(\mathcal{E}_0)$. For x_t such that $\bar{x}(\mathcal{E}_t) \leq x_t < x^*(\mathcal{E}_t)$, or equivalently $\bar{v}(\mathcal{E}_t) \leq V_t < 1$, we have that $\sigma_v = 0$, and so $dV_t = [\Phi(x_t) - \Phi(x_t^*)] dt < 0$, by the fact that Φ is increasing and that $x < x^*(\mathcal{E})$. Therefore, V_t enters the region below $\bar{v}(\mathcal{E}_t)$ in finite time when starting from any point $V_0 < 1$. And if $V_t \leq \bar{v}(\mathcal{E}_t)$, it can never exit this region. So V_t satisfies hypothesis (H1) of Lemma A.1.

It is easy to see that μ_v and σ_v^2 are continuous on $\{v < \bar{v}(\mathcal{E})\}$ and are only potentially infinite when $v = 0$. So V_t satisfies hypothesis (H2) of Lemma A.1.

Next, we showed earlier that $\sigma_x^2(\mathcal{E}, x) > \sigma_{x^*}^2(\mathcal{E})$ for all $x < \bar{x}(\mathcal{E})$, and so $\sigma_v^2(\mathcal{E}, v) > 0$ for all $v < \bar{v}(\mathcal{E})$. This verifies that V_t satisfies hypothesis (H3) of Lemma A.1.

Next, analyze the diffusion σ_v^2 asymptotically as $v \rightarrow 0$. We have that, for every $\mathcal{E}$,

$$\lim_{x \rightarrow -\infty} \sigma_x(\mathcal{E}, x)^2 = +\infty.$$

Therefore, by the fact that x^* and σ_{x^*} are bounded, we have

$$\lim_{v \rightarrow 0} \left(\frac{\sigma_v(\mathcal{E}, v)}{v} \right)^2 = \lim_{x \rightarrow -\infty} \left(\sigma_x(\mathcal{E}, x) - \sigma_{x^*}(\mathcal{E}) \right)^2 = +\infty > 0.$$

Hence, V_t satisfies hypothesis (H4) of Lemma A.1.

We next do a similar limiting analysis for the drift in order to check the ratio hypothesis (H5). Again, since x^* and σ_{x^*} are bounded, they will not matter in the limit as $v \rightarrow 0$, or equivalently $x \rightarrow -\infty$. Plugging (B.7) into the expression for μ_v , we have that

$$\begin{aligned}\theta &:= \lim_{v \rightarrow 0} \frac{2v\mu_v(\mathcal{E}, v)}{\sigma_v(\mathcal{E}, v)^2} = \lim_{x \rightarrow -\infty} \frac{2[\Phi(x) + \sigma_x^2(\mathcal{E}, x)]}{\sigma_x^2(\mathcal{E}, x)} = \lim_{x \rightarrow -\infty} \frac{2\Phi(x) + 2k[g(x) - \Phi(x)]}{k[g(x) - \Phi(x)]} \\ &= 2 + \frac{2}{k} \lim_{x \rightarrow -\infty} \frac{\Phi(x)}{g(x) - \Phi(x)} \geq 2 - \frac{2}{k} > 1,\end{aligned}$$

where we have used the fact that $\lim_{x \rightarrow -\infty} \frac{\Phi(x)}{g(x) - \Phi(x)} = -1 + \lim_{x \rightarrow -\infty} \frac{g(x)}{g(x) - \Phi(x)} \geq -1$, followed by the assumption $k > 2$. Consequently, hypothesis (H5) of Lemma A.1 holds.

This verifies all the hypotheses of Lemma A.1, proving that the construction is a valid non-explosive equilibrium.

Finally, we prove the claim that any volatility function is valid if it satisfies suitable boundary conditions. Instead of the σ_x^2 function in (B.7), consider any alternative function $\tilde{\sigma}_x^2$, which (a) coincides with σ_x^2 for $x \notin (-K, \bar{x}(\mathcal{E}) - K^{-1})$ for K arbitrarily large; and (b) is finite and exceeds $\sigma_{x^*}^2$ on $x \in (-K, \bar{x}(\mathcal{E}) - K^{-1})$. By inspection, the entire proof above remains valid. This proves statement (i) of the proposition.

Proof of statement (ii). We now prove that all equilibria with “excess volatility” are recessionary. Consider a general rule $\Phi(x)$ satisfying $\phi_x := \inf_x \Phi'(x) > 0$. Let $\sigma_{x,t}^2$ be any volatility process such that $\sigma_{x,t}^2 \geq \sigma_x^*(\mathcal{E}_t)^2$. We will study the “gap process” $\Delta_t := x_t - x_t^*$. Suppose, leading to contradiction that $\Delta_0 > 0$ was part of a non-explosive equilibrium.

The dynamics of Δ_t are

$$d\Delta_t = \left[\Phi(\Delta_t + x_t^*) - \Phi(x_t^*) + \frac{1}{2} \left(\sigma_{x,t}^2 - \sigma_x^*(\mathcal{E}_t)^2 \right) \right] dt + \left(\sigma_{x,t} - \sigma_x^*(\mathcal{E}_t) \right) dZ_t, \quad \Delta_0 > 0.$$

Consider the alternative process $\tilde{\Delta}_t$ which drops the excess variance from the drift:

$$d\tilde{\Delta}_t = \left[\Phi(\tilde{\Delta}_t + x_t^*) - \Phi(x_t^*) \right] dt + \left(\sigma_{x,t} - \sigma_x^*(\mathcal{E}_t) \right) dZ_t, \quad \tilde{\Delta}_0 = \Delta_0 > 0.$$

Define the stopping time

$$\tau_0 := \inf\{t > 0 : \tilde{\Delta}_t = 0\},$$

and put $T_0 := T \wedge \tau_0$. Then,

$$\mathbb{E}_0[e^{-\phi_x T_0} \tilde{\Delta}_{T_0}] = \tilde{\Delta}_0 + \mathbb{E}_0 \left[\int_0^{T_0} e^{-\phi_x t} \left(\Phi(\tilde{\Delta}_t + x_t^*) - \Phi(x_t^*) - \phi_x \tilde{\Delta}_t \right) dt \right] \geq \tilde{\Delta}_0 = \Delta_0,$$

since $\Phi(\tilde{\Delta} + x^*) - \Phi(x^*) \geq \phi_x \tilde{\Delta}$ for all $\tilde{\Delta} \geq 0$ (recall that $\phi_x > 0$ is the minimal slope of the general rule $\Phi(x)$). The left-hand-side can be written $e^{-\phi_x T} \mathbb{E}_0[\tilde{\Delta}_T \mathbf{1}_{\{T < \tau_0\}}]$, since $\tilde{\Delta}_{\tau_0} = 0$. Thus,

$$\mathbb{E}_0[\tilde{\Delta}_T \mathbf{1}_{\{T < \tau_0\}}] \geq e^{\phi_x T} \Delta_0,$$

which by taking $T \rightarrow \infty$ proves that $\limsup_{T \rightarrow \infty} \mathbb{E}_0[\tilde{\Delta}_T] = +\infty$ with positive probability. This implies that $\limsup_{T \rightarrow \infty} \mathbb{E}_0[\Delta_T] = +\infty$, since standard diffusion comparison theorems imply that $\Delta_T \geq \tilde{\Delta}_T$ almost-surely. Finally, this then implies that $\limsup_{T \rightarrow \infty} \mathbb{E}_0[x_T] = +\infty$, in violation of Condition 1, which proves the result. $\square$

C Uncertainty traps and inflation

This appendix collects the results for the case of partially-flexible prices, i.e., finite $\kappa > 0$.

C.1 General verification theorem

We begin with a general verification-type theorem, which establishes sufficient conditions for functions $\pi(\cdot)$ and $v(\cdot)$ to support a non-explosive sunspot equilibrium with $\pi_t = \pi(x_t)$ and $\sigma_{x,t}^2 = v(x_t)$.

Theorem C.1 (General construction with partially-flexible prices). *Consider the linearized Phillips curve (linear PC) with $\kappa > 0$, and the Taylor rule (MP) with $\bar{\tau} = \rho$, where Φ is continuous. For functions (v, f) , define*

$$\mathcal{T}_\Phi[v, f](x) := f'(x) \left[\Phi(x, f(x)) - f(x) + \frac{1}{2}v(x) \right] + \frac{1}{2}v(x)f''(x) - (\rho f(x) - \kappa x).$$

Fix $\bar{x} < \infty$. Suppose there exist functions $v : (-\infty, \bar{x}) \rightarrow (0, \infty)$ and $\pi : (-\infty, \bar{x}] \rightarrow \mathbb{R}$, together with a finite set of pasting points $\mathcal{P} \subset (-\infty, \bar{x})$, such that the following conditions hold.

- (i) *Regularity. The function π is continuous on $(-\infty, \bar{x}]$, C^1 on $(-\infty, \bar{x})$, and C^2 on each connected component of $(-\infty, \bar{x}) \setminus \mathcal{P}$. The function v is strictly positive, piecewise continuous, locally bounded, and locally bounded away from zero on compact subsets of $(-\infty, \bar{x})$. Finite jumps in v at points in $\mathcal{P}$ are allowed.*

(ii) Phillips-curve ODE. For every $x \in (-\infty, \bar{x}) \setminus \mathcal{P}$,

$$\mathcal{T}_\Phi[v, \pi](x) = 0.$$

(iii) Left-tail condition. Let $\mu(x) := \Phi(x, \pi(x)) - \pi(x) + \frac{1}{2}v(x)$. There exists $x_L < \bar{x}$, with $\mathcal{P} \subset (x_L, \bar{x})$, such that the left-tail scale and speed densities

$$s(x) := \exp \left\{ - \int_{x_0}^x \frac{2\mu(z)}{v(z)} dz \right\}, \quad m(x) := \frac{2}{v(x)s(x)}, \quad x_0 < x_L,$$

satisfy

$$\int_{-\infty}^{x_L} s(x) dx = \infty, \quad \int_{-\infty}^{x_L} \left(1 + x^2 + \pi(x)^2\right) m(x) dx < \infty. \quad (\text{LT})$$

(iv) Upper-boundary condition. Let $P := \pi(\bar{x})$ and $A_{\bar{x}} := \Phi(\bar{x}, P) - P$. Assume $A_{\bar{x}} < 0$. Moreover, as $x \uparrow \bar{x}$,

$$v(x) = a(\bar{x} - x)^\alpha + o((\bar{x} - x)^\alpha),$$

for some $a > 0$ and $\alpha \geq 1$, with $a \leq -2A_{\bar{x}}$ if $\alpha = 1$.

Then the processes generated by

$$dx_t = \mu(x_t)dt + \sqrt{v(x_t)} dZ_t, \quad \pi_t = \pi(x_t), \quad \sigma_{x,t}^2 = v(x_t),$$

constitute a non-explosive sunspot equilibrium, which is stationary and ergodic on $(-\infty, \bar{x})$.

Theorem C.1 does not prove existence. Instead, it provides conditions on $\pi(\cdot)$ and $v(\cdot) = \sigma_x^2(\cdot)$ that, if verified, deliver a non-explosive sunspot equilibrium with partially-flexible prices. For instance, we need to make sure that π satisfies the Phillips curve ODE, given a volatility function v . We also need to make sure the equilibrium is non-explosive, which is encoded in the critical left-tail condition (LT). At the upper limit, the upper-boundary condition says that volatility must vanish sufficiently quickly, as in the rigid-price limit. Additionally, the interest rate remains below the natural rate at this upper boundary, so that $\mu_x(\bar{x}) < 0$. These restrictions deliver the boundary behavior needed for a non-degenerate stationary density.

Below, we provide several constructions that satisfy the hypotheses of Theorem C.1. The first results specialize to linear Taylor rules. Then, we consider nonlinear rules.

C.2 Constructions with a linear Taylor rule

For the linear-rule results below, specialize the operator in Theorem C.1 to the Taylor rule (linear MP) and write

$$\mathcal{T}[g, f](x) := f'(x) \left[\phi_x x + (\phi_\pi - 1)f(x) + \frac{1}{2}g(x) \right] + \frac{1}{2}g(x)f''(x) - (\rho f(x) - \kappa x).$$

The results are organized into five lemmas and a corollary. Lemmas C.1–C.3 construct admissible outer regions for inflation $\pi(\cdot)$ and volatility $\sigma_x^2(\cdot)$: a deflationary left tail, an inflationary left tail, and an upper-boundary region. Corollary C.1 illustrates the degrees of freedom described in part (ii) of Theorem 2: it combines these concrete outer-region constructions with any admissible intermediate block to produce a global pair satisfying the conditions of Theorem C.1. Example C.1 presents one possible numerical construction of an admissible intermediate region, using an algorithm that retains several degrees of freedom. The existence claims in part (i) of Theorem 2 are delivered by the final two lemmas, Lemma C.4 and Lemma C.5. Lemma C.4 provides an admissible intermediate region in closed form for the deflationary left tail, yielding a tractable subclass of uncertainty traps and establishing the deflationary case in part (i). Lemma C.5 establishes existence of an admissible intermediate region for the inflationary left tail, completing part (i).

Lemma C.1 (Deflationary left-tail block). *Consider the linear Taylor rule (linear MP) with $\phi_x > 0$, $\phi_\pi > 1$, and $\bar{i} = \rho$. Let*

$$\pi_L(x) := \pi_0 + \pi_1 x, \quad 0 < \pi_1 < \min \left\{ \frac{\rho}{\phi_\pi - 1}, \frac{\kappa}{\rho} \right\},$$

and define v_L by

$$\mathcal{T}[v_L, \pi_L](x) = 0.$$

Then there exists $x_D < 0$ such that, for every $\underline{x} \leq x_D$, the pair (π_L, v_L) is an admissible deflationary left-tail block, in the sense of Theorem C.1, on $(-\infty, \underline{x}]$. In particular, $\pi_L(x) < 0$, $v_L(x) > 0$, and $v'_L(x) < 0$ for all $x \leq \underline{x}$; the Phillips-curve ODE holds on the block; and the left-tail scale and speed conditions (LT) are satisfied.

Lemma C.2 (Inflationary left-tail block). *Consider the linear Taylor rule (linear MP) with $\phi_x > 0$, $\phi_\pi > 1$, and $\bar{i} = \rho$. Let*

$$F(x) := \exp \left(-x - \frac{1}{1-x} \right), \quad \pi_L(x) := A_L F(x), \quad A_L > 0,$$

and define v_L by

$$\mathcal{T}[v_L, \pi_L](x) = 0.$$

Then there exists $x_I < 0$ such that, for every $\underline{x} \leq x_I$, the pair (π_L, v_L) is an admissible inflationary left-tail block, in the sense of Theorem C.1, on $(-\infty, \underline{x}]$. In particular, $\pi_L(x) > 0$, $v_L(x) > 0$, and $v'_L(x) < 0$ for all $x \leq \underline{x}$; the Phillips-curve ODE holds on the block; and the left-tail scale and speed conditions (LT) are satisfied.

Lemmas C.1 and C.2 are alternative ways to construct the left-tail block. These constructions reaffirm that uncertainty traps require conditional output volatility to be sufficiently countercyclical in deep recessions: $v'(x) < 0$ for x sufficiently negative. They also illustrate that uncertainty traps are compatible with either deflationary recessions or stagflationary episodes. In this sense, the sign of inflation is not central to the existence of uncertainty traps.

Lemma C.3 (Upper-boundary block). *Consider the linear Taylor rule (linear MP) with $\phi_x > 0$, $\phi_\pi > 1$, and $\bar{t} = \rho$. Fix $\bar{x} < \infty$ and choose $P \in \mathbb{R}$ such that $A_{\bar{x}} := \phi_x \bar{x} + (\phi_\pi - 1)P < 0$. Choose $a > 0$ and $\alpha \geq 1$, with $a \leq -2A_{\bar{x}}$ if $\alpha = 1$, and define $v_H(x) := a(\bar{x} - x)^\alpha$.*

Then there exists $\varepsilon > 0$ such that, for every $\hat{x} \in (\bar{x} - \varepsilon, \bar{x})$, there is a function π_H for which the pair (π_H, v_H) is an admissible upper-boundary block, in the sense of Theorem C.1, on $[\hat{x}, \bar{x})$. In particular,

$$\pi_H(\bar{x}) = P, \quad \lim_{x \uparrow \bar{x}} \pi'_H(x) = \frac{\rho P - \kappa \bar{x}}{A_{\bar{x}}},$$

the Phillips-curve ODE holds on $(\hat{x}, \bar{x})$, and the upper-boundary condition in Theorem C.1 is satisfied.

Corollary C.1 (Linear Taylor rules with partially-flexible prices). *Consider the linearized Phillips curve (linear PC) with $\kappa > 0$, and the linear Taylor rule (linear MP) with $\phi_x > 0$, $\phi_\pi > 1$, and $\bar{t} = \rho$.*

Fix a finite upper boundary $\bar{x}$. Take as the left-tail block either the deflationary block of Lemma C.1 or the inflationary block of Lemma C.2. Take as the upper-boundary block the block of Lemma C.3. Thus, for some $\underline{x} < \hat{x} < \bar{x}$, we have a left-tail pair (π_L, v_L) on $(-\infty, \underline{x}]$ and an upper-boundary pair (π_H, v_H) on $[\hat{x}, \bar{x})$. Moreover, the left-tail cutoff $\underline{x}$ can be chosen arbitrarily negative and the upper-boundary cutoff $\hat{x}$ can be chosen arbitrarily close to $\bar{x}$.

Suppose there exists an intermediate block $(\hat{\pi}, h)$ on $(\underline{x}, \hat{x})$, with $\hat{\pi}$ twice continuously differentiable and h continuous, such that

$$h(x) > 0 \quad \text{and} \quad \mathcal{T}[h, \hat{\pi}](x) = 0 \quad \text{for all } x \in (\underline{x}, \hat{x}),$$

and inflation pastes in value and slope: $\hat{\pi}(\underline{x}) = \pi_L(\underline{x})$, $\hat{\pi}'(\underline{x}) = \pi_L'(\underline{x})$, $\hat{\pi}(\hat{x}) = \pi_H(\hat{x})$, and $\hat{\pi}'(\hat{x}) = \pi_H'(\hat{x})$.

Define the pasted functions

$$v(x) := \begin{cases} v_L(x), & x \leq \underline{x}, \\ h(x), & \underline{x} < x < \hat{x}, \\ v_H(x), & \hat{x} \leq x < \bar{x}, \end{cases} \quad \pi(x) := \begin{cases} \pi_L(x), & x \leq \underline{x}, \\ \hat{\pi}(x), & \underline{x} < x < \hat{x}, \\ \pi_H(x), & \hat{x} \leq x \leq \bar{x}. \end{cases}$$

Then there exists a non-explosive sunspot equilibrium, stationary and ergodic on $(-\infty, \bar{x})$, with

$$\pi_t = \pi(x_t), \quad \sigma_{x,t}^2 = v(x_t).$$

Depending on the left-tail block chosen, the equilibrium has either a deflationary left tail or an inflationary left tail.

Following Corollary C.1, we now illustrate how the pasted functions required by the corollary can be generated in practice. The next example describes one shooting-and-pasting procedure for constructing the full equilibrium pair (π, σ_x^2) . The procedure is not meant to be exhaustive. It fixes the outer-block functional forms from Lemmas C.1–C.3 and uses a simple parametric family for volatility in the intermediate region, but still leaves several free parameters. Varying these parameters generates a continuum of numerical constructions, with the remaining constants recovered from the boundary and pasting conditions. In the parameterizations used for Figure 3 in the text, we choose the free parameters to produce reasonable-looking uncertainty traps.

Example C.1 (Numerical construction of sunspot equilibria). *This example illustrates one way to numerically construct the full pasted pair (π, σ_x^2) in Corollary C.1. The construction is not unique: using the outer-block functional forms from Lemmas C.1–C.3, we choose a simple parametric shape for intermediate volatility, set some parameters directly, and recover the remaining constants from boundary and pasting conditions. These free choices leave several degrees of freedom, generating a continuum of numerical constructions rather than a knife-edge intermediate block.*

Numerical algorithm. The support is $\{x < \bar{x}\}$, split into $x \leq \underline{x}$, $x \in (\underline{x}, \hat{x})$, and $x \in [\hat{x}, \bar{x})$. In this illustration,

$$\sigma_x^2(x) = \begin{cases} \tilde{\sigma}_x^2(x), & x \leq \underline{x}, \\ h_R + (h_L - h_R) \left(\frac{\hat{x} - x}{\hat{x} - \underline{x}} \right)^r, & x \in (\underline{x}, \hat{x}), \\ a(\bar{x} - x), & x \in [\hat{x}, \bar{x}), \end{cases} \quad \pi(x) = \begin{cases} \pi_L(x), & x \leq \underline{x}, \\ \tilde{\pi}(x), & x \in (\underline{x}, \bar{x}). \end{cases}$$

The construction proceeds as follows.

1. Choose $\underline{x} < \hat{x} < \bar{x}$, $a > 0$, $r > 0$, and $\pi(\bar{x}) = \bar{\pi}$. Let $A_{\bar{x}} := \phi_{\bar{x}}\bar{x} + (\phi_{\pi} - 1)\bar{\pi}$, and require $A_{\bar{x}} < 0$ and $a \leq -2A_{\bar{x}}$. In the deflationary-tail case, also choose $h_L > 0$.
2. Since $\sigma_x^2(\bar{x}) = 0$, the regular upper-boundary condition recovers $\pi'(\bar{x}) = \bar{\pi}'$, where $\bar{\pi}' = (\rho\bar{\pi} - \kappa\bar{x})/A_{\bar{x}}$. Set $h_R = a(\bar{x} - \hat{x})$, so volatility pastes continuously at $\hat{x}$.
3. Solve the Phillips-curve ODE backward on $(\underline{x}, \bar{x})$, using $\tilde{\pi}(\bar{x}) = \bar{\pi}$, $\tilde{\pi}'(\bar{x}) = \bar{\pi}'$, and the volatility profile above.
4. At $\underline{x}$, impose one of the left-tail templates: $\pi_L(x) = \pi_0 + \pi_1 x$ in the deflationary case, or $\pi_L(x) = A_L \exp(-x - 1/(1 - x))$ in the inflationary case. In the deflationary case, value and slope pasting recover (π_0, π_1) . In the inflationary case, value pasting recovers A_L , while h_L is chosen so that slope pasting holds.
5. Given the recovered left-tail inflation branch, construct $\tilde{\sigma}_x^2$ on $(-\infty, \underline{x}]$ from the algebraic equation $\mathcal{T}[\tilde{\sigma}_x^2, \pi_L](x) = 0$.
6. Verify admissibility. In the deflationary case, check $0 < \pi_1 < \min\{\rho/(\phi_{\pi} - 1), \kappa/\rho\}$. In the inflationary case, check $A_L > 0$ and $h_L > 0$. Intermediate-block positivity follows from $h_R > 0$, $h_L > 0$, and $r > 0$, while the left-tail sign, monotonicity, and non-explosion properties follow from Lemmas C.1 and C.2.

Parameter values for Figure 3. Figure 3 displays two numerical instances of the construction above. For both panels, $\rho = 0.04$, $\kappa = 0.02$, $\phi_x = 0.5$, and $\phi_{\pi} = 1.5$.

	<i>Deflationary left tail</i>	<i>Inflationary left tail</i>
$(\underline{x}, \hat{x}, \bar{x})$	$(-0.030, 0.005, 0.010)$	$(-6, -0.016, -0.006)$
(a, r)	$(0.0009, 0.80)$	$(0.00144, 2)$
$\bar{\pi}$	-0.011	0.0042
h_L	0.0005	997.6866
$\pi_L(x)$	$\pi_0 + \pi_1 x$	$A_L \exp\left(-x - \frac{1}{1-x}\right)$
<i>Recovered constants</i>	$(\pi_0, \pi_1) = (-0.0151, 0.0205)$	$A_L = 0.1179$

In the deflationary case, h_L is chosen directly and (π_0, π_1) are recovered from pasting. In the inflationary case, both h_L and A_L are recovered from the pasting conditions. Volatility need not be continuous at $\underline{x}$; the example in the left panel of Figure 3 features such a jump.

The numerical example above uses the pasting construction in Corollary C.1 to produce global inflation and volatility schedules satisfying the conditions in part (ii) of Theorem 2. To establish the existence claim in part (i), however, we also need theoretical results showing that such intermediate blocks can indeed be found. The next two lemmas provide these results. Lemma C.4 gives a closed-form deflationary subclass with an affine volatility function, corresponding to the “Ornstein-Uhlenbeck equilibria” studied in concurrent work by Lee and Dordal i Carreras (2026). Lemma C.5 then provides a local existence argument for an inflationary left tail, completing the proof that both deflationary and inflationary left tails are possible.

Lemma C.4 (Affine construction). *Consider the setting of Corollary C.1. Fix $0 < \pi_1 < \min\{\rho/(\phi_\pi - 1), \kappa/\rho\}$, choose $\pi_0 < 0$ satisfying condition (C.1), and let $\bar{x} < 0$ be the unique root of $G(x)$, where $G(x)$ is the affine deflationary left-tail volatility associated with $\pi_L(x) = \pi_0 + \pi_1 x$. Then the intermediate-block condition in Corollary C.1 is satisfied explicitly by taking $\hat{\pi}(x) = \pi_0 + \pi_1 x$ and $h(x) = G(x)$ on any intermediate interval $(\underline{x}, \hat{x}) \subset (-\infty, \bar{x})$ with an admissible deflationary left-tail cutoff $\underline{x}$. Hence the pasted construction satisfies Corollary C.1.*

In particular, the resulting equilibrium has the explicit affine form

$$\pi_t = \pi_0 + \pi_1 x_t, \quad \sigma_{x,t}^2 = G(x_t),$$

with state dynamics

$$dx_t = -\lambda_x(x_t - \omega_x) dt + \sqrt{G(x_t)} dZ_t, \quad x_t < \bar{x},$$

where $\lambda_x := \kappa/\pi_1 - \rho > 0$ and $\omega_x := \rho\pi_0/(\kappa - \rho\pi_1) < 0$.

The affine subclass is valuable because it delivers a closed-form characterization and provides a transparent illustration of existence. However, this subclass is not representative of all uncertainty trap equilibria. In particular, the affine case features (i) a volatility profile that is countercyclical throughout the state space; (ii) permanently negative output gaps and deflation. These are not generic properties. For one, Corollary C.1 shows instead that volatility only needs to be elevated in sufficiently deep recessions. It is completely permissible that volatility be low in mild recessions and spike up in sufficiently severe recessions (the left panel of Figure 3 features such a volatility profile). On the second point, permanent depressions and deflations are also not generic properties. As Corollary C.1 also shows, there can be inflations in the left tail, as well as output gaps above steady state (Figure 3 provides numerical examples with stagflationary episodes or deflationary expansions).

The affine construction establishes the deflationary case in part (i) of Theorem 2. The next lemma completes the existence argument by establishing the inflationary case.

Lemma C.5 (Inflationary construction). *In the setting of Corollary C.1, for $\bar{x} < 0$ sufficiently negative, there exist a choice of the inflationary left-tail block of Lemma C.2, thresholds $\underline{x} < \hat{x} < \bar{x}$, and an intermediate block $(\hat{\pi}, h)$ satisfying the positivity, Phillips-curve, and pasting conditions of that corollary. Hence the pasted construction supports a non-explosive stationary sunspot equilibrium with an inflationary left tail.*

C.3 Nonlinear Taylor rules

Finally, we show that uncertainty traps can also emerge under nonlinear, aggressive monetary-policy rules. The mechanism is qualitatively the same. Even when monetary policy reacts strongly to inflation and output, one can construct a sufficiently countercyclical volatility pattern in deep recessions such that precautionary saving dominates the intertemporal-substitution effect induced by lower interest rates. Indeed, Corollary C.2 shows that uncertainty traps persist under highly aggressive rules, including exponential and double-exponential response functions.

Corollary C.2 (Nonlinear Taylor rules). *Consider the linearized Phillips curve (linear PC) with $\kappa > 0$ and the Taylor rule (MP) with $\bar{r} = \rho$. Suppose monetary policy is given by either*

$$\Phi(x, \pi) = \phi_x \left(e^{\beta_x x} - e^{-\beta_x x} \right) + \phi_\pi \left(e^{\beta_\pi \pi} - e^{-\beta_\pi \pi} \right),$$

with $\phi_x, \phi_\pi, \beta_x, \beta_\pi > 0$ and $2\phi_\pi\beta_\pi > 1$, or

$$\Phi(x, \pi) = \phi_x \left(e^{e^{\beta_x x}} - e^{e^{-\beta_x x}} \right) + \phi_\pi \left(e^{e^{\beta_\pi \pi}} - e^{e^{-\beta_\pi \pi}} \right),$$

with $\phi_x, \phi_\pi, \beta_x, \beta_\pi > 0$ and $2e\phi_\pi\beta_\pi > 1$. Then the economy admits a continuum of non-explosive stationary sunspot equilibria, recursive in x_t .

Remark C.1. The nonlinear rules in Corollary C.2 are illustrative, not special. The proof also covers, for example, separable rules $\Phi(x, \pi) = F(x) + G(\pi)$ with $F_x > 0$, $G_\pi > 1$, and a sufficiently strong negative tail of $G(\pi) - \pi$. The exponential and double-exponential specifications are used as relatively simple examples because these tail inequalities are immediate.

C.4 Proofs

Proof of Theorem C.1. Let $\mu(x) := \Phi(x, \pi(x)) - \pi(x) + \frac{1}{2}v(x)$, as in the statement of the theorem. We first verify the equilibrium equations. Given

$$dx_t = \mu(x_t)dt + \sqrt{v(x_t)}dZ_t, \quad \pi_t = \pi(x_t), \quad \sigma_{x,t}^2 = v(x_t),$$

the dynamic IS equation holds by construction: $\mu(x_t) = \Phi(x_t, \pi_t) - \pi_t + \frac{1}{2}\sigma_{x,t}^2$.

We verify the Phillips curve. Since π is C^1 and piecewise C^2 , Itô–Tanaka’s formula has no local-time term at the pasting points: the coefficient of local time is proportional to the jump in π' , which is zero by assumption. Thus, away from the finite set $\mathcal{P}$,

$$d\pi_t = \left[\pi'(x_t)\mu(x_t) + \frac{1}{2}v(x_t)\pi''(x_t) \right] dt + \pi'(x_t)\sqrt{v(x_t)}dZ_t.$$

The drift equals $\rho\pi(x_t) - \kappa x_t$ because $\mathcal{T}_\Phi[v, \pi](x) = 0$ on each smooth block. The exceptional points in $\mathcal{P}$ do not affect the drift integral, since v is locally bounded away from zero near each such point and the diffusion spends zero Lebesgue time at singletons. Hence

$$d\pi_t = (\rho\pi_t - \kappa x_t)dt + \sigma_{\pi,t}dZ_t, \quad \sigma_{\pi,t} := \pi'(x_t)\sqrt{v(x_t)},$$

which is the linearized Phillips curve.

We now verify non-explosiveness and stationarity. By (LT), the left boundary $-\infty$ is unattainable and the speed measure has finite mass on the left tail. The same condition also gives finite left-tail second moments for x and $\pi(x)$.

Consider next the upper boundary. Let $u := \bar{x} - x$ and $\beta := -A_{\bar{x}} > 0$. Since

$P = \pi(\bar{x})$, $v(x) \rightarrow 0$, and Φ is continuous,

$$\mu(\bar{x} - u) = A_{\bar{x}} + o(1) = -\beta + o(1), \quad v(\bar{x} - u) = au^\alpha + o(u^\alpha)$$

as $u \downarrow 0$. Therefore the transformed process $u_t := \bar{x} - x_t$ has drift $\beta + o(1)$ and variance $au^\alpha + o(u^\alpha)$ near zero. Its scale density satisfies

$$s_u(u) = \exp \left\{ - \int_{u_0}^u \frac{2(\beta + o(1))}{ar^\alpha + o(r^\alpha)} dr \right\}.$$

If $\alpha > 1$, then

$$s_u(u) = \exp \left\{ \frac{2\beta}{a(\alpha - 1)} u^{1-\alpha} + o(u^{1-\alpha}) \right\},$$

and hence $\int_0 s_u(u) du = \infty$. If $\alpha = 1$, then $s_u(u) = u^{-2\beta/a+o(1)}$. The restriction $a \leq 2\beta$, equivalently $a \leq -2A_{\bar{x}}$, implies $\int_0 s_u(u) du = \infty$. Thus the upper boundary $\bar{x}$ is unattainable.

The speed density near the upper boundary is

$$m_u(u) = \frac{2}{\{au^\alpha + o(u^\alpha)\}s_u(u)}.$$

If $\alpha > 1$, the exponential growth of s_u dominates any power of u , so m_u is integrable near zero. If $\alpha = 1$, then

$$m_u(u) = u^{2\beta/a-1+o(1)},$$

which is integrable near zero because $\beta > 0$. Hence the speed measure has finite mass near $\bar{x}$.

On compact subintervals of $(-\infty, \bar{x})$, the speed density is locally integrable. Combining this fact with the left-tail condition (LT) and the upper-boundary estimates, the total speed mass is finite:

$$M := \int_{-\infty}^{\bar{x}} m(x) dx < \infty.$$

The normalized speed measure $m(x)dx/M$ is therefore an invariant probability distribution. Since both boundaries are unattainable, the standard scale-speed classification for one-dimensional diffusions (Rogers and Williams, 2000, Ch. V, Secs. 46–53) implies positive recurrence and ergodicity on $(-\infty, \bar{x})$.

Finally, the moment and transversality conditions required for non-explosiveness hold under the invariant distribution. The left-tail moment bounds follow from (LT); on compact subsets they follow from local integrability; and near $\bar{x}$, both x and $\pi(x)$ are

bounded. Thus

$$\mathbb{E}[x_t^2 + \pi_t^2] < \infty.$$

Moreover, since $x_t < \bar{x} < \infty$ almost surely, $\mathbb{E}[e^{(1+\varphi)x_t}] \leq e^{(1+\varphi)\bar{x}} < \infty$. Therefore, Condition 1 holds. The diffusion x_t is positive recurrent and converges in distribution to the unique invariant distribution $m(x)dx/M$. If initialized from this invariant distribution, the constructed processes constitute a stationary, ergodic, non-explosive sunspot equilibrium. $\square$

Proof of Lemma C.1. Let $\theta := \phi_\pi - 1$ to save notation. Since $\pi_L'' = 0$, the equation $\mathcal{T}[v_L, \pi_L] = 0$ gives

$$v_L(x) = 2\pi_0 \left(\frac{\rho}{\pi_1} - \theta \right) - 2 \left(\frac{\kappa}{\pi_1} + \phi_x + \theta\pi_1 - \rho \right) x.$$

Let

$$A := \frac{\rho}{\pi_1} - \theta > 0, \quad D := \frac{\kappa}{\pi_1} + \phi_x + \theta\pi_1 - \rho > 0.$$

Then $v_L(x) = 2\pi_0 A - 2Dx$, so $v_L'(x) = -2D < 0$ and $v_L(x) > 0$ for all sufficiently negative x . Since $\pi_1 > 0$, $\pi_L(x) = \pi_0 + \pi_1 x < 0$ for all sufficiently negative x as well. Thus there exists $x_D < 0$ such that, for every $\underline{x} \leq x_D$, $\pi_L(x) < 0$, $v_L(x) > 0$, and $v_L'(x) < 0$ for all $x \leq \underline{x}$. The Phillips-curve ODE holds by construction.

It remains to verify the left-tail scale and speed conditions. The drift induced by the block is

$$\mu_L(x) = \phi_x x + \theta\pi_L(x) + \frac{1}{2}v_L(x) = \frac{\rho\pi_L(x) - \kappa x}{\pi_1} = \frac{\rho\pi_0}{\pi_1} - \left(\frac{\kappa}{\pi_1} - \rho \right) x.$$

Let $\lambda := \kappa/\pi_1 - \rho > 0$. As $x \rightarrow -\infty$, $\mu_L(x) = \lambda|x| + O(1)$ and $v_L(x) = 2D|x| + O(1)$. Hence

$$\frac{2\mu_L(x)}{v_L(x)} \rightarrow \frac{\lambda}{D} > 0.$$

Therefore the scale density satisfies

$$s(x) = \exp \left\{ - \int_{x_0}^x \frac{2\mu_L(z)}{v_L(z)} dz \right\} = \exp \{ -cx + o(|x|) \}$$

for some $c > 0$, and so

$$\int_{-\infty}^{\underline{x}} s(x) dx = \infty.$$

The speed density satisfies

$$m(x) = \frac{2}{v_L(x)s(x)} = O\left(\frac{e^{cx/2}}{|x|}\right) \quad \text{as } x \rightarrow -\infty,$$

after reducing c if necessary. Since π_L is affine,

$$\int_{-\infty}^{\underline{x}} \left(1 + x^2 + \pi_L(x)^2\right) m(x) dx < \infty.$$

Thus the left-tail scale and speed conditions (LT) are satisfied, and (π_L, v_L) is an admissible deflationary left-tail block. $\square$

Proof of Lemma C.2. Let $\theta := \phi_\pi - 1$ to save notation. Let $\ell(x) := 1 + (1 - x)^{-2}$. Then $\pi'_L(x) = -\ell(x)\pi_L(x)$, and

$$\pi'_L(x) + \pi''_L(x) = \pi_L(x) \frac{x^2}{(1 - x)^4} > 0$$

for $x < 0$. Solving $\mathcal{T}[v_L, \pi_L] = 0$ gives

$$v_L(x) = \frac{2 [\theta \ell(x) \pi_L(x)^2 + \{\rho + \ell(x) \phi_x x\} \pi_L(x) - \kappa x]}{\pi_L(x) x^2 / (1 - x)^4}.$$

As $x \rightarrow -\infty$, $\ell(x) \rightarrow 1$ and $\pi_L(x) \sim A_L e^{-x}$. The dominant term in the numerator is $\theta \pi_L(x)^2$, and therefore $v_L(x) \sim 2\theta x^2 \pi_L(x) > 0$. Differentiating the same asymptotic expression gives $v'_L(x) \sim -2\theta x^2 \pi_L(x) < 0$. Since $\pi_L(x) > 0$ for all x , there exists $x_I < 0$ such that, for every $\underline{x} \leq x_I$, $\pi_L(x) > 0$, $v_L(x) > 0$, and $v'_L(x) < 0$ for all $x \leq \underline{x}$. The Phillips-curve ODE holds by construction.

It remains to verify the left-tail scale and speed conditions. The drift induced by the block is

$$\mu_L(x) = \phi_x x + \theta \pi_L(x) + \frac{1}{2} v_L(x).$$

Using $v_L(x) \sim 2\theta x^2 \pi_L(x)$, we obtain

$$\mu_L(x) \sim \theta x^2 \pi_L(x), \quad \frac{2\mu_L(x)}{v_L(x)} \rightarrow 1 \quad \text{as } x \rightarrow -\infty.$$

Hence the scale density satisfies

$$s(x) = \exp \left\{ - \int_{x_0}^x \frac{2\mu_L(z)}{v_L(z)} dz \right\} = e^{-x+o(|x|)},$$

so $\int_{-\infty}^x s(x) dx = \infty$. The speed density satisfies

$$m(x) = \frac{2}{v_L(x)s(x)} \sim \frac{Ce^{2x}}{x^2}$$

for some constant $C > 0$. Since $\pi_L(x)^2 \sim A_L^2 e^{-2x}$, we have $\pi_L(x)^2 m(x) = O(x^{-2})$. Therefore

$$\int_{-\infty}^x \left(1 + x^2 + \pi_L(x)^2\right) m(x) dx < \infty.$$

The left-tail scale and speed conditions (LT) are satisfied, so (π_L, v_L) is an admissible inflationary left-tail block. $\square$

Proof of Lemma C.3. Let $\theta := \phi_\pi - 1$ to save notation. Set $u := \bar{x} - x$ and $p(u) := \pi_H(\bar{x} - u)$. The equation $\mathcal{T}[v_H, \pi_H] = 0$ becomes

$$\frac{1}{2}au^\alpha p''(u) - H(u, p(u))p'(u) = R(u, p(u)),$$

where

$$H(u, p) := \phi_x(\bar{x} - u) + \theta p + \frac{1}{2}au^\alpha, \quad R(u, p) := \rho p - \kappa(\bar{x} - u).$$

Since $H(0, P) = A_{\bar{x}} < 0$, H is bounded away from zero near $(0, P)$. The regular boundary slope is obtained by setting $u = 0$ in the reduced equation:

$$p'(0) = -\frac{R(0, P)}{A_{\bar{x}}}, \quad \text{or equivalently} \quad \lim_{x \uparrow \bar{x}} \pi'_H(x) = \frac{\rho P - \kappa \bar{x}}{A_{\bar{x}}}.$$

We next solve the singular initial-value problem. Let $\psi(u, p) := -R(u, p)/H(u, p)$ and write $p'(u) = \psi(u, p(u)) + z(u)$. Substitution gives the singular Volterra system

$$\begin{aligned} p(u) &= P + \int_0^u \{\psi(v, p(v)) + z(v)\} dv, \\ z(u) &= - \int_0^u K_p(u, v) [\psi_u(v, p(v)) + \psi_p(v, p(v))\{\psi(v, p(v)) + z(v)\}] dv, \end{aligned}$$

where

$$K_p(u, v) := \exp \left(\int_v^u \frac{2H(r, p(r))}{ar^\alpha} dr \right).$$

Choose $\eta > 0$ and $\varepsilon > 0$ so that $H(u, p) \leq -\eta$ whenever $0 \leq u < \varepsilon$ and $|p - P| \leq \eta$. On such a neighborhood, ψ and its first derivatives are bounded and Lipschitz. Moreover,

uniformly over the ball $\|p - P\|_\infty \leq \eta$,

$$\sup_{0 < u < \varepsilon} \int_0^u K_p(u, v) dv \rightarrow 0 \quad \text{as } \varepsilon \downarrow 0.$$

For $\alpha = 1$, this follows from the power bound $K_p(u, v) \leq (v/u)^{2\eta/a}$; for $\alpha > 1$, from the exponential decay as $v \downarrow 0$. The same estimates, with one extra factor $\int_v^u r^{-\alpha} dr$ (or $\log(u/v)$ when $\alpha = 1$), give the corresponding Lipschitz bound for K_p as a function of p . Hence the displayed Volterra operator is a contraction, in the sup norm, on a small ball in $C[0, \varepsilon] \times C[0, \varepsilon]$, with $p(0) = P$ and $z(0) = 0$, after reducing ε . Thus there is a unique local solution p on $[0, \varepsilon)$ with $p(0) = P$ and the boundary slope above. The same bounds give $z(u) = O(u)$, and therefore

$$p(u) = P + p'(0)u + O(u^2), \quad p'(u) = p'(0) + O(u).$$

Defining $\pi_H(x) := p(\bar{x} - x)$ gives a solution on $(\bar{x} - \varepsilon, \bar{x}]$, and hence on $[\hat{x}, \bar{x})$ for any $\hat{x} \in (\bar{x} - \varepsilon, \bar{x})$.

Finally, $v_H(x) = a(\bar{x} - x)^\alpha$, $A_{\bar{x}} < 0$, and $a \leq -2A_{\bar{x}}$ when $\alpha = 1$. These are exactly the upper-boundary conditions required by Theorem C.1. Thus (π_H, v_H) is an admissible upper-boundary block. $\square$

Proof of Corollary C.1. By Lemma C.1 or Lemma C.2, the left-tail block satisfies the Phillips-curve ODE, positivity of volatility, and the required left-boundary scale and speed conditions. The same lemmas also imply that the left-tail cutoff $\underline{x}$ can be taken arbitrarily far to the left. Similarly, Lemma C.3 gives a local upper-boundary solution on $(\bar{x} - \varepsilon, \bar{x})$, so $\hat{x}$ can be chosen arbitrarily close to $\bar{x}$.

The upper-boundary block satisfies the Phillips-curve ODE and the upper-boundary non-attainability conditions by Lemma C.3. By assumption, the intermediate block satisfies the ODE and pastes inflation in value and slope at $\underline{x}$ and $\hat{x}$. Hence the pasted pair (π, v) satisfies the hypotheses of Theorem C.1. The theorem delivers a stationary, ergodic, non-explosive sunspot equilibrium with $\pi_t = \pi(x_t)$ and $\sigma_{x,t}^2 = v(x_t)$. $\square$

Proof of Lemma C.4. Let $\theta := \phi_\pi - 1$, $A := \rho/\pi_1 - \theta > 0$, and $D := \kappa/\pi_1 + \phi_x + \theta\pi_1 - \rho > 0$. The affine volatility associated with $\pi_L(x) = \pi_0 + \pi_1 x$ is

$$G(x) := 2\pi_0 A - 2Dx.$$

The condition on π_0 is

$$-\pi_0 \geq \frac{\pi_1 D^2}{\rho \phi_x + \theta \kappa}. \quad (\text{C.1})$$

Since $G(\bar{x}) = 0$, its unique root is $\bar{x} = \pi_0 A / D < 0$, and $G(x) = 2D(\bar{x} - x) > 0$ for all $x < \bar{x}$.

Take any admissible deflationary cutoff $\underline{x}$ from Lemma C.1 and any $\hat{x} \in (\underline{x}, \bar{x})$. Set $\hat{\pi}(x) = \pi_0 + \pi_1 x$ and $h(x) = G(x)$ on $(\underline{x}, \hat{x})$, and use the affine upper-boundary block $\pi_H(x) = \pi_0 + \pi_1 x$, $v_H(x) = G(x)$ on $[\hat{x}, \bar{x})$. Then $h > 0$, the value and slope pasting conditions hold exactly, and the intermediate ODE holds because

$$\pi_1 \left[\phi_x x + \theta(\pi_0 + \pi_1 x) + \frac{1}{2} G(x) \right] = \rho(\pi_0 + \pi_1 x) - \kappa x.$$

It remains only to check that this affine upper-boundary block satisfies the upper-boundary conditions. Let $P := \pi_0 + \pi_1 \bar{x}$. Then

$$A_{\bar{x}} := \phi_x \bar{x} + \theta P = \frac{\pi_0}{\pi_1 D} (\rho \phi_x + \theta \kappa) < 0.$$

Since $G(x) = 2D(\bar{x} - x)$, the affine upper block has $a = 2D$ and $\alpha = 1$. Condition (C.1) is equivalent to $2D \leq -2A_{\bar{x}}$. Evaluating the displayed affine ODE at $x = \bar{x}$ gives $\pi_1 A_{\bar{x}} = \rho P - \kappa \bar{x}$, so the affine block also has the regular boundary slope of Lemma C.3. By the local uniqueness in that lemma, it is the corresponding upper-boundary solution. Hence Lemma C.3 applies. Together with Lemma C.1, Corollary C.1 delivers the equilibrium.

Finally, substituting $\pi(x) = \pi_0 + \pi_1 x$ and $v(x) = G(x)$ into the IS drift gives

$$\mu_x(x) = \phi_x x + \theta(\pi_0 + \pi_1 x) + \frac{1}{2} G(x) = \frac{\rho \pi_0}{\pi_1} + \left(\rho - \frac{\kappa}{\pi_1} \right) x = -\lambda_x (x - \omega_x),$$

with $\lambda_x = \kappa / \pi_1 - \rho > 0$ and $\omega_x = \rho \pi_0 / (\kappa - \rho \pi_1) < 0$. □

Proof of Corollary C.2. We prove the result by constructing, for each sufficiently large M , a pair (π, v) satisfying the hypotheses of Theorem C.1. The construction uses an affine inflation schedule $\pi(x) = L_M(x)$ on the whole state space $(-\infty, \bar{x})$, with $\bar{x} = -M$, and defines volatility residually from the Phillips-curve ODE. We then verify positivity of volatility, the upper-boundary condition, and the left-tail scale and speed conditions.

Let $H(x, \pi) := \Phi(x, \pi) - \pi$. In the exponential case,

$$H_\pi(x, \pi) = \phi_\pi \beta_\pi \left(e^{\beta_\pi \pi} + e^{-\beta_\pi \pi} \right) - 1 \geq 2\phi_\pi \beta_\pi - 1 > 0,$$

and $H_x > 0$. In the double-exponential case,

$$H_\pi(x, \pi) = \phi_\pi \beta_\pi \left(e^{\beta_\pi \pi} e^{e^{\beta_\pi \pi}} + e^{-\beta_\pi \pi} e^{-e^{\beta_\pi \pi}} \right) - 1 \geq 2e\phi_\pi \beta_\pi - 1 > 0,$$

and again $H_x > 0$. Since $H(0, 0) = 0$, it follows that $H(x, \pi) < 0$ whenever $x < 0$ and $\pi < 0$.

Choose $q > 1$ such that $r := \beta_\pi q \kappa / \rho > \beta_x$. For $M > 0$, set $\bar{x} := -M$ and $y := -q\kappa M / \rho$. Then $H(\bar{x}, y) < 0$. Define

$$\pi_1(M) := \frac{\rho y - \kappa \bar{x}}{H(\bar{x}, y)} = \frac{-\kappa(q-1)M}{H(\bar{x}, y)} > 0, \quad \pi_0(M) := y - \pi_1(M)\bar{x},$$

and $L_M(x) := \pi_0(M) + \pi_1(M)x$. Because $r > \beta_x$, the negative inflation component of $H(\bar{x}, y)$ dominates the output component as $M \rightarrow \infty$. Hence $-H(\bar{x}, y) \rightarrow \infty$ faster than M , so $\pi_1(M) \rightarrow 0$. Therefore, for M sufficiently large, $0 < \pi_1(M) < \kappa/\rho$ and $\pi_0(M) < 0$.

For such M , define

$$v(x) := 2 \left[\frac{\rho L_M(x) - \kappa x}{\pi_1(M)} - H(x, L_M(x)) \right], \quad x < \bar{x}.$$

Since $L'_M = \pi_1(M)$ and $L''_M = 0$, this definition gives $\mathcal{T}_\Phi[v, L_M](x) = 0$. Also, $v(\bar{x}) = 0$ by the definition of $\pi_1(M)$, and

$$v'(x) = 2 \left[\rho - \frac{\kappa}{\pi_1(M)} - H_x(x, L_M(x)) - \pi_1(M) H_\pi(x, L_M(x)) \right] < 0.$$

Thus $v(x) > 0$ for all $x < \bar{x}$.

Near $\bar{x}$, write $v(x) = a(\bar{x} - x) + O((\bar{x} - x)^2)$, where $a := -v'(\bar{x}) > 0$. The upper-boundary drift is $A_{\bar{x}} = H(\bar{x}, y) < 0$. Moreover,

$$\frac{a}{-2A_{\bar{x}}} = \frac{\kappa/\pi_1(M) - \rho + H_x(\bar{x}, y) + \pi_1(M)H_\pi(\bar{x}, y)}{-H(\bar{x}, y)} \rightarrow 0.$$

Indeed, $\kappa/\pi_1(M)$ is of order $-H(\bar{x}, y)/M$, the output derivative is asymptotically dominated because $r > \beta_x$, and the term $\pi_1(M)H_\pi(\bar{x}, y)$ is negligible relative to $-H(\bar{x}, y)$ in both the exponential and double-exponential cases. Thus $a \leq -2A_{\bar{x}}$ for all sufficiently large M .

It remains to verify the left-tail condition of Theorem C.1. The drift induced by the

ODE is

$$\mu(x) = H(x, L_M(x)) + \frac{1}{2}v(x) = \frac{\rho L_M(x) - \kappa x}{\pi_1(M)} = \frac{\rho \pi_0(M)}{\pi_1(M)} + \left(\rho - \frac{\kappa}{\pi_1(M)} \right) x.$$

Thus $\mu(x)$ grows only linearly as $x \rightarrow -\infty$, while $-H(x, L_M(x))$, and hence $v(x)$, grows at least exponentially in the exponential case and at least double-exponentially in the double-exponential case. Therefore $2\mu(x)/v(x)$ is integrable at $-\infty$, the scale density is bounded above and below by positive constants in the far left tail, and the speed density inherits the exponential or double-exponential decay of $1/v$. Hence the scale density has infinite integral at the left boundary, and the speed density has finite $(1 + x^2 + L_M(x)^2)$ -moment.

We have verified all hypotheses of Theorem C.1: regularity and the Phillips-curve ODE hold by construction, $v > 0$ on $(-\infty, \bar{x})$, the upper-boundary condition holds, and the left-tail scale and speed conditions hold. The theorem therefore delivers a non-explosive stationary sunspot equilibrium. Since the construction works for every sufficiently large M , varying M generates a continuum of equilibria. $\square$

Proof of Lemma C.5. Write $\theta := \phi_\pi - 1$, let $F(x) := \exp\{-x - (1 - x)^{-1}\}$ and $\ell(x) := 1 + (1 - x)^{-2}$, and set $\bar{x} = -M$. Choose P as the larger positive root of

$$\ell(\bar{x})\theta P^2 + \{\ell(\bar{x})\phi_x \bar{x} + \rho\}P - \kappa \bar{x} = 0. \quad (\text{C.2})$$

For M large, both roots are positive. If $P_+(M)$ and $P_-(M)$ denote the larger and smaller roots, then $P_-(M) = \kappa/\phi_x + O(M^{-1})$ and $P_+(M) = (\phi_x/\theta)M + O(1)$. Taking $P = P_+(M)$, we have $A_{\bar{x}} := \phi_x \bar{x} + \theta P = -\rho/\ell(\bar{x}) - \theta P_-(M) < 0$. Set $A_L := P/F(\bar{x})$ and $\pi_L(x) := A_L F(x)$. Since $\pi'_L(\bar{x}) = -\ell(\bar{x})P$, (C.2) implies $\pi_L(\bar{x}) = P$ and $\pi'_L(\bar{x}) = (\rho P - \kappa \bar{x})/A_{\bar{x}}$. Thus the inflationary branch has the boundary value and the regular upper-boundary slope required at $\bar{x}$.

Let v_L be defined by $\mathcal{T}[v_L, \pi_L] = 0$. Lemma C.2 gives the left-tail scale and speed conditions for this branch. It remains to verify that the same branch is admissible up to the matched boundary. Put $y := -x$, $\ell_y := 1 + (1 + y)^{-2}$, and $\Pi_M(y) := \pi_L(-y) = P_+(M) \exp\{y - M + (1 + M)^{-1} - (1 + y)^{-1}\}$. Then $\Pi_M(y)/y$ is increasing on $[M, \infty)$, with $\Pi_M(y)/y \geq P_+(M)/M = \phi_x/\theta + O(M^{-1})$, and the residual formula is

$$v_L(-y) = 2 \frac{(1 + y)^4}{y^2} W_M(y), \quad W_M(y) := \theta \ell_y \Pi_M(y) + \rho - \phi_x y \ell_y + \frac{\kappa y}{\Pi_M(y)}.$$

Equation (C.2) gives $W_M(M) = 0$. Moreover,

$$W'_M(y) = \theta \Pi_M(y)(\ell_y^2 + \ell'_y) - \phi_x(\ell_y + y\ell'_y) + \frac{\kappa(1 - y\ell_y)}{\Pi_M(y)}, \quad \ell'_y = -2(1 + y)^{-3}.$$

The lower bound on $\Pi_M(y)/y$ implies $W'_M(y) > 0$ on $[M, 2M]$ for M sufficiently large, while the exponential growth of Π_M implies $W'_M(y) > 0$ on $[2M, \infty)$. Hence $W_M(y) > 0$ for all $y > M$, so $v_L(x) > 0$ for all $x < \bar{x}$. Since the prefactor $(1 + y)^4/y^2$ is increasing for large y , increasing M if necessary also gives $v'_L(x) < 0$ for all $x < \bar{x}$. Together with Lemma C.2, this makes (π_L, v_L) an admissible inflationary left-tail block for any cutoff $\underline{x} < \bar{x}$.

Because $A_{\bar{x}} < 0$, Lemma C.3 gives an upper-boundary block (π_H, v_H) with boundary value P . Choose any $a > 0$ and $\alpha > 1$, and take $v_H(x) = a(\bar{x} - x)^\alpha$. The same lemma gives $\pi_H(\bar{x}) = P$ and $\lim_{x \uparrow \bar{x}} \pi'_H(x) = (\rho P - \kappa \bar{x})/A_{\bar{x}} = \pi'_L(\bar{x})$. Therefore, using the local expansion from the proof of Lemma C.3, we have $\pi_H(\bar{x} - z) - \pi_L(\bar{x} - z) = O(z^2)$ and $\pi'_H(\bar{x} - z) - \pi'_L(\bar{x} - z) = O(z)$ as $z \downarrow 0$.

We now construct the intermediate block. Since $v_L(\bar{x}) = 0$ and $v'_L(x) < 0$ to the left of $\bar{x}$, there are $c_M > 0$ and $\eta > 0$ such that $v_L(x) \geq c_M(\bar{x} - x)$ on $[\bar{x} - \eta, \bar{x})$. Choose M large enough that the slope margin c_M dominates the Hermite constants below. For $0 < \delta < \eta/2$, set $\hat{x} := \bar{x} - \delta$ and $\underline{x} := \bar{x} - \eta$. Let $\hat{\pi}$ be the cubic Hermite interpolant on $[\underline{x}, \hat{x}]$ matching π_L in value and slope at $\underline{x}$, and matching π_H in value and slope at $\hat{x}$. Writing $E_\delta := \hat{\pi} - \pi_L$, the endpoint estimates above give $E_\delta(\underline{x}) = E'_\delta(\underline{x}) = 0$, $E_\delta(\hat{x}) = O(\delta^2)$, and $E'_\delta(\hat{x}) = O(\delta)$. The explicit cubic-Hermite formula therefore implies $\|E_\delta\|_{C^2([\underline{x}, \hat{x}])} \leq C_M \delta$.

Define h residually by

$$h(x) = \frac{2[\rho \hat{\pi}(x) - \kappa x - \hat{\pi}'(x)\{\phi_x x + \theta \hat{\pi}(x)\}]}{\hat{\pi}'(x) + \hat{\pi}''(x)}.$$

For the left-tail branch, $\pi'_L(x) + \pi''_L(x) = \pi_L(x)x^2/(1 - x)^4 > 0$. Hence, for δ small, the denominator $\hat{\pi}' + \hat{\pi}''$ remains positive on $[\underline{x}, \hat{x}]$. By continuity of the residual map in (π, π', π'') , the Hermite bound implies $|h(x) - v_L(x)| \leq C_M \delta$ on $[\underline{x}, \hat{x}]$, after possibly increasing C_M . Since $\bar{x} - x \geq \delta$ on this interval and M was chosen so that $C_M < c_M$, we have $h(x) \geq v_L(x) - C_M \delta \geq (c_M - C_M)\delta > 0$.

The value and slope pasting conditions hold by construction, and $\mathcal{T}[h, \hat{\pi}] = 0$ holds by the definition of h . Therefore the intermediate-block condition in Corollary C.1 is satisfied, and that corollary delivers the claimed equilibrium. $\square$

D Discrete-time model

This appendix shows that the paper's main results are not specific to continuous time. We develop a discrete-time counterpart of the baseline model that keeps the household Euler equation nonlinear, while using the linearized Phillips curve and a linear Taylor rule. In the absence of government debt—or with passive fiscal policy—the Taylor principle fails to select a unique equilibrium: the economy admits non-explosive sunspot equilibria with heightened volatility in deep recessions, i.e., uncertainty traps. Active fiscal policy, by contrast, disciplines the economy's response to aggregate shocks and rules out the self-fulfilling countercyclical volatility that sustains these traps. Thus, neither the multiplicity result under passive fiscal policy nor the disciplining role of active fiscal policy relies on the continuous-time formulation. All omitted proofs are collected at the end of this appendix.

D.1 Environment

We consider a canonical New Keynesian economy with complete markets and nominal rigidities. Since the environment is standard, we keep the model description succinct and refer the reader to [Gali \(2015\)](#) for further details.

Time is discrete, $t = 0, 1, 2, \dots$. Let P_t be the price level, $\Pi_t = P_t/P_{t-1}$ gross inflation, and I_t the gross nominal interest rate. Define $\pi_t := \log \Pi_t$, $i_t := \log I_t$, and $x_t := \log(Y_t/Y^*)$, where $Y^* > 0$ is the constant level of natural output. Let $\beta := e^{-\rho} \in (0, 1)$. The only aggregate shock is an i.i.d. sunspot ε_t , with atomless distribution F , continuous support, and no direct effect on preferences, technology, or policy fundamentals.

Households. The household chooses $\{C_t, L_t, B_t\}_{t \geq 0}$, where B_t is the nominal face value of one-period bonds purchased at t , to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - \frac{L_t^{1+\varphi}}{1+\varphi} \right], \quad \varphi > 0,$$

subject to $P_t C_t + \frac{B_t}{I_t} = W_t L_t + B_{t-1} + P_t \mathcal{D}_t - P_t T_t.$

Here $\mathcal{D}_t$ denotes real firm dividends and T_t real lump-sum taxes net of transfers. Without a fiscal authority, the bond is in zero net supply; when government debt is introduced, the same B_t is the face value of outstanding government liabilities. The main result of

households' optimization problem is their Euler equation, or nonlinear IS curve:

$$1 = e^{i_t - \rho + x_t} \mathbb{E}_t [e^{-x_{t+1} - \pi_{t+1}}]. \quad (\text{IS-dt})$$

Firms. Intermediate firm j faces demand $Y_t(j) = (P_t(j)/P_t)^{-\epsilon} Y_t$, with $\epsilon > 1$, and produces with $Y_t(j) = L_t(j)$. Firm-level log inflation is $\pi_t(j) := \log(P_t(j)/P_{t-1}(j))$. The Rotemberg adjustment cost is $(\vartheta/2)\pi_t(j)^2 Y_t$, with $\vartheta > 0$, and is assumed to be non-pecuniary or rebated back to consumers. Given the stochastic discount factor $M_{t,s}$ between periods t and $s \geq t$, firm j chooses $\{P_s(j), L_s(j)\}_{s \geq t}$ to maximize

$$\mathbb{E}_t \sum_{s=t}^{\infty} M_{t,s} \left[\frac{P_s(j)}{P_s} Y_s(j) - \frac{W_s}{P_s} L_s(j) - \frac{\vartheta}{2} \pi_s(j)^2 Y_s \right],$$

subject to demand and production. In symmetric equilibrium, $P_t(j) = P_t$, $Y_t(j) = L_t(j) = Y_t$, and $\pi_t(j) = \pi_t$. The natural level of output Y^* satisfies $(Y^*)^{1+\varphi} = (\epsilon - 1)/\epsilon$. The linearized implication of optimal price setting is

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t, \quad \text{where } \kappa := \vartheta^{-1}(\epsilon - 1)(1 + \varphi) > 0. \quad (\text{NKPC-dt})$$

Monetary policy. Assume monetary policy uses a linear Taylor rule with target rate ρ :

$$i_t = \rho + \phi_x x_t + \phi_\pi \pi_t. \quad (\text{MP-dt})$$

We define a non-explosive equilibrium by adapted processes $\{x_t, \pi_t, i_t\}_{t \geq 0}$ satisfying the nonlinear IS ([IS-dt](#)), the linearized NKPC ([NKPC-dt](#)), monetary rule ([MP-dt](#)), and a non-explosiveness condition (Condition 2 below) that rules out both finite-time and asymptotic explosions. This condition is sufficient to guarantee agents' transversality conditions and to ensure that the households' and firms' optimization problems are well-posed, in the sense that the associated payoffs are strictly greater than $-\infty$.²³

²³Indeed, household utility is bounded below by the discounted sum of terms controlled by $1 + |x_t| + e^{(1+\varphi)x_t}$. For firms, symmetric dividends are $\mathcal{D}_t = Y_t - S_t Y_t - (\vartheta/2)\pi_t^2 Y_t$, whose negative part is bounded, up to constants, by $Y_t[e^{(1+\varphi)x_t} + \pi_t^2]$. Since $M_{t,t+j} Y_{t+j} = \beta^j Y_t$, Condition 2 implies $\mathbb{E}_t \sum_{j \geq 0} M_{t,t+j} \mathcal{D}_{t+j}^- < \infty$. Finally, the uniform bound in ([D.1](#)) gives the discounted tail limits for $|x_T|$, $e^{(1+\varphi)x_T}$, and π_T^2 , so the relevant household and firm tail/TVC terms vanish.

Condition 2 (Non-explosiveness). *An allocation is non-explosive if, for each t ,*

$$\begin{aligned} \mathbb{P}_t (|x_{t+j}| + |\pi_{t+j}| < \infty \text{ for all } j \geq 0) &= 1 \\ \sup_{j \geq 0} \mathbb{E}_t \left[|x_{t+j}| + e^{(1+\varphi)x_{t+j}} + \pi_{t+j}^2 \right] &< \infty, \end{aligned} \quad (\text{D.1})$$

D.2 Uncertainty traps

The following lemma shows that uncertainty traps can arise even under interest-rate rules that satisfy the Taylor principle. As in the main text, uncertainty traps feature heightened volatility in deep recessions.

Proposition D.1 (Discrete-time uncertainty traps). *Consider the discrete-time model with the exact nonlinear IS equation (IS-dt), the linearized NKPC (NKPC-dt), and the linear Taylor rule (MP-dt) with $\phi_\pi > 1$ and $\phi_x > 0$. Then, there exists a nonempty interval $\mathcal{A} \subset (\kappa, \kappa/(1 - \beta))$ such that, for each $a \in \mathcal{A}$, there is a non-explosive equilibrium with*

$$\pi_t = ax_t, \quad x_{t+1} = Q_a(x_t, \varepsilon_{t+1}), \quad \text{Var}_t(x_{t+1}) = -v_a x_t$$

for some function Q_a and some positive constant v_a . This non-explosive equilibrium is recessionary, deflationary, and features countercyclical volatility, in the sense that $x_t \leq 0$ and $\pi_t \leq 0$ for all $t \geq 0$, while $\text{Var}_t(x_{t+1})$ is strictly decreasing in x_t .

There are a few differences between this result and our continuous-time characterization. Most importantly, the discrete-time construction imposes a specific structure on the stochastic process for x_t throughout its domain. Indeed, the discrete-time construction sets inflation and conditional volatility to be linear in the log output gap everywhere. By contrast, in continuous time, recall that $\sigma_x^2(x)$ is restricted only as $x \rightarrow -\infty$ or $x \rightarrow 0$. The additional flexibility away from these boundaries can be important when studying the quantitative properties of uncertainty traps. This distinction highlights the key benefit of working in a continuous-time environment: all the analysis is highly tractable and allows us to present a complete characterization of self-fulfilling volatility.

D.3 Government debt and fiscal policy

Introduce one-period nominal government debt. Let B_{t-1} be nominal debt inherited from period $t - 1$. Real primary surpluses are $S_t = s_t Y_t$, for some surplus-to-GDP

process s_t to be defined below. The government's flow budget constraint is

$$\frac{B_t}{I_t} + P_t S_t = B_{t-1}.$$

Let $\ell_t := B_{t-1}/(P_t Y_t)$ denote the market value of inherited debt relative to output. We may combine the flow budget constraint with the household Euler equation to obtain the useful equation²⁴

$$\ell_t = s_t + \beta \mathbb{E}_t \ell_{t+1}. \quad (\text{D.2})$$

In addition to (D.2), the government no-Ponzi condition $\lim_{T \rightarrow \infty} \mathbb{E}_t[\beta^T \frac{Y_t}{Y_{t+T}} \frac{B_{t+T-1}}{P_{t+T}}] = 0$ must hold, or equivalently

$$\lim_{T \rightarrow \infty} \beta^T \mathbb{E}_t \ell_{t+T} = 0 \quad (\text{D.3})$$

must also hold. We may iterate (D.2) forward and use the no-Ponzi condition (D.3) to get the familiar debt valuation equation

$$\ell_t = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j s_{t+j}. \quad (\text{GD-dt})$$

We consider the class of fiscal rules

$$s_t = \bar{s} + \gamma \ell_t.$$

A passive fiscal rule is one that guarantees the government no-Ponzi condition, so that the debt-valuation equation (GD-dt) follows automatically from the flow budget constraint rather than imposing an additional restriction on equilibrium. By contrast, an active fiscal rule is one that does not automatically guarantee that (GD-dt) holds for any path of output and inflation.

In our specification, all rules with $0 < \gamma < 1$ are passive. Indeed, substituting $s_t = \bar{s} + \gamma \ell_t$ into the recursion $\ell_t = s_t + \beta \mathbb{E}_t \ell_{t+1}$ gives $\beta \mathbb{E}_t \ell_{t+1} = (1 - \gamma) \ell_t - \bar{s}$. Iterating yields

$$\beta^T \mathbb{E}_t \ell_{t+T} = (1 - \gamma)^T \ell_t - \bar{s} \sum_{j=0}^{T-1} (1 - \gamma)^{T-1-j} \beta^j,$$

²⁴Dividing the flow budget constraint by $P_t Y_t$ yields $B_t/(I_t P_t Y_t) = \ell_t - s_t$. Multiplying the household Euler equation $1 = \beta \mathbb{E}_t[(I_t/\Pi_{t+1})(Y_t/Y_{t+1})]$ by $B_t/(P_t Y_t)$ gives $B_t/(I_t P_t Y_t) = \beta \mathbb{E}_t[B_t/(P_{t+1} Y_{t+1})] = \beta \mathbb{E}_t \ell_{t+1}$. Combining these two results, we obtain (D.2).

which converges to 0 as $T \rightarrow \infty$ for every finite ℓ_t , since $\beta, 1 - \gamma \in (0, 1)$. Thus (GD-dt) is implied by the flow budget constraint and the fiscal rule, and it does not impose an additional restriction on equilibrium. The equilibrium characterization is therefore the same as if there was no debt or fiscal policies (the fiscal rule and the government budget constraint only determine the path of primary surpluses and government debt that supports the monetary allocation).

On the other hand, a traditional example of active fiscal policy is $\gamma = 0$, so that $s_t = \bar{s}$. That is, the surplus-to-GDP ratio is exogenous (and here, for simplicity, a constant). Then (GD-dt) becomes $\ell_t = \bar{s}/(1 - \beta)$, which is an additional equilibrium restriction pinning down the market value of debt-to-GDP. We may use this restriction to rule out uncertainty traps.

Proposition D.2 (Active fiscal eliminates uncertainty traps). *In the active-fiscal economy with $\gamma = 0$, the uncertainty-trap dynamics of Proposition D.1 are not equilibria.*

D.4 Proofs

Proof of Proposition D.1. We first set up the non-empty interval $\mathcal{A}$ and define some important constants. Let $m_a := (a - \kappa)/(\beta a)$ and $C_a := 1 + \phi_x + \phi_\pi a$. As $a \rightarrow \kappa$, notice that $m_a \rightarrow 0$, while $C_a \rightarrow 1 + \phi_x + \phi_\pi \kappa > 0$. Hence, there exists a nonempty interval $\mathcal{A} \subset (\kappa, \kappa/(1 - \beta))$ such that $m_a \in (0, 1)$ and $C_a > m_a(1 + a)$ for all $a \in \mathcal{A}$. For each $a \in \mathcal{A}$, choose $r_a > 1 + a$ satisfying

$$m_a r_a \log \left(\frac{r_a}{r_a - (1 + a)} \right) = C_a,$$

which is possible because the left-hand side is continuous in $r_a > 1 + a$, converges to $m_a(1 + a)$ as $r_a \rightarrow \infty$, and diverges to $+\infty$ as $r_a \downarrow 1 + a$.

We will now present a construction in which the negative output gap, $y_t := -x_t$, has a Gamma distribution. In particular, we demonstrate a construction in which

$$y_{t+1} \mid y_t \sim \text{Gamma}(m_a r_a y_t, r_a),$$

where $m_a r_a y_t$ is the “shape parameter” and r_a is the “rate parameter” of the Gamma distribution. To do this, let $G_{k,r}$ denote the cdf of a Gamma distribution with shape parameter $k > 0$ and rate parameter $r > 0$. Recall that F denotes the distribution

function of the sunspot shock ε_{t+1} . For $y > 0$, define $\tilde{Q}_a$ by

$$\tilde{Q}_a(y, \varepsilon) = \begin{cases} G_{m_a r_a y, r_a}^{-1}(F(\varepsilon)), & \text{if } y > 0; \\ 0, & \text{if } y = 0. \end{cases}$$

In that case, we have that

$$\begin{aligned} \mathbb{P}[y_{t+1} < q \mid y_t] &= \mathbb{P}[\tilde{Q}_a(y_t, \varepsilon_{t+1}) < q \mid y_t] \\ &= \mathbb{P}[F(\varepsilon_{t+1}) < G_{m_a r_a y_t, r_a}(q) \mid y_t] = G_{m_a r_a y_t, r_a}(q) \end{aligned}$$

as desired. Given this construction for the negative output gap y , we define the transition function of the output gap $x = -y$ by $Q_a(x, \varepsilon) := -\tilde{Q}_a(-x, \varepsilon)$.

Now, we verify the equilibrium equations hold. First, using the formula for the mean of a Gamma random variable, we have $\mathbb{E}_t[y_{t+1}] = m_a y_t$, so $\mathbb{E}_t[x_{t+1}] = m_a x_t$. Since $m_a = (a - \kappa)/(\beta a)$, (NKPC-dt) holds with $\pi_t = a x_t$. Moreover, the Gamma moment-generating function gives

$$\mathbb{E}_t[e^{(1+a)y_{t+1}}] = \left(\frac{r_a}{r_a - (1+a)} \right)^{m_a r_a y_t} = e^{C_a y_t},$$

so $\mathbb{E}_t[e^{-x_{t+1} - \pi_{t+1}}] = e^{-C_a x_t}$. Combining the Taylor rule (MP-dt) with $\pi_t = a x_t$ yields $i_t - \rho + x_t = (1 + \phi_x + \phi_\pi a) x_t = C_a x_t$. Combining this with the previous result, we have $e^{i_t - \rho + x_t} \mathbb{E}_t[e^{-x_{t+1} - \pi_{t+1}}] = e^{C_a x_t} e^{-C_a x_t} = 1$, which verifies (IS-dt).

The recessionary and deflationary properties (i.e., $x_t \leq 0$ and $\pi_t \leq 0$) are immediate from the construction above. From the Gamma distribution, the variance formula is

$$\text{Var}_t(x_{t+1}) = \text{Var}_t(y_{t+1}) = \frac{m_a r_a y_t}{r_a^2} = -\frac{m_a}{r_a} x_t,$$

so the constant $\nu_a > 0$ is given by $\nu_a = m_a / r_a$.

It remains to check Condition 2. Since $e^{(1+\varphi)x_t} \leq 1$, $|x_t| = y_t$, and $\pi_t^2 = a^2 y_t^2$, it suffices to show that $\sup_{j \geq 0} \mathbb{E}_t[y_{t+j}^2] < \infty$. First, compute

$$\mathbb{E}_t[y_{t+1}^2] = \text{Var}_t(y_{t+1}) + \mathbb{E}_t[y_{t+1}]^2 = \frac{m_a}{r_a} y_t + m_a^2 y_t^2$$

Using this result and the j -horizon mean $\mathbb{E}_t[y_{t+j}] = m_a^j y_t$, we have

$$\mathbb{E}_t[y_{t+j}^2] = \mathbb{E}_t[\mathbb{E}_{t+j-1}[y_{t+j}^2]] = \mathbb{E}_t\left[\frac{m_a}{r_a} y_{t+j-1} + m_a^2 y_{t+j-1}^2\right] = \frac{m_a^j}{r_a} y_t + m_a^2 \mathbb{E}_t[y_{t+j-1}^2]$$

Since this holds for all $j \geq 1$, we can work backwards j steps to get

$$\begin{aligned} \mathbb{E}_t[y_{t+j}^2] &= \frac{m_a^j}{r_a} y_t + m_a^2 \left(\frac{m_a^{j-1}}{r_a} y_t + m_a^2 \mathbb{E}_t[y_{t+j-2}^2] \right) \\ &= \dots = \left(\frac{m_a^j}{r_a} + \frac{m_a^{j+1}}{r_a} + \dots + \frac{m_a^{2j-1}}{r_a} \right) y_t + m_a^{2j} y_t^2 \end{aligned}$$

Recalling the fact that $m_a \in (0, 1)$, we can bound the first term by its infinite sequence:

$$\mathbb{E}_t[y_{t+j}^2] \leq \frac{m_a^j}{r_a} \frac{1}{1 - m_a} y_t + m_a^{2j} y_t^2 < \frac{1}{r_a(1 - m_a)} y_t + y_t^2$$

Hence, $\sup_{j \geq 0} \mathbb{E}_t[y_{t+j}^2] < \infty$. □

Proof of Proposition D.2. First, recall that for this fiscal policy the government debt valuation equation (GD-dt) is equivalently $\ell_t = \bar{s}/(1 - \beta)$. Now, we rewrite this condition in terms of pre-determined debt and current output and inflation. Define $b_{t-1} := \log(B_{t-1}/(P_{t-1}Y^*))$ and $\bar{b} := \log(\frac{\bar{s}}{1-\beta})$. In this notation, we have $\ell_t = e^{b_{t-1} - \pi_t - x_t}$ and so the restriction $\ell_t = \bar{s}/(1 - \beta)$ can be rewritten as

$$\pi_t + x_t = b_{t-1} - \bar{b}. \tag{D.4}$$

This immediately proves that the uncertainty traps of Proposition D.1 are no longer valid equilibria. Indeed, if $\pi_t = ax_t$, then (D.4) implies that $x_t = \frac{b_{t-1} - \bar{b}}{1+a}$, which is pre-determined as of date $t - 1$. Thus, it cannot be subject to time- t shocks. □

E Extension with idiosyncratic risk

We flesh out some details of the idiosyncratic risk extension discussed in Section 2.4.4. There is a continuum of agents of two types, labeled A and B. Recall the Brownian motion shock $d\tilde{Z}_t$ directly redistributes consumption across types. When $d\tilde{Z}_t > 0$, an amount $Y_t \nu(\omega_t, x_t) d\tilde{Z}_t$ of consumption is transferred from type-B to type-A agents; when $d\tilde{Z}_t < 0$ the transfer operates in the opposite direction. Here, C_t denotes aggregate output or consumption, while $\nu(\omega_t, x_t)$ is an exogenous function of the consumption

share of type-A agents ω_t and the output gap x_t . We consider the following functional form

$$v(\omega_t, x_t) = \sqrt{\omega_t(1 - \omega_t)}v(x_t) \quad (\text{E.1})$$

where $x_t = \log(Y_t/Y^*)$ denotes output gap. Consumption goods received through this redistribution must be consumed immediately. Within each type, the charge or transfer is allocated proportionally to the individual agents. Finally, there are no assets that allow agents to insure against $\tilde{Z}_t$ uncertainty.

We also allow a shift in the monetary policy target, described explicitly later, to keep consistency of the monetary policy rule with full stabilization, i.e., $x_t = \pi_t = 0$. Apart from this, the rest of the model is identical to the baseline specification, including the presence of the non-fundamental Brownian motion dZ_t .

Equilibrium characterization. The model aggregates within type, so we can work with a representative agent for each type. Write agent type i 's consumption dynamics

$$\frac{dc_{i,t}}{c_{i,t}} = \mu_{c_i,t}dt + \sigma_{c_i,t}dZ_t + \left(\mathbf{1}_{\{i=A\}} - \mathbf{1}_{\{i=B\}}\right) \frac{v(\omega_t, x_t)Y_t}{c_{i,t}}d\tilde{Z}_t. \quad (\text{E.2})$$

The transfers due to $\tilde{Z}$ are exogenous, and the agent is only allowed to optimally pick μ_{c_i} and σ_{c_i} . To solve this problem, we must consider a different “shadow SDF” for each type of agent given that markets are incomplete with respect to $\tilde{Z}_t$. The shadow SDF for agent type i follows

$$\frac{dM_{i,t}}{M_{i,t}} = -r_tdt - h_t dZ_t + v_{m_i,t}d\tilde{Z}_t \quad (\text{E.3})$$

where r_t is the risk-free real interest rate, h_t is the risk price associated to Z_t , and $v_{m_i,t}$ is a shadow risk price associated to the transfer shock $\tilde{Z}_t$.

Given logarithmic preferences, the FOC is $M_{i,t} = \lambda e^{-\rho t} c_{i,t}^{-1}$. By Itô's lemma,

$$\mu_{c_i,t} = r_t - \rho + h_t^2 + v(\omega_t, x_t)^2 \quad (\text{E.4})$$

$$\sigma_{c_i,t} = h_t \quad (\text{E.5})$$

$$\left(\mathbf{1}_{\{i=A\}} - \mathbf{1}_{\{i=B\}}\right) v(\omega_t, x_t)(Y_t/c_{t,i}) = -v_{m_i,t} \quad (\text{E.6})$$

Market clearing for goods is given by

$$Y_t = c_{A,t} + c_{B,t} \quad (\text{E.7})$$

and so

$$\frac{dY_t}{Y_t} = \omega_t \frac{dc_{A,t}}{c_{A,t}} + (1 - \omega_t) \frac{dc_{B,t}}{c_{B,t}} \quad (\text{E.8})$$

The dynamics of Y_t take the generic form

$$\frac{dY_t}{Y_t} = \mu_{y,t} dt + \sigma_{y,t} dZ_t + \nu_{y,t} d\tilde{Z}_t \quad (\text{E.9})$$

for some drift and diffusion coefficients. Substituting in the optimal consumption dynamics derived above into (E.8), we obtain

$$\begin{aligned} \mu_{y,t} &= r_t - \rho + h_t^2 + \omega_t \left(\frac{v(\omega_t, x_t)}{\omega_t} \right)^2 + (1 - \omega_t) \left(\frac{v(\omega_t, x_t)}{1 - \omega_t} \right)^2 \\ &= r_t - \rho + \sigma_{y,t}^2 + v(x_t)^2 \\ \sigma_{y,t} &= h_t \\ \nu_{y,t} &= v(\omega_t, x_t) - v(\omega_t, x_t) = 0 \end{aligned}$$

which verifies that the transfer shock does not affect aggregate dynamics directly. Given $x_t = \log(Y_t/Y^*)$, we can write the dynamic IS curve as

$$\mu_{x,t} = r_t - \rho + v(x_t)^2 + \frac{1}{2} \sigma_{x,t}^2 \quad (\text{E.10})$$

We consider the following monetary policy rule

$$\iota_t = \rho - v(0)^2 + \Phi(x_t, \pi_t) \quad (\text{E.11})$$

which has the target rate $\bar{\iota} = \rho - v(0)^2$. Then, using (E.11) inside (E.10), we have the final result

$$\mu_{x,t} = \Phi(x_t, \pi_t) - \pi_t + v(x_t)^2 - v(0)^2 + \frac{1}{2} \sigma_{x,t}^2 \quad (\text{E.12})$$

This proves equation (16) in the main text. Together with the Phillips curve, equation (E.12) characterizes the equilibrium.

F Taylor Principle in the non-stochastic model

To distinguish our main results, we also review the standard *deterministic* multiplicity in NK models and how aggressive monetary policy rules can eliminate this indeterminacy. For these results, we start by using a linear Taylor rule and the linearized Phillips curve (as in the literature). After that, we generalize these existing results to nonlinear Phillips curves and nonlinear Taylor rules as well.

F.1 Linearized Phillips curve

We study a deterministic setting. In this setting, recall that the MSV equilibrium is $x = \pi = 0$ forever, provided monetary policy sets the target rate at $\bar{r} = \rho$.

Can there exist other deterministic equilibria? As is well known, the answer to this question hinges on the stability/instability properties of the equilibrium dynamical system for (x_t, π_t) . We will review this analysis here. First, we specialize to the policy rule ([linear MP](#)) with the target rate $\bar{r} = \rho$. Combining ([linear MP](#)) with ([IS](#)), the dynamics of x_t are given by

$$\dot{x}_t = \phi_x x_t + (\phi_\pi - 1)\pi_t. \quad (\text{F.1})$$

The IS curve is linear in a deterministic equilibrium with a linear Taylor rule. We also consider here the linear Philips curve ([linear PC](#)) as in most of the existing literature.

The typical determinacy analysis picks an aggressive Taylor rule that renders the above system unstable. The system ([F.1](#)) and ([linear PC](#)) can be written in matrix form as

$$\begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \end{bmatrix} = \mathcal{A}_{\text{linear}} \begin{bmatrix} x_t \\ \pi_t \end{bmatrix}, \quad \text{where} \quad \mathcal{A}_{\text{linear}} := \begin{bmatrix} \phi_x & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix}. \quad (\text{F.2})$$

The eigenvalues of $\mathcal{A}_{\text{linear}}$ are both strictly positive, and the system unstable, if $\phi_x > -\rho$ and $\phi_\pi > 1 - \rho\phi_x/\kappa$. This is the continuous-time version of the eigenvalue conditions in [Blanchard and Kahn \(1980\)](#).

By contrast, if either parameter condition is violated, then the system has one or two stable eigenvalues. In such case, there are a continuum of non-explosive equilibria, which one can index by the initial conditions (x_0, π_0) . As the explicit parameter conditions make clear, instability occurs when monetary policy is sufficiently aggressive (i.e., “active”) in responding to the output gap and inflation, whereas stability occurs when the response function is less aggressive (i.e., “passive”).

Proposition F.1. *Consider the linearized Phillips curve (linear PC) and monetary policy rule (linear MP) with $\bar{r} = \rho$. Restrict attention to deterministic equilibria ($\sigma_x = \sigma_\pi = 0$). If $\phi_x > -\rho$ and $\phi_\pi > 1 - \rho\phi_x/\kappa$, the only deterministic non-explosive equilibrium is $(x_t, \pi_t) = (0, 0)$ forever—the MSV solution. If either condition is violated, then a continuum of deterministic non-explosive equilibria exist.*

Proof of Proposition F.1. This result is a special case of the proof of Proposition H.1 in Appendix H. The eigenvalues (λ_1, λ_2) of $\mathcal{A}_{\text{linear}}$ are the key. Asymptotic instability of this system is guaranteed if $\text{Re}(\lambda_1), \text{Re}(\lambda_2) > 0$. This holds if and only if $\det(\mathcal{A}_{\text{linear}}) > 0$ and $\text{tr}(\mathcal{A}_{\text{linear}}) > 0$, which is equivalent to $\phi_x > -\rho$ and $\phi_\pi > 1 - \rho\phi_x/\kappa$. $\square$

Remark F.1 (Nonlinear Phillips curve). *We have used the linearized Phillips curve here for simplicity and exposition. We analyze the nonlinear Phillips curve in Appendix F.2, and the conclusion is similar to Proposition F.1 but the proof is more complicated.*

Remark F.2 (Explosive equilibria). *A key clause is the requirement that equilibria satisfy Condition 1, ruling out explosions. What if asymptotic explosions were permitted, while finite-time explosions were ruled out? It turns out that, by adopting an aggressive nonlinear Taylor rule, monetary policy can force an explosion in finite-time, and hence select a unique equilibrium. We analyze this situation in Appendix F.3. In that sense, the spirit of Proposition F.1 is preserved even under broader notions of equilibrium.*

F.2 Nonlinear Phillips curve

This section briefly explores the stability properties of the nonlinear Phillips curve, in contrast to the linearized version used oftentimes in the paper. We will do this only in the context of deterministic equilibria, for simplicity. For convenience, we repeat this nonlinear equation here:

$$\dot{\pi}_t = \rho\pi_t - \kappa \left(\frac{e^{(1+\varphi)x_t} - 1}{1 + \varphi} \right). \quad (\text{F.3})$$

We also repeat the IS curve after substituting the linear Taylor rule with target rate $\bar{r} = \rho$:

$$\dot{x}_t = \phi_x x_t + (\phi_\pi - 1)\pi_t. \quad (\text{F.4})$$

A deterministic non-explosive equilibrium in this environment is (x_t, π_t) that satisfy (F.3)-(F.4) and asymptotic non-explosion Condition 1.

The nonlinearity of the Phillips curve does not change the basic determinacy result of Proposition F.1, as we show next (although our proof requires stronger assumptions on the Taylor rule to ensure global determinacy).

Proposition F.2. *Consider the system (F.3)-(F.4) with $\phi_x > \rho$ and $\phi_\pi > 1$. Then, the only initial pair (x_0, π_0) consistent with a deterministic non-explosive equilibrium is $(x_0, \pi_0) = (0, 0)$. Any other initial pair diverges, but only asymptotically (i.e., not in finite time).*

Proof of Proposition F.2. Define $f(x) := \frac{e^{(1+\varphi)x} - 1}{1+\varphi}$. From (F.3)-(F.4), the steady state solves

$$-\phi_x x = (\phi_\pi - 1)\kappa\rho^{-1}f(x)$$

The two sides of this equation have opposite slopes in x , so the unique solution is $x = 0$, proving the unique steady state is $(x, \pi) = (0, 0)$. The steady state is locally unstable, by the same linearized eigenvalue analysis leading to Proposition F.1. By the local stable manifold theorem, we have that the unique stable solution to the dynamics is in fact this steady state. We now prove that any non-explosive equilibrium (satisfying Condition 1) must have $(x_t, \pi_t) = (0, 0)$ for all t . Assume not, i.e., assume, leading to contradiction, that $x_t \in [\underline{x}, \bar{x}]$ for all $t > 0$, where $\underline{x} < 0 < \bar{x}$.

First, from (F.3),

$$e^{-\rho t}\pi_t - \pi_0 = -\kappa \int_0^t e^{-\rho s} f(x_s) ds \quad (\text{F.5})$$

Substituting (F.5) into (F.4), we have

$$\dot{x}_t = \phi_x x_t + (\phi_\pi - 1) \left[e^{\phi_x t} \pi_0 - \frac{\kappa}{\rho} \int_0^t \rho e^{\rho(t-u)} f(x_u) du \right] \quad (\text{F.6})$$

Under the boundedness assumption, we may bound $f(\underline{x}) \leq f(x_t) \leq f(\bar{x})$, which when plugging into (F.6) leads to

$$\underbrace{\phi_x x_t + (\phi_\pi - 1) \left[e^{\phi_x t} \pi_0 - \frac{\kappa}{\rho} (e^{\rho t} - 1) f(\bar{x}) \right]}_{:=L_t} \leq \dot{x}_t \leq \underbrace{\phi_x x_t + (\phi_\pi - 1) \left[e^{\phi_x t} \pi_0 - \frac{\kappa}{\rho} (e^{\rho t} - 1) f(\underline{x}) \right]}_{:=U_t}$$

If $\pi_0 > 0$, then $L_t, U_t \rightarrow +\infty$ as $t \rightarrow \infty$ for every possible value of $x_t \in [\underline{x}, \bar{x}]$. On the other hand, if $\pi_0 < 0$, then $L_t, U_t \rightarrow -\infty$ as $t \rightarrow \infty$ for every possible value of $x_t \in [\underline{x}, \bar{x}]$. Hence, $\pi_0 > 0$ implies $x_T > \bar{x}$ for some $T > 0$, while $\pi_0 < 0$ implies $x_T < \underline{x}$ for some $T > 0$. This contradicts the bounded set $x_t \in [\underline{x}, \bar{x}]$, which implies $\pi_0 = 0$ is required.

However, since time 0 is arbitrary in this analysis, and the entire argument could be shifted forward in time, we in fact require $\pi_t = 0$ for all $t \geq 0$. Going back to equation (F.3), we then have that $x_t = 0$ for all $t \geq 0$. $\square$

F.3 Very nonlinear Taylor rules and finite-time explosions

Suppose we would like to allow deterministic equilibria that explode asymptotically, in violation of Condition 1. For instance, [Cochrane \(2011\)](#) considers some types of asymptotically exploding equilibria in his argument for non-uniqueness. In that case, is the spirit of Proposition F.1 still true, i.e., do there exist Taylor rules which can eliminate indeterminacies? The answer is yes, but a “nuclear Taylor rule” is required to force explosion in finite time.

In particular, let us dispense with the linear rule ([linear MP](#)). Suppose the response function ([MP](#)) takes the nonlinear form

$$\Phi(x, \pi) = \frac{\phi_x}{2}(e^x - e^{-x}) + \pi \quad (\text{F.7})$$

with $\phi_x > 0$ and suppose the target rate is again the flexible-price rate $\bar{r} = \rho$. Note that the log-linearized version of (F.7) renders the linear Taylor rule ([linear MP](#)) with $\phi_\pi = 1$.

Combining (F.7) with ([IS](#)), the dynamics of x_t are given by

$$\dot{x}_t = \frac{\phi_x}{2}(e^{x_t} - e^{-x_t}) \quad (\text{F.8})$$

This ODE has solution

$$x_t = \log \left(\frac{1 - Ke^{\phi_x t}}{1 + Ke^{\phi_x t}} \right)$$

where $K = \frac{1-e^{x_0}}{1+e^{x_0}}$. This process diverges in *finite time* for any $x_0 \neq 0$: it explodes at time $T = -\phi_x^{-1} \log(|K|)$. Hence, we have proved by construction the following result.

Proposition F.3. *Taylor rules exist such that any deterministic equilibrium has $x_t = 0$ forever.*

The analysis above abstracts from any feedback effects from inflation to output gap by setting a monetary policy rule with $\phi_\pi = 1$. This serves two purposes. First, it emphasizes the focus on self-fulfilling demand and not inflation per se. Equilibrium characterization requires the output gap to remain bounded for any finite horizon. There is no such requirement for inflation (e.g., hyperinflation might be an equilibrium outcome). Second, it simplifies the analysis and illustrates the point with examples that

permit closed form solutions. As an additional benefit, Proposition F.3 holds for either the linearized or non-linear Phillips curves.

Determinacy extends beyond the particular response function (F.7) that has exactly a one-for-one inflation response. In particular, consider inflation sensitivities of more than one-for-one, such as

$$\Phi(x, \pi) = \frac{\phi_x}{2}(e^x - e^{-x}) + \phi_\pi \pi, \quad \phi_x > 0, \phi_\pi > 1. \quad (\text{F.9})$$

While more challenging technically to analyze, this rule also selects the zero output gap equilibrium $x_t = 0$. We demonstrate this result formally next.

Under rule (F.9), the dynamical system for (x_t, π_t) is

$$\dot{\pi}_t = \rho \pi_t - \kappa f(x_t) \quad (\text{F.10})$$

$$\dot{x}_t = \frac{\phi_x}{2}(e^{x_t} - e^{-x_t}) + (\phi_\pi - 1)\pi_t \quad (\text{F.11})$$

where $f(x) := (1 + \varphi)^{-1}[e^{(1+\varphi)x} - 1]$.

Proposition F.4. *Consider the system (F.10)-(F.11) with $\phi_x > 0$ and $\phi_\pi > 1$. Then, $(x_t, \pi_t) = (0, 0)$ is the unique equilibrium that does not explode in finite time.*

Proof of Proposition F.4. Suppose the solution $(x_t(\phi_\pi), \pi_t(\phi_\pi))_{t \geq 0}$ associated to some $\phi_\pi > 1$ (which is unique prior to an explosion by the standard ODE uniqueness theorem) did not explode in finite time. In that case, because the solution is continuous in ϕ_π (again, standard ODE theorems ensure this), it follows that the solution $(x_t(\tilde{\phi}_\pi), \pi_t(\tilde{\phi}_\pi))_{t \geq 0}$ associated with $\tilde{\phi}_\pi < \phi_\pi$ also does not explode in finite time. Continuity requires this: otherwise, the two solutions would be infinitely far apart at some finite time T when one of the solutions does explode. But Proposition F.3 has already shown that $(x_t(1), \pi_t(1))_{t \geq 0}$ is explosive in finite time, a contradiction. $\square$

G Inflation dynamics under Rotemberg

Here, we generalize the sticky-price model of Rotemberg (1982) to our environment. Since firms in our economy are ex-ante identical, they will have identical utilization and price-setting incentives, allowing us to study a representative firm's problem and a symmetric equilibrium.

To set up the representative intermediate-goods-producer problem, let l_t denote the firm's hired labor, at some equilibrium wage W_t . The firm produces $y_t = l_t$. The firm

makes its price choice p_t , internalizing its demand $y_t = (p_t/P_t)^{-\varepsilon}Y_t$, where P_t and Y_t are the aggregate price and output. This demand curve comes from an underlying Dixit-Stiglitz structure with CES preferences (with substitution elasticity $\varepsilon > 1$) and monopolistic competition in the intermediate goods sector.

Letting M_t denote the real SDF process, the representative firm solves

$$\sup_{p,l} \mathbb{E} \left[\int_0^\infty M_t \left(\frac{p_t}{P_t} y_t - \frac{W_t l_t}{P_t} - \frac{1}{2\eta} \left(\frac{1}{dt} \frac{dp_t}{p_t} \right)^2 Y_t \right) dt \right] \quad (\text{G.1})$$

$$\text{subject to } y_t = (p_t/P_t)^{-\varepsilon} Y_t \quad (\text{G.2})$$

$$y_t = l_t \quad (\text{G.3})$$

The quadratic price adjustment cost in (G.1) has a penalty parameter η . As $\eta \rightarrow 0$ ($\eta \rightarrow \infty$), prices become permanently rigid (flexible). We assume that this price adjustment cost is purely non-pecuniary for simplicity (this means that adjustment costs do not affect the resource constraint). Alternatively, we could redistribute these adjustment costs lump-sum to the representative household.

Before solving the problem, we can immediately note the following property: price changes are necessarily absolutely continuous (“order dt ”). Indeed, the adjustment cost per unit of time is a function of price changes per unit of time, i.e., $\frac{1}{dt} \frac{dp_t}{p_t}$. If prices were to change faster than dt , say with the Brownian motion dZ_t , then $\frac{1}{dt} \frac{dp_t}{p_t}$ would be unbounded almost-surely (because Brownian motion is nowhere-differentiable), leading to infinite adjustment costs. Consequently, we know that $\frac{1}{dt} \frac{dp_t}{p_t} = \frac{\dot{p}_t}{p_t}$ for some $\dot{p}_t$.

The firm’s optimal price sequence solves a dynamic optimization problem. Substituting the demand curve from (G.2) and the production function from (G.3), we may rewrite problem (G.1) as

$$\sup_{\dot{p}} \mathbb{E}_t \left[\int_t^\infty \frac{M_s Y_s}{M_t Y_t} \left(\left(\frac{p_s}{P_s} \right)^{1-\varepsilon} - \frac{W_s}{P_s} \left(\frac{p_s}{P_s} \right)^{-\varepsilon} - \frac{1}{2\eta} \left(\frac{\dot{p}_s}{p_s} \right)^2 \right) ds \right].$$

Furthermore, note that in the log utility model used in the text, we have $M_t Y_t = e^{-\rho t}$. Letting J denote this firm’s value function, the HJB equation is

$$0 = \sup_{\dot{p}_t} \left\{ \left(\frac{p_t}{P_t} \right)^{1-\varepsilon} - \frac{W_t}{P_t} \left(\frac{p_t}{P_t} \right)^{-\varepsilon} - \frac{1}{2\eta} \left(\frac{\dot{p}_t}{p_t} \right)^2 - \rho J_t + \frac{1}{dt} \mathbb{E}_t [dJ_t] \right\}$$

The firm value function follows a process of the form

$$dJ_t = [\mu_{J,t} + \dot{p}_t \frac{\partial}{\partial p} J_t] dt + \sigma_{J,t} dZ_t,$$

where $\mu_{J,t}$ and $\sigma_{J,t}$ are only functions of aggregate states (not the individual price). The only part that the firm can affect is $\dot{p}_t \frac{\partial}{\partial p} J_t$. Plugging these results back into the HJB equation and taking the FOC, we have

$$0 = -\frac{1}{\eta} \left(\frac{\dot{p}_t}{p_t} \right) \frac{1}{p_t} + \frac{\partial}{\partial p} J_t \quad (\text{G.4})$$

Differentiating the HJB equation with respect to the state variable p_t , we have the envelope condition

$$(\varepsilon - 1) \left(\frac{p_t}{P_t} \right)^{-\varepsilon} \frac{1}{P_t} - \varepsilon \frac{W_t}{P_t} \left(\frac{p_t}{P_t} \right)^{-\varepsilon-1} \frac{1}{P_t} = \frac{1}{\eta} \left(\frac{\dot{p}_t}{p_t} \right)^2 \frac{1}{p_t} - \rho \frac{\partial}{\partial p} J_t + \frac{1}{dt} \mathbb{E}_t \left[d \left(\frac{\partial}{\partial p} J_t \right) \right], \quad (\text{G.5})$$

where the last term uses the stochastic Fubini theorem. Combining equations (G.4) and (G.5), we have

$$\eta(\varepsilon - 1) \left(\frac{p_t}{P_t} \right)^{-\varepsilon} \frac{1}{P_t} - \eta \varepsilon \frac{W_t}{P_t} \left(\frac{p_t}{P_t} \right)^{-\varepsilon-1} \frac{1}{P_t} = \left(\frac{\dot{p}_t}{p_t} \right)^2 \frac{1}{p_t} - \rho \left(\frac{\dot{p}_t}{p_t} \right) \frac{1}{p_t} + \frac{1}{dt} \mathbb{E}_t \left[d \left(\left(\frac{\dot{p}_t}{p_t} \right) \frac{1}{p_t} \right) \right] \quad (\text{G.6})$$

At this point, define the firm-level inflation rate $\pi_t := \dot{p}_t / p_t$, note that $\mathbb{E}_t \left[d \left(\pi_t \frac{1}{p_t} \right) \right] = \frac{1}{p_t} \mathbb{E}_t [d\pi_t] - \frac{1}{p_t} \pi_t^2 dt$, and use the symmetry assumption $p_t = P_t$ in (G.6) to get

$$\eta(\varepsilon - 1) - \eta \varepsilon \frac{W_t}{P_t} = -\rho \pi_t + \frac{1}{dt} \mathbb{E}_t [d\pi_t]. \quad (\text{G.7})$$

Equation (G.7) is the continuous-time stochastic Phillips curve, with π_t interpreted also as the aggregate inflation rate (given a symmetric equilibrium).

Finally, note that the firm's optimization problem also requires the following transversality condition (see Theorem 9.1 of Fleming and Soner (2006)):

$$\lim_{T \rightarrow \infty} \mathbb{E}_t [M_T Y_T J_T] = 0.$$

In a symmetric equilibrium ($p = P$), and using the log utility result $M_t Y_t = e^{-\rho t}$, we

have that

$$M_T Y_T J_T = \mathbb{E}_T \left[\int_T^\infty e^{-\rho t} \left(1 - (Y^*)^{1+\varphi} e^{(1+\varphi)x_t} - \frac{1}{2\eta} \pi_t^2 \right) dt \right]$$

Take expectations and the limit $T \rightarrow \infty$. A sufficient condition for the firm transversality condition is that

$$\int_0^\infty e^{-\rho t} \mathbb{E} \left[e^{(1+\varphi)x_t} + \pi_t^2 \right] dt < \infty,$$

and that the corresponding discounted terminal terms vanish:

$$\lim_{T \rightarrow \infty} \mathbb{E}_t [e^{(1+\varphi)x_T - \rho T}] = 0 \quad (\text{G.8})$$

$$\lim_{T \rightarrow \infty} \mathbb{E}_t [e^{-\rho T} \pi_T^2] = 0. \quad (\text{G.9})$$

Condition 1 implies the integrability requirement above. It also implies (G.8)–(G.9), because $\mathbb{E}[e^{(1+\varphi)x_T}]$ and $\mathbb{E}[\pi_T^2]$ are bounded along the tail. Hence the firm transversality condition holds.

H Equilibrium with fiscal dominance

Theorems 4 only provides a “local” result, i.e., that $\sigma_x = 0$, without characterizing the full dynamic equilibrium. They also show that inflation is not determined from the debt valuation equation alone. Monetary policy is needed to pin inflation down.

For tractability, we specialize here to a quasi-linear setting that corresponds to Example B.1 in Section B.1. In particular, this section assumes a linearized Phillips curve (linear PC), the linear Taylor rule (linear MP), and an Ornstein-Uhlenbeck process for monetary shocks $d\mathcal{E}_t = -\zeta \mathcal{E}_t dt + dZ_t$, which is the continuous-time version of an AR(1). Section B.1 presents a closed-form solution for the MSV equilibrium in this setting.

Here with active fiscal policy, we will obtain a different solution. For the fiscal side, we assume the surplus-to-output has dynamics following $ds_t = \lambda_s(\bar{s} - s_t) + \varsigma_{s,t} \cdot dZ_t$, for some arbitrary volatility process $\varsigma_{s,t}$. We also assume debt is short-term as in the baseline specification. Finally, to keep the analysis simple, we assume monetary policy adopts the time-varying target rate $\bar{r}_t = \rho - \frac{1}{2}|\varsigma_{x,t}|^2$, where ς_x is the endogenous sensitivity of x to the fiscal shocks dZ . Because Theorem 4 pins down $\varsigma_{x,t}$ uniquely, its inclusion in the target rate is conceptually distinct from the “risk targeting” we studied in Section 3.1.

The present target rate only serves as a normalization, so that the economy fluctuates around $(x, \pi) = (0, 0)$ rather than inheriting a non-zero steady-state.

Writing the equilibrium dynamics in vector form, with $F_t := (x_t, \pi_t, \mathcal{E}_t)'$, we have

$$dF_t = \mathcal{A}F_t dt + \mathcal{B}_t dZ_t + \mathcal{C}_t d\mathcal{Z}_t,$$

where $\mathcal{A} := \begin{bmatrix} \phi_x & \phi_\pi - 1 & \sigma \\ -\kappa & \rho & 0 \\ 0 & 0 & -\zeta \end{bmatrix}$, $\mathcal{B}_t := \begin{bmatrix} 0 \\ \sigma_{\pi,t} \\ 1 \end{bmatrix}$, and $\mathcal{C}_t := \begin{bmatrix} \zeta'_{x,t} \\ \zeta'_{\pi,t} \\ 0 \end{bmatrix}$

Notice that the first entry of $\mathcal{B}_t$ is zero, because of Theorem 4. Theorem 4 also restricts the first entry of surplus shock loadings $\mathcal{C}_t$. We follow a relatively standard analysis by doing a spectral decomposition of the transition matrix $\mathcal{A} = V\Lambda V^{-1}$, and analyzing the rotated system $\tilde{F}_t := V^{-1}F_t$. By integrating the system $d\tilde{F}_t = \Lambda\tilde{F}_t dt + V^{-1}\mathcal{B}_t dZ_t + V^{-1}\mathcal{C}_t \cdot d\mathcal{Z}_t$, we obtain

$$\mathbb{E}_0 \tilde{F}_t = \exp(\Lambda t) \tilde{F}_0. \quad (\text{H.1})$$

The rest is a familiar stability analysis of (H.1), in view of the non-explosion Condition 1. One eigenvalue of $\mathcal{A}$ is $-\zeta$, corresponding to the exogenous monetary shock. There are three cases regarding the other two eigenvalues that correspond to the dynamics of (x, π) : both eigenvalues have positive real parts, the eigenvalues have opposite signs, or both eigenvalues have negative real parts. Pursuing this analysis, we then obtain the following generalization of some familiar results.

Proposition H.1. *Consider the economy with: the linearized Phillips curve (linear PC); the linear Taylor rule (B.1) with target rate $\bar{\pi}_t = \rho - \frac{1}{2}|\zeta_{x,t}|^2$ and monetary shocks $\sigma\mathcal{E}_t$, where $\mathcal{E}_t$ follows the process $d\mathcal{E}_t = -\zeta\mathcal{E}_t dt + dZ_t$; and the surplus dynamics $ds_t = \lambda_s(\bar{s} - s_t) + \zeta_{s,t} \cdot d\mathcal{Z}_t$. A non-explosive equilibrium takes one of three forms, ignoring knife-edge cases for the parameters:*

1. *If $\rho + \phi_x > 0$ and $\rho\phi_x + \kappa(\phi_\pi - 1) > 0$, then equilibrium generically fails to exist.*
2. *If $\rho\phi_x + \kappa(\phi_\pi - 1) < 0$, the unique equilibrium features*

$$\pi_t = \frac{1}{\phi_\pi - 1} \left[\beta x_t + \frac{\beta(\rho + \zeta) - \kappa(\phi_\pi - 1)}{(\phi_x + \zeta)(\rho + \zeta) + \kappa(\phi_\pi - 1)} \sigma \mathcal{E}_t \right],$$

$$\text{where } \beta := \frac{1}{2}(\rho - \phi_x - \sqrt{(\rho - \phi_x)^2 - 4\kappa(\phi_\pi - 1)}).$$

3. *If $\rho + \phi_x < 0$ and $\rho\phi_x + \kappa(\phi_\pi - 1) > 0$, then the equilibrium is not unique.*

In all cases, the output gap is pinned down by fiscal states $(\frac{B_t}{P_t}, s_t)$ as $x_t = \log(\frac{B_t/P_t}{\Psi(s_t)Y^*})$ for some function $\Psi(s)$.

Proposition [H.1](#) is reminiscent of the large literature of fiscal theory of the price level (FTPL) in linearized NK models. Equilibrium cannot exist with both “active fiscal” and “active money” regimes (case 1). Equilibrium exists and is unique when “passive money” is paired with “active fiscal” (case 2). Passive money just means that $\rho\phi_x + \kappa(\phi_\pi - 1) < 0$, which says that monetary policy is not too aggressive. These results echo [Leeper \(1991\)](#). A finding which differs slightly from the literature is our case 3: monetary policy that acts super aggressively against inflation but acts counterintuitively to output induces non-uniqueness, despite active fiscal policy. This case can have self-fulfilling inflation dynamics because monetary policy induces globally stable dynamics through its rule.

Of these three cases, the interesting case is the active-fiscal passive-money regime (case 2), which delivers a unique equilibrium. There are two important takeaways. First, FTPL ensures uniqueness for a broad range of monetary policy rules and not some knife-edge rule. A unique equilibrium is achieved even under an interest rate peg, which fits into case 2 by $\phi_x = \phi_\pi = \sigma = 0$. Second, the FTPL equilibrium is observationally distinct from the self-fulfilling stochastic equilibria under the Taylor principle. Under FTPL, the output gap x_t is pinned down as a function of the real debt balance B_t/P_t and the primary surplus level s_t . All output gap volatility is tied to fiscal states and not directly to monetary shocks. By contrast, the volatile equilibria possible under the Taylor principle feature an entire class of possible output gap dynamics which are all decoupled from fiscal states.

Proof of Proposition [H.1](#). First, note that the spectral decomposition of $\mathcal{A} = V\Lambda V^{-1}$ is

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \quad \text{and} \quad V = \begin{bmatrix} v(\lambda_1) & v(\lambda_2) & v(\lambda_3) \end{bmatrix},$$

where the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ and the corresponding eigenvectors $v(\lambda_1), v(\lambda_2), v(\lambda_3)$ are

$$\begin{aligned} \lambda_1 &= \frac{1}{2} \left[\rho + \phi_x + \sqrt{(\rho - \phi_x)^2 - 4\kappa(\phi_\pi - 1)} \right] \\ \lambda_2 &= \frac{1}{2} \left[\rho + \phi_x - \sqrt{(\rho - \phi_x)^2 - 4\kappa(\phi_\pi - 1)} \right] \\ \lambda_3 &= -\zeta. \end{aligned}$$

and

$$v(\lambda_1) = \begin{pmatrix} \frac{\phi_\pi - 1}{\lambda_1 - \phi_x} \\ 1 \\ 0 \end{pmatrix}, \quad v(\lambda_2) = \begin{pmatrix} \frac{\phi_\pi - 1}{\lambda_2 - \phi_x} \\ 1 \\ 0 \end{pmatrix}, \quad v(\lambda_3) = \begin{pmatrix} -(\rho + \zeta)\sigma \\ -\kappa\sigma \\ (\phi_x + \zeta)(\rho + \zeta) + \kappa(\phi_\pi - 1) \end{pmatrix}$$

Recall equation (H.1) that

$$\mathbb{E}_0[\tilde{F}_t] = \exp(\Lambda t) \tilde{F}_0, \quad (\text{H.2})$$

where $\tilde{F}_t = V^{-1}F_t$ is a rotated version of the state $F_t = (x_t, \pi_t, \mathcal{E}_t)'$, and where

$$V^{-1} = \frac{1}{\lambda_2 - \lambda_1} \begin{bmatrix} \frac{(\lambda_1 - \phi_x)(\lambda_2 - \phi_x)}{\phi_\pi - 1} & -(\lambda_1 - \phi_x) & \frac{\lambda_1 - \phi_x}{\phi_\pi - 1} \frac{(\lambda_2 - \phi_x)(\rho + \zeta) - \kappa(\phi_\pi - 1)}{(\phi_x + \zeta)(\rho + \zeta) + \kappa(\phi_\pi - 1)} \sigma \\ -\frac{(\lambda_1 - \phi_x)(\lambda_2 - \phi_x)}{\phi_\pi - 1} & \lambda_2 - \phi_x & -\frac{\lambda_2 - \phi_x}{\phi_\pi - 1} \frac{(\lambda_1 - \phi_x)(\rho + \zeta) - \kappa(\phi_\pi - 1)}{(\phi_x + \zeta)(\rho + \zeta) + \kappa(\phi_\pi - 1)} \sigma \\ 0 & 0 & \frac{\lambda_2 - \lambda_1}{(\phi_x + \zeta)(\rho + \zeta) + \kappa(\phi_\pi - 1)} \end{bmatrix}.$$

In equation (H.2), $\exp(\Lambda t)$ refers to element-by-element exponentiation of Λ .

Let's consider the three cases of the proposition, using Condition 1 to kill explosive solutions to (H.2):

1. *Case 1:* $\rho + \phi_x > 0$ and $\rho\phi_x + \kappa(\phi_\pi - 1) > 0$.

In this case, $\text{Re}(\lambda_1), \text{Re}(\lambda_2) > 0$. Therefore, all non-explosive solutions to (H.2) must satisfy $\tilde{F}_t^{(1)} = \tilde{F}_t^{(2)} = 0$. Using the expression for V^{-1} , this implies that

$$x_t = -\frac{(\rho + \zeta)\sigma}{(\phi_x + \zeta)(\rho + \zeta) + \kappa(\phi_\pi - 1)} \mathcal{E}_t \quad \text{and} \quad \pi_t = -\frac{\kappa\sigma}{(\phi_x + \zeta)(\rho + \zeta) + \kappa(\phi_\pi - 1)} \mathcal{E}_t$$

2. *Case 2:* $\rho\phi_x + \kappa(\phi_\pi - 1) < 0$.

In this case, both eigenvalues are real and have opposite signs: $\lambda_1 > 0 > \lambda_2$. Therefore, all non-explosive solutions to (H.2) must satisfy $\tilde{F}_t^{(1)} = 0$, which using the expression for V^{-1} implies

$$\pi_t = \frac{\lambda_2 - \phi_x}{\phi_\pi - 1} x_t + \frac{(\lambda_2 - \phi_x)(\rho + \zeta) - \kappa(\phi_\pi - 1)}{(\phi_x + \zeta)(\rho + \zeta) + \kappa(\phi_\pi - 1)} \frac{\sigma}{\phi_\pi - 1} \mathcal{E}_t. \quad (\text{H.3})$$

Given $\sigma_{x,t} = 0$ from Theorem 4, this then implies $\sigma_{\pi,t} = 0$ as well.

3. *Case 3:* $\rho + \phi_x < 0$ and $\rho\phi_x + \kappa(\phi_\pi - 1) > 0$.

In this case, $\text{Re}(\lambda_1), \text{Re}(\lambda_2) < 0$, meaning all initial conditions to (H.2) are non-explosive. Therefore, any $\tilde{F}_0$ corresponds to a valid equilibrium.

In all cases, we note that x_0 and $\varsigma_{x,t}$ are pinned down by (GD) at $t = 0$ and at $t > 0$, respectively. Indeed, equation (GD) says that $b_t = \mathbb{E}_t[\int_t^\infty e^{-\rho(u-t)} s_u du]$. Using $ds_t = \lambda_s[s_t - \bar{s}]dt + \varsigma_{s,t} \cdot dZ_t$ to compute the present-value on the right-hand-side, we obtain

$$b_t = b(s_t) := \frac{\bar{s}}{\rho} + \frac{s_t - \bar{s}}{\rho + \lambda_s},$$

which is exogenous. On the other hand, recall the definition of real debt-to-GDP $b_t = B_t/P_t Y_t = e^{-x_t} B_t/P_t Y^*$. Comparing these two expressions, we obtain

$$x_t = \log\left(\frac{B_t/P_t}{b(s_t)Y^*}\right). \quad (\text{H.4})$$

Applying Itô's formula to this expression, we can also obtain the sensitivity to fiscal shocks:

$$\varsigma_{x,t} = -\frac{\rho}{\lambda_s \bar{s} + \rho s_t} \varsigma_{s,t}. \quad (\text{H.5})$$

Thus, x_t and $\varsigma_{x,t}$ are pinned down by B_t/P_t and s_t .

The remaining claims to prove are the existence/uniqueness statements. In Case 1, the equilibrium fails to exist generically, because $x_t \propto \sigma \mathcal{E}_t$ cannot be consistent with (H.4) and (H.5). In Case 2, the equilibrium is unique, because the initial condition $\mathcal{E}_0$ is given exogenously, x_0 is pinned down by (H.4), and so π_0 is pinned down by (H.3), and because the surplus shock exposures $\varsigma_{x,t}$ and $\varsigma_{\pi,t} = \frac{\lambda_2 - \phi_x}{\phi_\pi - 1} \varsigma_{x,t}$ are pinned down by (H.5). In Case 3, the equilibrium is not unique because, although x_0 is pinned down by (H.4), π_0 is not pinned down. Furthermore, π_t can have arbitrary non-fiscal volatility $\sigma_{\pi,t}$, despite the fact that $\sigma_{x,t} = 0$. $\square$